Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
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DEFINITIONS AND ABBREVIATIONS

Document	Information and the medium on which it is contained (ISO 9000:2015)
Employer	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	Ma'aden Project Management Team
'MMGP Project'	Mansourah Massarah Gold Processing Project
Attribute	Information element that describes a document or tag.
Supplier	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	Quality Assurance
QC	Quality Control
Works	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of Grinding & Classification (3220) - Relining Machine Structure Foundation. The member sizes and reinforcements are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD Pro. The STAAD modelling has been done with beam & plate elements. Plate mat with subgrade modulus has been considered as per L&T approved Soil Report. Base Pressure check has been done from STAAD Pro output.

For strength calculation Bending moment & Shear Force values are taken from STAAD output & designed in the excel sheet.Isometric views of STAAD model are attached in Attachment-2.

4. DESIGN REQUIREMENTS

4.1. Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A02 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

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4.2. Materials Property

Density of Material
- Density of Dry Soil : 1.9 T/m^3
- Density of Concrete : 2.5 T/m^3
- Density of Water : 1.0 T/m^3

Concrete Strength of 28 days
- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade
- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

Grout Grade
- Grout shall be cementitious under steel column bases of Strength 40 MPa

4.3. Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	Minimum Cove (mm)
Exposed to Soil	Foundation Bottom & Side	75
	Foundation Top	50
	Column & Beam	50
Exposed to Weather	Column	50
	Beam	50
	Slab	40

4.5. Allowable Soil Bearing Capacity

Recommended net allowable bearing capacity as per L&T soil report MD-216-3000-EG-CV-RPT-1008-A02, bore hole BH-02.

BH-no	Depth of foundation below FGL(m)	Recommended net allowable bearing capacity(KN/m2)	Vertical modulus of subgrage reaction(KN/m2/m)
BH-02	1.5	750	90000

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5. SAFETY FACTOR AGAINST STABILITY

5.1. Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0
5.2. Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

6. LOAD CALCULATIONS

6.1. Primary Load Cases

The primary load values are calculated in the excel sheet. The primary loads are mention as given below.

1 DEAD LOAD (DL)
2 LIVE LOAD (LL)
3 OPERATING FLUID LOAD (FL)
4 EQUIPMENT DEAD LOAD (EQDL)
5 EQUIPMENT LIVE LOAD (EQLL)
6 STARTING BELT TENSION (SBT)
7 WIND LOAD (+)X (WL +X)
8 WIND LOAD (-)X (WL -X)
9 WIND LOAD (+)Z (WL +Z)
10 WIND LOAD (-)Z (WL -Z)
11 SEISMIC LOAD (SL +X)
12 SEISMIC LOAD (SL -X)
13 SEISMIC LOAD (SL +Z)
14 SEISMIC LOAD (SL -Z)
15 TEMPERATURE LOAD (TL)
16 PIPE LATERAL LOAD (PL)
17 EQUIPMENT TEST LOAD (EQTL)
18 MONORAIL LOAD (ML1)
19 MONORAIL LOAD (ML2)
20 MONORAIL LOAD (ML3)
21 MONORAIL LOAD (ML4)
22 MONORAIL LOAD (ML5)
23 MONORAIL LOAD (ML6)
24 MONORAIL LOAD (ML7)
25 MONORAIL LOAD (ML8)
26 MONORAIL LOAD (ML9)
27 MONORAIL LOAD (ML10)
28 MONORAIL LOAD (ML11)

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6.2. Primary Load Calculation

1 Dead Load (DL)

a. Selfweight of structural frame

b. Reactions from Steel Structure

c. 300mm thick RCC floor. Floor load = 0.30*25 = 7.5 KN/m^2

d. Load from 200 thick RCC wall

Height of wall = 5.75 m

UDL on tie beam = =5.75 x 0.2 x 25 = 28.75 KN/m

e. Load from 300 thick RCC wall

Height of wall = 5.75 m

UDL on tie beam = =5.75 x 0.3 x 25 = 43.125 KN/m

f. Dead Load from 200thick RCC stair

Width of stair = 1 m

Inclined length of stair flight = 4 m

UDL on beam = =4.0 x 0.2 x 25 x 1.0/1.25 = 16 KN/m

2 Live Load (LL)

a. Reactions from Steel Structure

b. Live load intensity on floor (other than stack area)= 5 KN/m^2

c. Live load intensity on floor (stack area)

Unit weight of each stack = 40 KN

No of stack = 100 nos

Total load = 4000 KN

Pressure on floor = 4000/14*6.4 = 44.6 KN/m^2

d. Live load from stair

Live load intensity = 4 KN/m^2

Plan length of stair flight = 3.2 m

UDL on beam = =3.2 x 4 x 1.0/1.25 = 10.24 KN/m

3 Equipment dead Load (EQDL)

a. Reactions from Steel Structure

b. Load from Relining Machine

From rock box = 285 KN

From Winch = 10 KN

168 KN

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4 Wind load calculation

Design wind pressure on structure (p) = $qz*G*C_f$ *(SBC:301-2007 , Eq. No. 7.2-22)*

Design Parameters :

Basic wind speed (V) =	155 Km/h	*(As per Design Basis)*
Structure Type =	Main wind force resisting system	
Wind Directionality Factor (K_d) =	0.85	*(SBC:301-2007 , Table 6.4-1)*
Exposure category =	C	*(As per Design Basis)*
Topographic Factor (K_{zt}) =	1	*(SBC:301-2007 , Eq. No. 6.4-1)*
The building and structure classification category =	II	*(SBC:301-2007 , Table 1.6-1)*
Importance factor (I) =	1	*(SBC:301-2007 , Table 6.5-1)*
Enclosure classification =	Enclosed Building	
Mean Roof Height (h) =	6.55 m	
Type of structure =	Rigid Structure	

Calculation of Velocity Pressure (qz) :

Velocity Pressure at height z, $qz = 0.0473*10^{-3}*Kz*Kzt*Kd*V^2*I$ KN/m^2 *(SBC:301-2007 , Eq. No. 7.2-12)*

Kz : Velocity pressure exposure coefficient *(SBC:301-2007 , Table 7.2-2)*

for Exposure category = C and Case-1&2

z (m)	Unit	0-5	6	8
K_z	m	0.86	0.9	0.95
Corresponding Velocity Pressure :				
qz	KN/m^2	0.831	0.869	0.918

Calculation of Gust effect factor (G) :

Wind along X-Direction :

Horizontal dimension of building measured parallel to wind direction, L = 21.85 m

Horizontal dimension of building measured normal to wind direction, B = 26 m

For Rigid Structure, Gust effect factor (Gx) = 0.85 *(SBC:301-2007 , CL.No. 7.2.7.1)*

Wind along Z-Direction :

Horizontal dimension of building measured parallel to wind direction, L = 26 m

Horizontal dimension of building measured normal to wind direction, B = 21.85 m

For Rigid Structure, Gust effect factor (Gx) = 0.85 *(SBC:301-2007 , CL.No. 7.2.7.1)*

Calculation of Net Force Coefficient (C_f) :

At ground level, v = 0.3 (Wind along X-Direction) Cf = 1.2 *(SBC:301-2007 , Fig 7.2-16)*

At ground level, v = 0.252 (Wind along Z-Direction) Cf = 1.2 *(SBC:301-2007 , Fig 7.2-16)*

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Design wind pressure on structure along X-Direction :

Height =	0-5m	6m	8m	
p =	0.848	0.886	0.936	KN/m^2

Design wind pressure on structure along Z-Direction :

Height =	0-5m	6m	8m	
p =	0.848	0.886	0.936	KN/m^2

Wind Load along X-Direction :

WIND DIRECTION	COLUMN GRID	EFFECTIVE WIDTH (m)	WIND PRESSURE P (KN/m^2)	WIND LOAD (KN/m)
WL(+X)	3220/C-8	3.55	0.886	3.145
	3220/D-8	9.45		8.373
	3220/F-8	9.45		8.373
	3220/G-8	3.55		3.145
WL(-X)	3220/C-11	3.55		3.145
	3220/D-11	6.325		5.604
	3220/E-11	6.25		5.538
	3220/F-11	6.325		5.604
	3220/G-11	3.55		3.145

Wind Load along Z-Direction :

WIND DIRECTION	COLUMN GRID	EFFECTIVE WIDTH (m)	WIND PRESSURE P (KN/m^2)	WIND LOAD (KN/m)
WL(+Z)	3220/G-8	3.85	0.886	3.411
	3220/G-9	7		6.202
	3220/G-10	7.075		6.268
	3220/G-11	3.925		3.478
WL(-Z)	3220/C-8	3.85		3.411
	3220/C-9	7		6.202
	3220/C-10	7.075		6.268
	3220/C-11	3.925		3.478

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Seismic parameters :

For Ordinary reinforced concrete moment frames

Site Class :	B		Table 14.1.1
Occupancy Category :	II		Table 1.6-1
Spectral Acceleration, S_S :	0.1 g		
Spectral Acceleration, S_1 :	0.02 g		
Importance Factor ,I :	1		Table 9.5
Response Modification Coefficient ,R :		2.5	Table 10.2

Site Coefficient :

F_a :	1	Table 9.4.3a
F_v :	1	Table 9.4.3b

Maximum Spectral Response Accelerations for Short and 1-second Periods :

$S_{MS} = F_a S_s$:	0.1	Eq 9.4.3-1
$S_{M1} = F_v S_1$:	0.02	Eq 9.4.3-2

Design Spectral Response Accelerations for Short and 1-second Periods :

$S_{DS} = 2/3 S_{MS}$:	0.067	Eq 9.4.4-1
$S_{D1} = 2/3 S_{M1}$:	0.013	Eq 9.4.4-2

Response Modification Coefficient ,R :	2.5	Table 10.2
Period Coefficient, Ct :	0.044	Table 10.9.3.2

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6.3. Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
201	DL+EQDL+FL
202	DL+EQDL+FL+LL+EQLL+TL
203	DL+EQDL+FL+LL+SBT+TL
204	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
205	DL+EQDL+FL+0.75(LL+SBT)+0.75TL
206	DL+EQDL+FL+PL
207	DL+EQDL+FL+PL+LL+EQLL+TL
208	DL+EQDL+FL+PL+LL+SBT+TL
209	DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL
210	DL+EQDL+FL+PL+0.75(LL+SBT)+0.75TL
211	DL+EQDL+FL+WL(+X)
212	DL+EQDL+FL+WL(-X)
213	DL+EQDL+FL+WL(+Z)
214	DL+EQDL+FL+WL(-Z)
215	DL+EQDL+FL+0.7SL(+X)
216	DL+EQDL+FL+0.7SL(-X)
217	DL+EQDL+FL+0.7SL(+Z)
218	DL+EQDL+FL+0.7SL(-Z)
219	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
220	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
221	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
222	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
223	DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
224	DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
225	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
226	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
227	0.6(DL+EQDL)+WL(+X)
228	0.6(DL+EQDL)+WL(-X)
229	0.6(DL+EQDL)+WL(+Z)
230	0.6(DL+EQDL)+WL(-Z)
231	0.6(DL+EQDL)+0.7SL(+X)
232	0.6(DL+EQDL)+0.7SL(-X)
233	0.6(DL+EQDL)+0.7SL(+Z)
234	0.6(DL+EQDL)+0.7SL(-Z)
235	DL+EQTL
236	DL+EQDL+FL+LL+EQLL+ML1
237	DL+EQDL+FL+LL+EQLL+ML2
238	DL+EQDL+FL+LL+EQLL+ML3
239	DL+EQDL+FL+LL+EQLL+ML4
240	DL+EQDL+FL+LL+EQLL+ML5
241	DL+EQDL+FL+LL+EQLL+ML6
242	DL+EQDL+FL+LL+EQLL+ML7
243	DL+EQDL+FL+LL+EQLL+ML8
244	DL+EQDL+FL+LL+EQLL+ML9
245	DL+EQDL+FL+LL+EQLL+ML10
246	DL+EQDL+FL+LL+EQLL+ML11

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6.4 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
101	1.4(DL+EQDL+FL)
102	1.4(DL+EQDL+FL)+1.7(LL+EQLL)
103	1.4(DL+EQDL+FL)+1.7(LL+SBT)
104	1.4(DL+EQDL+FL+TL)+1.7(LL+EQLL)
105	1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)
106	1.4(DL+EQDL+FL+PL)
107	1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
108	1.4(DL+EQDL+FL+PL)+1.7(LL+SBT)
109	1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)
110	1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
111	1.2(DL+EQDL)+0.8WL(+X)
112	1.2(DL+EQDL)+0.8WL(-X)
113	1.2(DL+EQDL)+0.8WL(+Z)
114	1.2(DL+EQDL)+0.8WL(-Z)
115	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
116	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
117	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
118	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
119	1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0.5FL
120	1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.5FL
121	1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0.5FL
122	1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.5FL
123	0.9(DL+EQDL)+1.6WL(+X)
124	0.9(DL+EQDL)+1.6WL(-X)
125	0.9(DL+EQDL)+1.6WL(+Z)
126	0.9(DL+EQDL)+1.6WL(-Z)
127	0.9(DL+EQDL)+1.0SL(+X)
128	0.9(DL+EQDL)+1.0SL(-X)
129	0.9(DL+EQDL)+1.0SL(+Z)
130	0.9(DL+EQDL)+1.0SL(-Z)
131	1.2DL+EQTL
132	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML1
133	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML2
134	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML3
135	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
136	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML5
137	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML6
138	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML7
139	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML8
140	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML9
141	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML10
142	1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML11

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7. DESIGN CALCULATIONS

7.1. Design of Stair

A) Input Data :

Specified compressive strength of concrete, f_c'	=	35	Mpa
Specified yield strength of reinforcement, f_y	=	420	Mpa
Nornal weight concrete, λ	=	1	
Unit weight of concrete, γ_{conc}	=	25	KN/m^3
Effective span in longer direction (l_y)	=	6400	mm
Effective span in shorter direction (S)	=	1200	mm

B) Design Consideration :

Dia of main reinforcement in short direction		= 16	mm
Dia of main reinforcement in long direction		= 16	mm
Consider overall depth, h		= 250	mm
Clear cover to main reinforcement	[table 20.6.1.3.1 SBC 304:2018]	= 40	mm
Effective depth in shorter direction, d_s		= 202	mm

C) Load Data :

i)Live Load intensity w_{LL} = 4 KN/m^2 (refer 6.2 Primary Load Calculation)

ii) Dead load calculation :

Self weight of slab = w_{DL} = 250 /1000 * 25 = 6.25 KN/m^2

Total load intensity, w = w_{LL} + w_{DL} = 10.25 KN/m^2

D) Structural Design :

Ratio of short span to long span ,m = S/L_y = 0.19 < 0.5, one way slab

Bending Moment $M_s = wS^2/10$ = 10.25 * 6.4 ^2/ 10 = 41.98 KN-m

Partial safety factor ,Υ_m = 1.4

E) Check for flexure :-

Factored moment in short direction ,M_{uS} = 58.78 KN-m

β_1 = 0.8 steel strain ε_t = 0.005 and φ = 0.9

now, $C = 0.85f_c'ba$; $T = A_sf_y$; $M_u = \phi M_n = \varphi f_y A_s(d_s - a/2)$; $C = T$

C = 0.85 * 35 * 1000 * a ; T = 420 * A_s ; a = 0.014 *A_s

M_{uS} = 0.9 * 420 *(202 - 0.014 A_s/2)*A_s = 58.78 * 10^6

$A_{s\ required}$ = 792 mm^2

$A_{s,min}$ = 0.18 / 100 * 1000 * 250 = 450 mm^2 [Table 8.6.1.1 SBC 304:18]

Spacing required = 201 / 792 * 1000 = 253.9 mm c/c

Maximum spacing = = 450 mm c/c [cl 8.7.2.2 SBC 304:2018]

provide 16 φ main bar @ 200 mm c/c A_s provided = 1005.0 mm^2

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Check for Tension controlled section :

$\varepsilon_t = \varepsilon_c / c\ (d_s - c)$ $\quad c = 0.014 * 1005.0 / 0.800 = 17.7$ mm

$\varepsilon_t = 0.003 *(202 - 17.7)/ 17.74 = 0.031 > 0.005$ tension controlled

F) Check for shear :

Shear force at a distance d from the face of supporting beam

Width of supporting beam,b = 400 mm

Effective depth in shorter direction ,d_s= 202 mm

UDL along shorter direction considering 1m width, w= 10.25 KN/m

Factored shear,V_u=1.4w*(0.5S-0.5b-d_s)=

1.4 * 10.3 *(0.5 * 1.2 - 0.5 * 0.2 - 0.202)= 4.28 KN

$V_{c1} = (0.16 \lambda \sqrt{f_c'} + 17 \rho_w V_u d_s / M_{us}) b_w d_s$

$V_{c2} = (0.16 \lambda \sqrt{f_c'} + 17 \rho_w) b_w d_s$

$V_{c3} = 0.29 \lambda \sqrt{f_c'}\ b_w d_s$

V_c = Minimum of V_{c1}, V_{c2} and V_{c3}

A_s=	1005	mm^2	M_{us}=	58.78 KN-m	= 191 KN
ρ_w=	0.0050		b_w =	1000 mm	= 208 KN
λ=	1		d_s =	202 mm	= 347 KN
[table 22.5.5.1 SBC 304 : 2018]					= 191 KN

$V_u \leq \varphi (V_c + 0.66\sqrt{f_c'} b_w d_s)$ = 735.1 KN ok [cl 22.5.5.1 SBC 304 : 2018]

φV_c = = 144 KN >4.28KN

Remarks : shear reinforcement not reqd

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7.2. Design of Roof Slab

Design of 200mm thick roof slab

A) Input Data :

Specified compressive strength of concrete, f_c'	=	35	Mpa
Specified yield strength of reinforcement, f_y	=	420	Mpa
Nornal weight concrete, λ	=	1	
Unit weight of concrete, γ_{conc}	=	25	KN/m^3
Effective span in longer direction (l_y)	=	7150	mm
Effective span in shorter direction (S)	=	3200	mm

B) Design Consideration :

Dia of main reinforcement in short direction	=	10	mm
Dia of main reinforcement in long direction	=	10	mm
Consider overall depth, h	=	200	mm
Clear cover to main reinforcement [table 20.6.1.3.1 SBC 304:2018]	=	40	mm
Effective depth in shorter direction, d_s	=	155	mm

C) Load Data :

i)Live Load intensity w_{LL} = 5 KN/m^2 (refer 6.2 Primary Load Calculation)

ii) Dead load calculation :

Self weight of slab = w_{DL} = 200 /1000 * 25 = 5 KN/m^2

Total load intensity, w = w_{LL} + w_{DL} = 10.00 KN/m^2

D) Structural Design :

Ratio of short span to long span ,m = S/L_y	=	0.45	< 0.5, one way slab
Bending Moment $M_s = wS^2/10$ = 10.00 * 3.2 ^2/ 10	=	10.24	KN-m
Partial safety factor ,Υ_m	=	1.4	

E) Check for flexure :-

Factored moment in short direction ,M_{uS} = 14.34 KN-m

β_1 = 0.8 steel strain ε_t = 0.005 and φ = 0.9

now, $C = 0.85f_c'ba$; $T = A_sf_y$; $M_u = \phi M_n = \varphi f_y A_s(d_s - a/2)$; $C = T$

C = 0.85 * 35 * 1000 * a ; T = 420 * A_s ; a = 0.014 *A_s

M_{uS} = 0.9 * 420 *(155 - 0.014 A_s/2)*A_s = 14.34 * 10^6

$A_{s\ required}$ = 247 mm^2

$A_{s,min}$	=	0.18	/	100	*	1000	*	200	= 360 mm^2 [Table 8.6.1.1 SBC 304:18]
Spacing required	=	79	/	360	*	1000			= 219.4 mm c/c
Maximum spacing	=								= 400 mm c/c [cl 8.7.2.2 SBC 304:2018]

provide 10 φ main bar @ 200 mm c/c A_s provided = 395.0 mm^2

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Check for Tension controlled section :

$\varepsilon_t = \varepsilon_c / c\ (d_s - c)$ $\quad c = 0.014 * 395.0 / 0.800 = 6.97$ mm

$\varepsilon_t = 0.003 *(155 - 6.97)/ 6.97 = 0.064 > 0.005$ tension controlled

F) Check for shear :

Shear force at a distance d from the face of supporting beam

Width of supporting beam,b = 400 mm

Effective depth in shorter direction ,d_s= 155 mm

UDL along shorter direction considering 1m width, w= 10.00 KN/m

Factored shear,V_u=1.4w*(0.5S-0.5b-d_s)=

1.4 * 10.0 *(0.5 * 3.2 - 0.5 * 0.2 - 0.155)= 18.8 KN

$V_{c1} = (0.16 \lambda \sqrt{f_c'} + 17 \rho_w V_u d_s / M_{us}) b_w d_s$

$V_{c2} = (0.16 \lambda \sqrt{f_c'} + 17 \rho_w) b_w d_s$

$V_{c3} = 0.29 \lambda \sqrt{f_c'}\ b_w d_s$

V_c = Minimum of V_{c1}, V_{c2} and V_{c3}

A_s= 395 mm^2 M_{us}= 14.34 KN-m

ρ_w= 0.0025 b_w = 1000 mm

λ= 1 d_s = 155 mm

[table 22.5.5.1 SBC 304 : 2018]

V_{c1} = 148 KN

V_{c2} = 153 KN

V_{c3} = 266 KN

V_c = 148 KN

$V_u \leq \varphi (V_c + 0.66\sqrt{f_c'} b_w d_s)$ = 565.0 KN ok [cl 22.5.5.1 SBC 304 : 2018]

φV_c = 111 KN >18.83KN

Remarks : shear reinforcement not reqd

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Design of 300mm thick roof slab

A) Input Data :

Specified compressive strength of concrete, f_c'	=	35	Mpa
Specified yield strength of reinforcement, f_y	=	420	Mpa
Nornal weight concrete, λ	=	1	
Unit weight of concrete, γ_{conc}	=	25	KN/m^3
Effective span in longer direction (l_y)	=	7150	mm
Effective span in shorter direction (S)	=	3200	mm

B) Design Consideration :

Dia of main reinforcement in short direction	=	16	mm
Dia of main reinforcement in long direction	=	16	mm
Consider overall depth, h	=	300	mm
Clear cover to main reinforcement [table 20.6.1.3.1 SBC 304:2018]	=	40	mm
Effective depth in shorter direction, d_s	=	252	mm

C) Load Data :

i)Live Load intensity w_{LL} = 44.6 KN/m^2 (refer 6.2 Primary Load Calculation)

ii) Dead load calculation :

Self weight of slab = w_{DL} = 300 /1000 * 25 = 7.5 KN/m^2

Total load intensity, w = w_{LL} + w_{DL} = 52.10 KN/m^2

D) Structural Design :

Ratio of short span to long span ,m = S/L_y	=	0.45	< 0.5, one way slab
Bending Moment $M_s = wS^2/10$ = 52.10 * 3.2 ^2/ 10	=	53.35	KN-m
Partial safety factor ,Υ_m	=	1.4	

E) Check for flexure :-

Factored moment in short direction ,M_{uS} = 74.69 KN-m

β_1 = 0.8 steel strain ε_t = 0.005 and φ = 0.9

now, $C = 0.85f_c'ba$; $T = A_sf_y$; $M_u = \phi M_n = \varphi f_y A_s(d_s - a/2)$; $C = T$

$C = 0.85 * 35 * 1000 * a$; $T = 420 * A_s$; $a = 0.014 * A_s$

$M_{uS} = 0.9 * 420 *(252 - 0.014\ A_s/2)*A_s = 74.69 * 10^6$

$A_{s\ required}$ = 802 mm^2

$A_{s,min}$	= 0.18 / 100 * 1000 * 300	=	540	mm^2	[Table 8.6.1.1 SBC 304:18]
Spacing required	= 201 / 802 * 1000	=	250.6	mm c/c	
Maximum spacing	=	=	450	mm c/c	[cl 8.7.2.2 SBC 304:2018]

provide 16 φ main bar @ 200 mm c/c A_s provided = 1005.0 mm^2

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Check for Tension controlled section :

$\varepsilon_t = \varepsilon_c / c\ (d_s - c)$ $\quad c = 0.014 * 1005.0 / 0.800 = 17.7$ mm

$\varepsilon_t = 0.003 *(252 - 17.7)/ 17.74 = 0.040 > 0.005$ tension controlled

F) Check for shear :

Shear force at a distance d from the face of supporting beam

Width of supporting beam,b = 400 mm

Effective depth in shorter direction ,d_s= 252 mm

UDL along shorter direction considering 1m width, w= 52.10 KN/m

Factored shear,V_u=1.4w*(0.5S-0.5b-d_s)=

1.4 * 52.1 *(0.5 * 3.2 - 0.5 * 0.2 - 0.252)= 91 KN

$V_{c1} = (0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d_s / M_{us}) b_w d_s$	A_s= 1005 mm^2	M_{us}=	74.69 KN-m	=	244	KN
$V_{c2} = (0.16\lambda\sqrt{f_c'} + 17\rho_w) b_w d_s$	ρ_w= 0.0040	b_w =	1000 mm	=	256	KN
$V_{c3} = 0.29\lambda\sqrt{f_c'}\ b_w d_s$	λ= 1	d_s =	252 mm	=	432	KN
V_c = Minimum of V_{c1}, V_{c2} and V_{c3}	[table 22.5.5.1 SBC 304 : 2018]			=	244	KN

$V_u \le \varphi (V_c + 0.66\sqrt{f_c'} b_w d_s)$ = 920.8 KN ok [cl 22.5.5.1 SBC 304 : 2018]

φV_c = = 183 KN >91.03KN

Remarks : shear reinforcement not reqd

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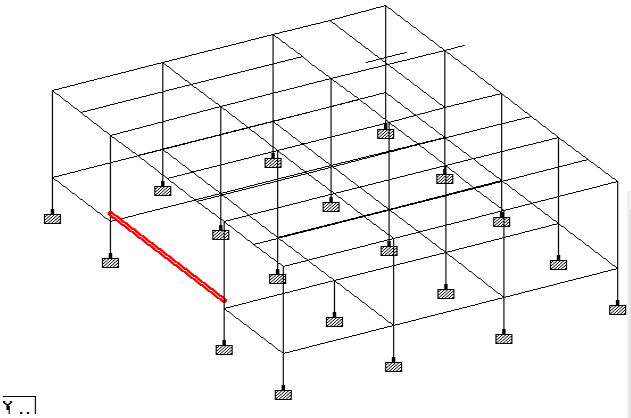
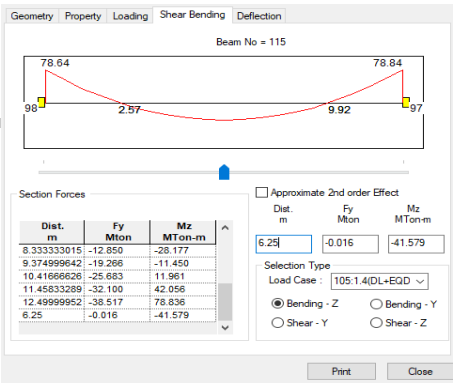
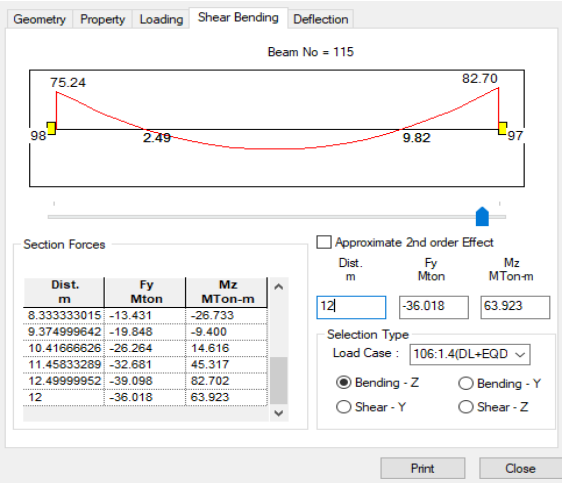
7.3. Design of Beams

Design of beam : TB-1

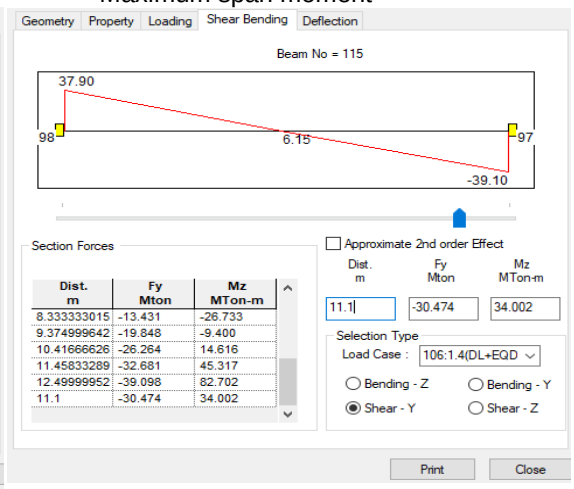




Maximum span moment



Moment at the face of column



Shear at the distance d from face of column

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Design of beam = TB-1

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G24: _TB_1									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	115	98	-2.5	32.9	0	0	0	66.4	131 1.2DL+EQTL
Min Fx	115	98	-9.8	38.4	0	0	0	78.4	132 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML1
Max Fy	115	98	-9.6	38.5	0	0	0	78.6	103 1.4(DL+EQDL+FL)+1.7(LL+SBT)
Min Fy	115	97	-9	-39	0	0	-0	82.7	106 1.4(DL+EQDL+FL+PL)
Max Fz	115	98	-5.8	25.4	0	0	0	54.8	128 0.9(DL+EQDL)+1.0SL(-X)
Min Fz	115	98	-9.6	38.3	0	0	0	77.3	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Max Mx	115	98	-5.9	24.5	0	0	0	49.2	124 0.9(DL+EQDL)+1.6WL(-X)
Min Mx	115	98	-9.8	38.4	0	0	0	78.4	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Max My	115	98	-9.6	38.3	0	0	0	77.3	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Min My	115	97	-9.6	-39	0	0	-0	80.2	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Max Mz	115	97	-9	-39	0	0	-0	82.7	106 1.4(DL+EQDL+FL+PL)
Min Mz	115	98	-5.3	23.2	0	0	0	40.8	125 0.9(DL+EQDL)+1.6WL(+Z)

Max support Top Moment = 64 t-m at the face of the beam

Max span Bottom Moment = -41.6 t-m at the face of the beam

Max Shear = 31 t Corresponding Moment = 34 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete,f_c' =	35	Mpa	
Specified yield strength of main reinforcement,f_y =	420	Mpa	
Yield strength of transverse reinforcement,f_{yt} =	420	Mpa	
Beam width,b =	300	mm	
Beam depth,h =	1000	mm	
Cover =	50	mm	
Dia of stirrups,d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.047	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 64.00 t-m
Effective depth ,d = 905.0 mm
Area of steel required ,Ast,required = 1972 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 956 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 905 mm^2

Provide 3 nos 25 dia bar althrough at 1st layer
Provide 3 nos 20 dia at 2nd layer
Ast,provided = 2416 mm^2 > 1972 mm^2 ok

Max span Bottom Moment = 41.6 t-m
Effective depth ,d = 907.5 mm
Area of steel required ,Ast,required = 1253 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd 959 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 908 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer
Provide 3 nos 20 dia at 2nd layer
Ast,provided = 1886 mm^2 > 1253 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 31 t Corresponding Moment, Mu = 34 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 28.4 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 29 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 46.7 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 21.3 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 12.9 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.34 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.26* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.25* mm^2/m } table 9.6.3.3 SBC 304 : 2018
Provide 2L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 0.79 mm^2/m ok

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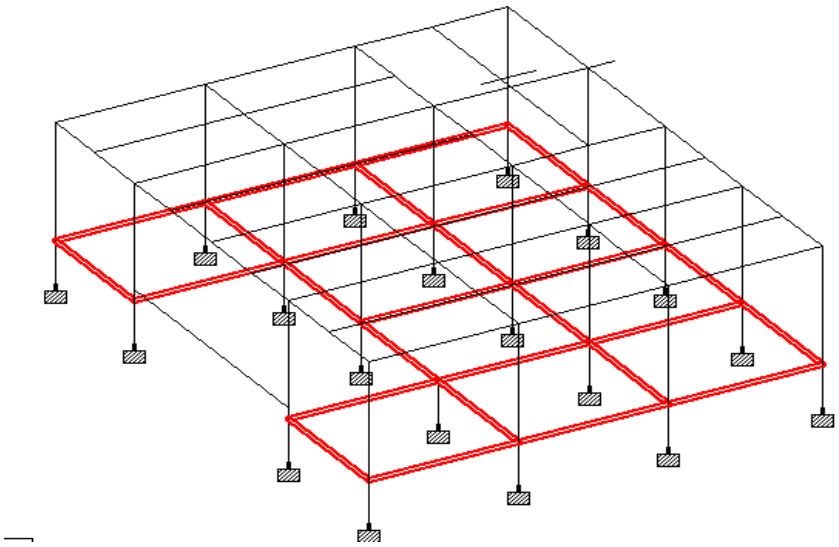
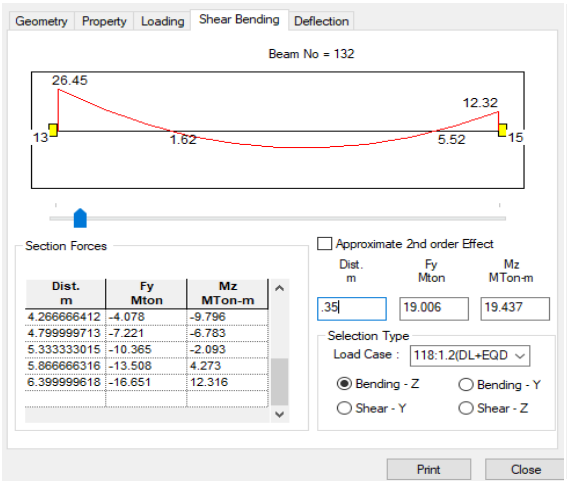
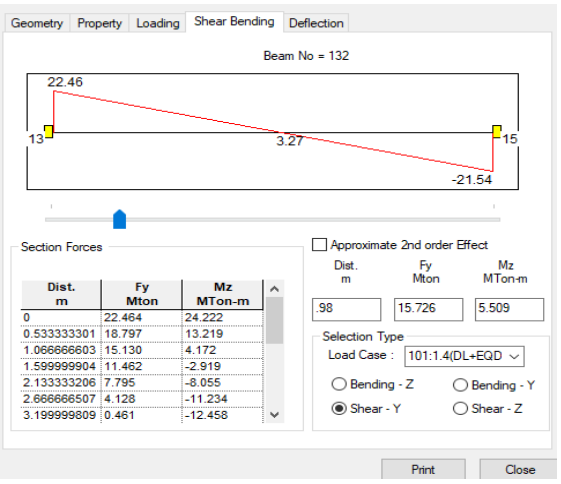
Design of beam : TB-2





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = TB-2

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G3: _TB_2									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	116	6	4.24	3.17	0.05	0	-0.14	4.14	103 1.2(DL+EQDL+FL)+1.6(LL+SBT)
Min Fx	130	19	-31	0.38	0.05	0	-0.17	-0.38	107 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
Max Fy	132	13	3.97	22.5	-0	0	0.01	24.2	101 1.4(DL+EQDL+FL)
Min Fy	125	13	3.33	-22	-0.01	0	-0.02	24.2	106 1.4(DL+EQDL+FL+PL)
Max Fz	129	18	-3.2	12.9	0.12	0	-0.44	14	108 1.2(DL+EQDL+FL+PL)+1.6(LL+SBT)
Min Fz	123	9	-27	2.07	-0.1	0	0.31	3.71	110 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+SBT)
Max Mx	129	18	-3.2	12.9	0.12	0	-0.44	14	108 1.2(DL+EQDL+FL+PL)+1.6(LL+SBT)
Min Mx	133	17	-13	14.6	-0.09	0	0.31	17.8	135 1.2(DL+EQDL+FL)+1.6(LL+EQLL)+1.6ML4
Max My	129	17	-3.2	-14	0.12	0	0.32	16	108 1.2(DL+EQDL+FL+PL)+1.6(LL+SBT)
Min My	129	18	-3.2	12.9	0.12	0	-0.44	14	108 1.2(DL+EQDL+FL+PL)+1.6(LL+SBT)
Max Mz	132	13	3.59	21.1	-0	0	0.01	26.5	118 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
Min Mz	116	9	2.02	1.01	0.03	0	0.08	-8.18	118 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)

Max support Top Moment = 19.5 t-m at the face of the beam

Max span Bottom Moment = -12.5 t-m at the face of the beam

Max Shear = 16 t Corresponding Moment = 6 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	300	mm	
Beam depth,h =	700	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.047	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 19.50 t-m
Effective depth ,d = 630.0 mm
Area of steel required ,Ast,required = 846 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 666 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 630 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer

Ast,provided = 943 mm^2 > 846 mm^2 ok

Max span Bottom Moment = 12.5 t-m
Effective depth ,d = 630.0 mm
Area of steel required ,Ast,required = 534 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd 666 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 630 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer

Ast,provided = 943 mm^2 > 666 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 16 t Corresponding Moment, Mu = 6 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 20.6 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 19.5 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 32.4 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 14.6 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 1.84 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.07 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ 0.26 mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ 0.25 mm^2/m }
Provide 2L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 0.79 mm^2/m ok

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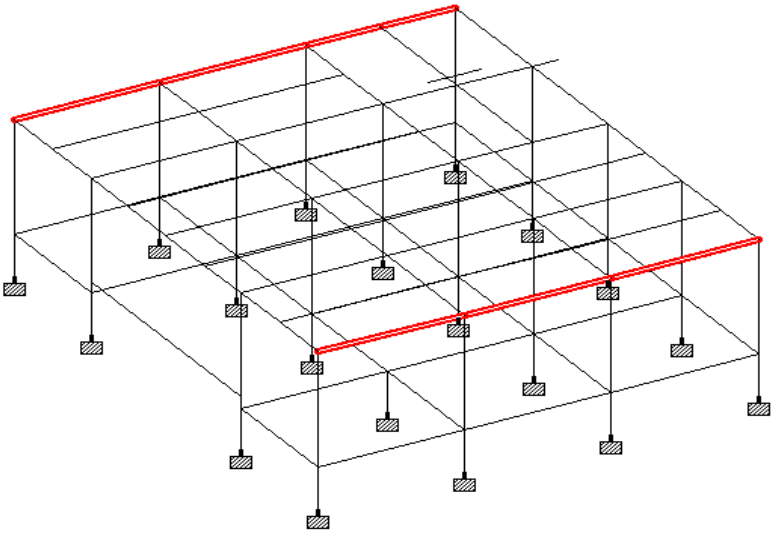
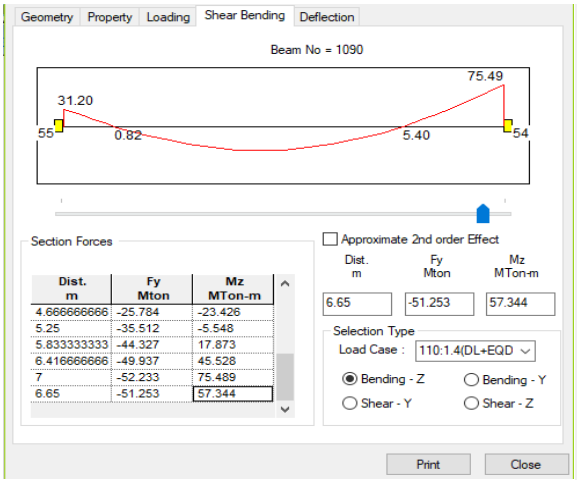
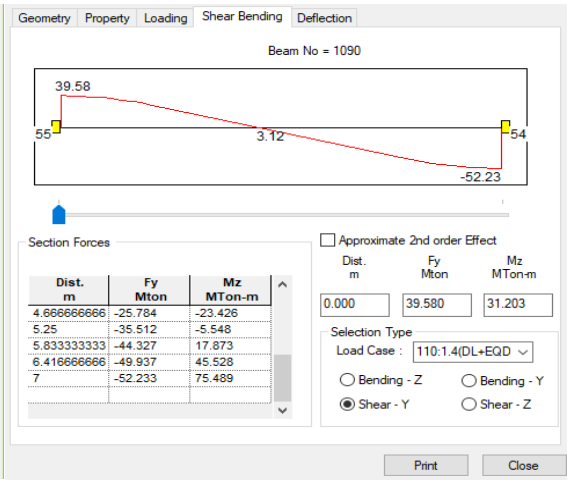
Design of beam : FB-1





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-1

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G4: _FB_1

	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1001	39	0	8.57	0	0	0	8.81	101 1.4(DL+EQDL+FL)
Min Fx	1001	39	0	8.57	0	0	0	8.81	101 1.4(DL+EQDL+FL)
Max Fy	1091	54	0	46.7	0	0	0	61	137 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML6
Min Fy	1090	54	0	-52	0	0	0	75.5	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Max Fz	1004	65	0	-13	0	0	0	-37.2	126 0.9(DL+EQDL)+1.6WL(-Z)
Min Fz	1003	41	0	30.4	0	0	0	43.9	118 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
Max Mx	1004	65	0	-19	0	0	0	-52.2	118 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
Min Mx	1092	52	0	14.6	0	0	0	26.9	137 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML6
Max My	1004	42	0	-18	0	0	0	19.3	126 0.9(DL+EQDL)+1.6WL(-Z)
Min My	1004	65	0	-13	0	0	0	-37.2	126 0.9(DL+EQDL)+1.6WL(-Z)
Max Mz	1090	54	0	-52	0	0	0	75.5	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Min Mz	1003	65	0	28.8	0	0	0	-67	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)

Max support Top Moment = 57.5 t-m at the face of the beam

Max span Bottom Moment = -67.0 t-m at the face of the beam

Max Shear = 40 t Corresponding Moment = 31.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 57.50 t-m
Effective depth ,d = 705.0 mm
Area of steel required ,Ast,required = 2289 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 993 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 940 mm^2

Provide 4 nos 25 dia bar althrough at 1st layer
Provide 4 nos 20 dia at 2nd layer
Ast,provided = 3221 mm^2 > 2289 mm^2 ok

Max span Bottom Moment = 67 t-m
Effective depth ,d = 705.0 mm
Area of steel required ,Ast,required = 2694 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd 993 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 940 mm^2

Provide 4 nos 25 dia bar althrough at 1st layer
Provide 4 nos 20 dia at 2nd layer
Ast,provided = 3221 mm^2 > 2694 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 40 t Corresponding Moment, Mu = 31.5 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 31.6 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 32.2 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 48.4 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 23.7 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 21.7 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.73 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.35* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.33* mm^2/m } table 9.6.3.3 SBC 304 : 2018
Provide 4L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 1.57 mm^2/m ok

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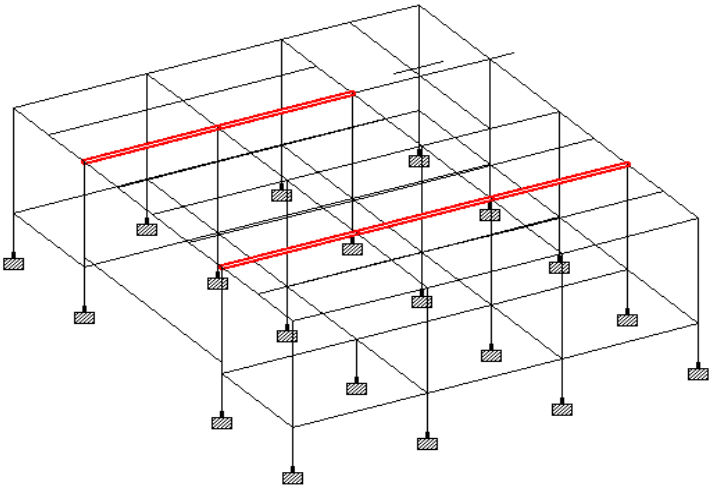
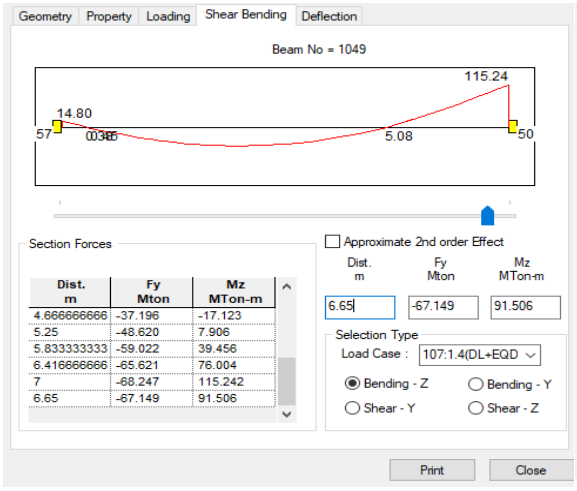
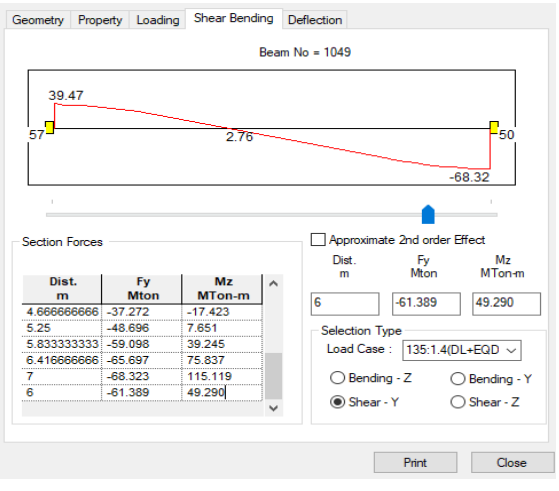
Design of beam : FB-2





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-2

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G5: _FB_2

	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1018	43	0	13.1	0	0	0	14.5	101 1.4(DL+EQDL+FL)
Min Fx	1018	43	0	13.1	0	0	0	14.5	101 1.4(DL+EQDL+FL)
Max Fy	1048	56	0	64.7	0	0	0	92	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)
Min Fy	1049	50	0	-68	0	0	0	113	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)
Max Fz	1018	43	0	6.02	0	0	0	-0	127 0.9(DL+EQDL)+1.0SL(+X)
Min Fz	1050	50	0	16.7	0	0	0	31.6	118 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL
Max Mx	1018	43	0	18.6	0	0	0	16.1	107 1.4(DL+EQDL+FL+PL)+1.7(LL+E(
Min Mx	1048	56	0	64.7	0	0	0	92	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)
Max My	1050	51	0	-10	0	0	0	7.99	117 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL
Min My	1050	50	0	17	0	0	0	32.9	117 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL
Max Mz	1049	50	0	-68	0	0	0	113	107 1.4(DL+EQDL+FL+PL)+1.7(LL+E(
Min Mz	1048	57	0	-6.7	0	0	0	-5.21	131 1.2DL+EQTL

Max support Top Moment = 92 t-m at the face of the beam

Max span Bottom Moment = -60.4 t-m at the face of the beam

Max Shear = 61.5 t Corresponding Moment = 49.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 92.00 t-m

Effective depth ,d = 700.5 mm

Area of steel required ,Ast,required = 3847 mm^2 $As = M_u / \varphi f_y (d - a/2)$

$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 987 mm^2 cl 9.6.1.2 SBC 304 : 2018

(b) $1.4/f_y$bd 934 mm^2

Provide 4 nos 25 dia bar althrough at 1st layer

Provide 4 nos 25 dia at 2nd layer

Ast,provided = 3929 mm^2 > 3847 mm^2 ok

Max span Bottom Moment = 60.4 t-m

Effective depth ,d = 700.5 mm

Area of steel required ,Ast,required = 2428 mm^2 $As = M_u / \varphi f_y (d - a/2)$

$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd 987 mm^2 cl 9.6.1.2 SBC 304 : 2018

(b) $1.4/f_y$bd 934 mm^2

Provide 4 nos 25 dia bar althrough at 1st layer

Provide 4 nos 25 dia at 2nd layer

Ast,provided = 3929 mm^2 > 2428 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 61.5 t Corresponding Moment, Mu = 49.5 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 32.3 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 33.2 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 48.1 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 24.3 t

Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 49.7 t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 1.69 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.35* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.33* mm^2/m } table 9.6.3.3 SBC 304 : 2018

Provide 4L-12dia stirrups @200 mm c/c

therefore, A_v/s provided = 2.26 mm^2/m ok

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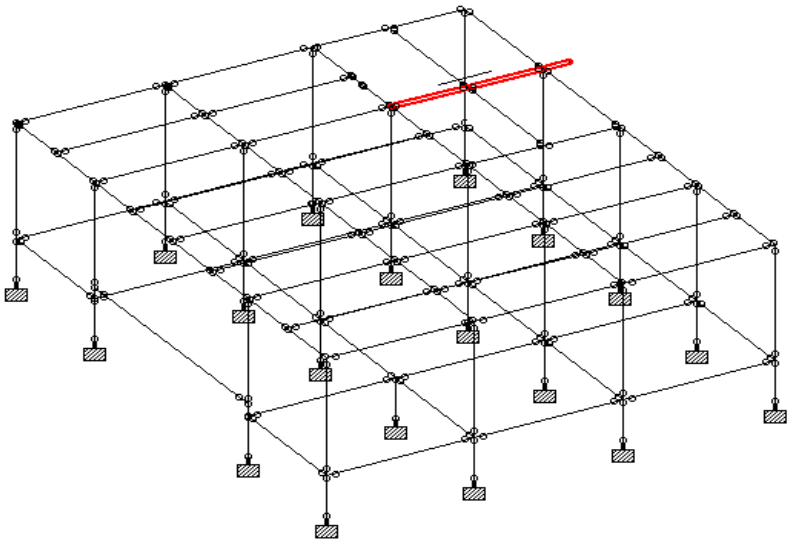
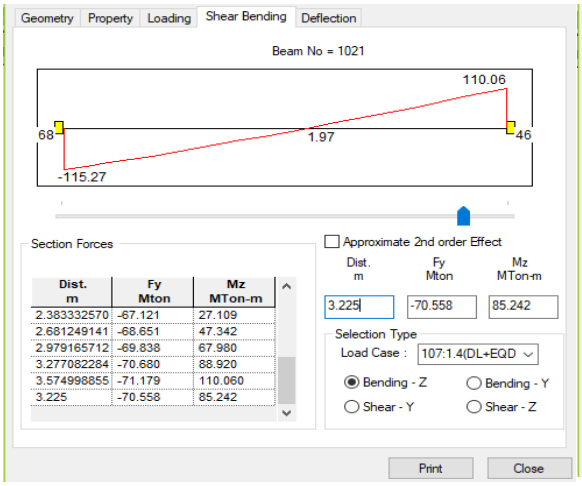
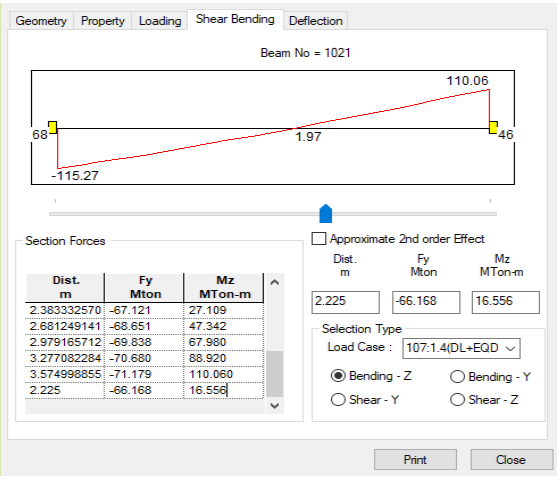
Design of beam : FB-2A





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-2A

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G6: _FB_2A

	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1022	46	1.7	63.4	-0.04	0	0.05	75.1	140 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML9
Min Fx	1022	46	-0.2	23.3	-0	0	0.01	27.4	123 0.9(DL+EQDL)+1.6WL(+X)
Max Fy	1020	45	0	65.7	0	0	0	90	104 1.4(DL+EQDL+FL+TL)+1.7(LL+EQLL)
Min Fy	1021	46	0	-71	0	0	0	110	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Max Fz	1022	46	0.22	14.4	0.34	0	-0.42	16.3	126 0.9(DL+EQDL)+1.6WL(-Z)
Min Fz	1022	46	0.74	23.9	-0.36	0	0.45	27.2	117 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
Max Mx	1020	45	0	37	0	0	0	52.9	125 0.9(DL+EQDL)+1.6WL(+Z)
Min Mx	1020	45	0	51.6	0	0	0	72.4	118 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
Max My	1022	46	0.74	23.9	-0.36	0	0.45	27.2	117 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
Min My	1022	46	0.22	14.4	0.34	0	-0.42	16.3	126 0.9(DL+EQDL)+1.6WL(-Z)
Max Mz	1021	46	0	-71	0	0	0	110	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Min Mz	1021	68	0	-54	0	0	0	-116	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)

Max support Top Moment = 85.5 t-m at the face of the beam

Max span Bottom Moment = -116 t-m at the face of the beam

Max Shear = 66.5 t Corresponding Moment = 17 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	200	mm	
Dia of stirrups, d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 85.50 t-m
Effective depth ,d = 547.0 mm
Area of steel required ,Ast,required = 4914 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 770 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 729 mm^2

Provide 4 nos 25 dia bar althrough at 1st layer
Provide 4 nos 32 dia at 2nd layer
Ast,provided = 5183 mm^2 > 4914 mm^2 ok

Max span Bottom Moment = 116 t-m
Effective depth ,d = 697.0 mm
Area of steel required ,Ast,required = 5051 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd 982 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 929 mm^2

Provide 4 nos 25 dia bar althrough at 1st layer
Provide 4 nos 32 dia at 2nd layer
Ast,provided = 5183 mm^2 > 5051 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 66.5 t Corresponding Moment, Mu = 17 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 50.4 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 35.2 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 47.8 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 26.4 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 53.5 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 1.83 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.35* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.33* mm^2/m } table 9.6.3.3 SBC 304 : 2018
Provide 4L-12dia stirrups @200 mm c/c

therefore, A_v/s provided = 2.26 mm^2/m ok

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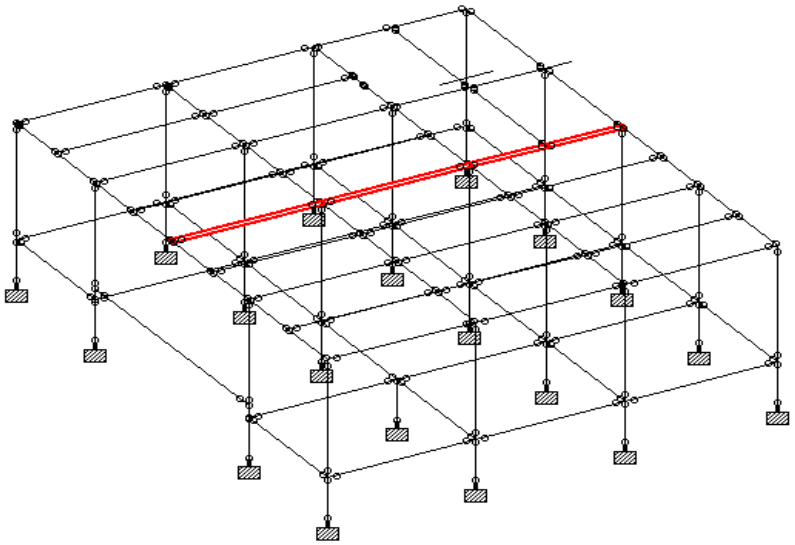
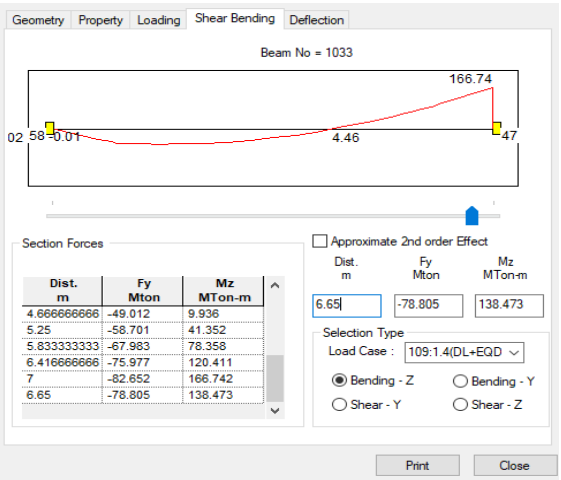
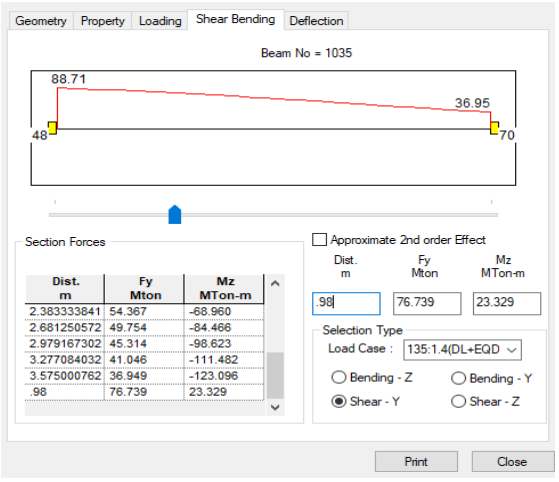
Design of beam : FB-3







Moment at the face of column

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Design of beam = FB-3

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G7: _FB_3									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1033	58	0	58.5	0	-0.01	0	-0	101 1.4(DL+EQDL+FL)
Min Fx	1033	58	0	58.5	0	-0.01	0	-0	101 1.4(DL+EQDL+FL)
Max Fy	1035	48	0	88.5	0	0.03	0	103	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Fy	1033	47	0	-83	0	-0.02	0	167	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)
Max Fz	1035	48	0	67.6	0	0.02	0	72	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Min Fz	1035	48	0	51	0	0.01	0	56.7	130 0.9(DL+EQDL)+1.0SL(-Z)
Max Mx	1035	48	0	87.5	0	0.03	0	99.8	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Min Mx	1036	70	0	-16	0	-0.02	0	-124	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)
Max My	1036	49	0	-37	0	-0.01	0	10.6	126 0.9(DL+EQDL)+1.6WL(-Z)
Min My	1033	47	0	-47	0	-0	0	96.1	126 0.9(DL+EQDL)+1.6WL(-Z)
Max Mz	1033	47	0	-83	0	-0.02	0	167	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)
Min Mz	1035	70	0	36.2	0	0.03	0	-124	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)

Max support Top Moment = 139 t-m at the face of the beam

Max span Bottom Moment = -124 t-m at the face of the beam

Max Shear = 77 t Corresponding Moment = 23.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	4300	mm	
Beam depth,h =	700	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.003	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 139.00 t-m
Effective depth ,d = 630.0 mm
Area of steel required ,Ast,required = 5928 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 26 nos 20 dia bar althrough at 1st layer

Ast,provided = 8171 mm^2 > 5928 mm^2 ok

Max span Bottom Moment = 124 t-m
Effective depth ,d = 630.0 mm
Area of steel required ,Ast,required = 5289 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 26 nos 20 dia bar althrough at 1st layer

Ast,provided = 8171 mm^2 > 5289 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 77 t Corresponding Moment, Mu = 23.5 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 285 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 270 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 465 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 203 t
Remarks : provide minimum shear reinforcement

$V_s = V_u/\varphi - V_c$ - t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = - mm^2/m

$A_{v,min}/s >$ max of $\begin{cases} (a)\ 0.062\sqrt{f_c'}b/f_{yt} \\ (b)\ 0.35b/f_{yt} \end{cases}$ *3.76* mm^2/m table 9.6.3.3 SBC 304 : 2018
3.58 mm^2/m

Provide 10L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 3.93 mm^2/m ok

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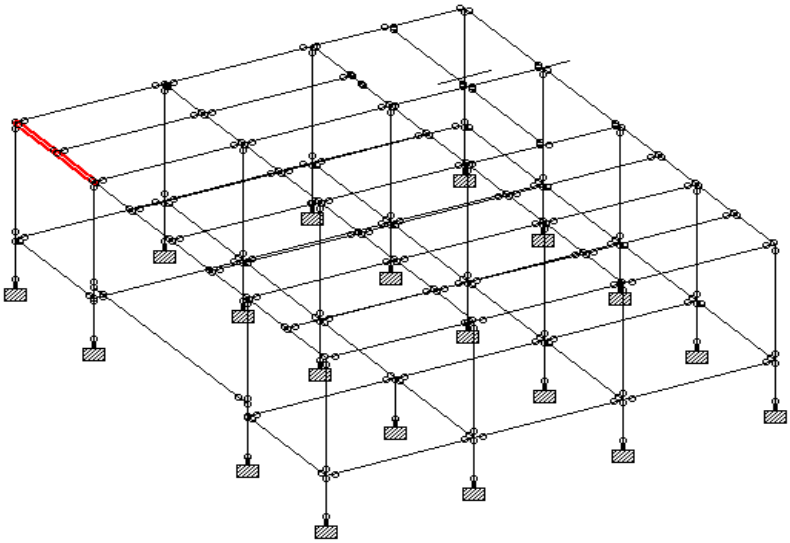
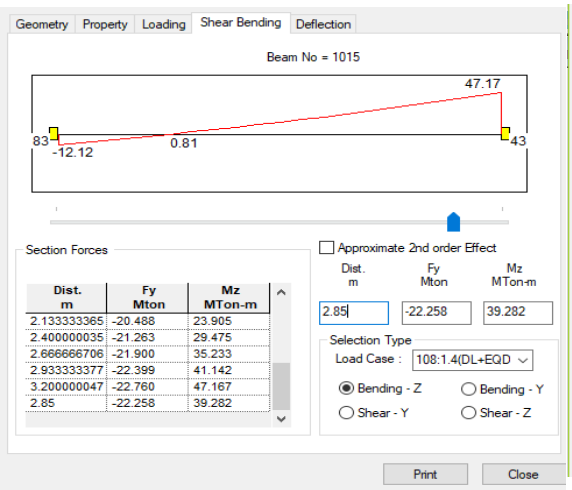
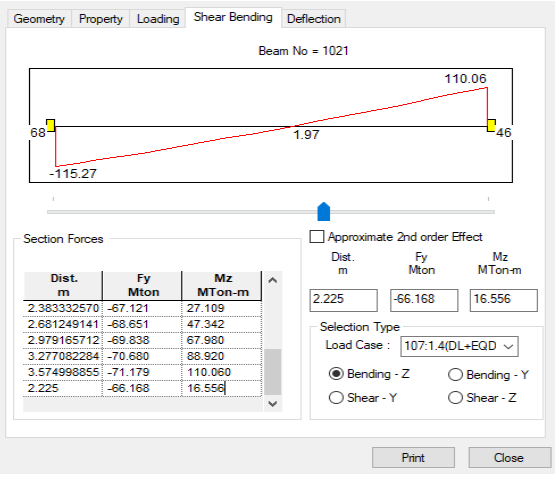
Design of beam : FB-4





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-4

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G17: _FB_4									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1008	61	0.16	8.09	0	0	0	6.08	122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.
Min Fx	1008	61	-0.2	4.82	0	0	0	-3.19	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Fy	1008	61	0	10.4	0	0	0	4.28	103 1.4(DL+EQDL+FL)+1.7(LL+SBT)
Min Fy	1015	43	0	-26	0	-0	0	54.1	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Max Fz	1008	61	0	5.99	0.12	0	-0.18	0.33	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fz	1008	61	0	6.92	-0.12	0	0.18	2.57	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Max Mx	1008	61	0	10.4	0	0	0	3.21	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Mx	1015	83	0	-16	0	-0	0	-12.9	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Max My	1008	61	0	6.92	-0.12	0	0.18	2.57	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Min My	1008	61	0	3.41	0.12	0	-0.18	-0.54	127 0.9(DL+EQDL)+1.0SL(+X)
Max Mz	1015	43	0	-26	0	-0	0	54.6	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Min Mz	1015	83	0	-16	0	-0	0	-12.9	140 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML9

Max support Top Moment = 39.5 t-m at the face of the beam

Max span Bottom Moment = -12.9 t-m at the face of the beam

Max Shear = 66.5 t Corresponding Moment = 17 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	600	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.024	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 39.50 t-m

Effective depth ,d = 730.0 mm

Area of steel required ,Ast,required = 1466 mm^2 $As = M_u / \varphi f_y (d - a/2)$

$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 1542 mm^2 cl 9.6.1.2 SBC 304 : 2018

(b) $1.4/f_y$bd 1460 mm^2

Provide 6 nos 20 dia bar althrough at 1st layer

Ast,provided = 1886 mm^2 > 1542 mm^2 ok

Max span Bottom Moment = 12.9 t-m

Effective depth ,d = 730.0 mm

Area of steel required ,Ast,required = 473 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 20 dia bar althrough at 1st layer

Ast,provided = 1886 mm^2 > 473 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 66.5 t Corresponding Moment, Mu = 17 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 50.6 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 44.7 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 75.1 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 33.5 t

Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 44 t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 1.44 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.52* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.5* mm^2/m } table 9.6.3.3 SBC 304 : 2018

Provide 4L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 1.57 mm^2/m ok

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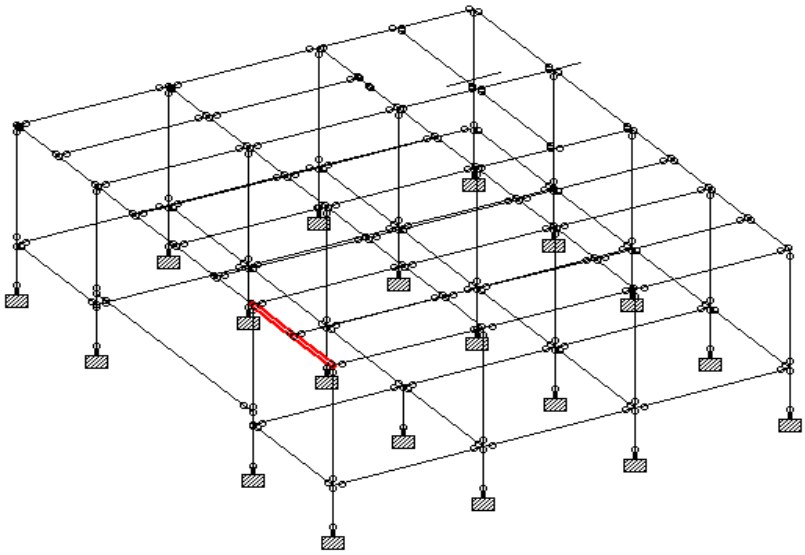
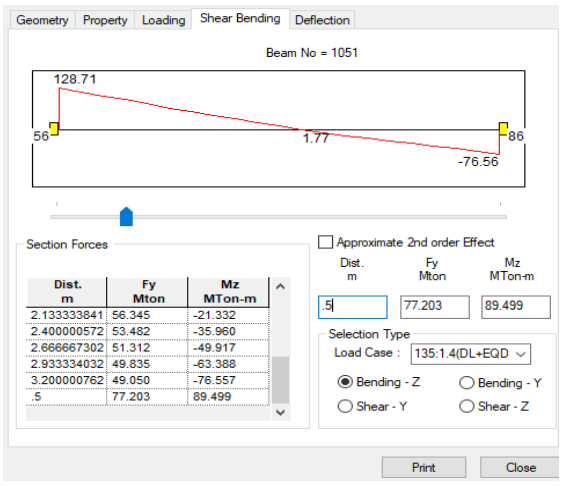
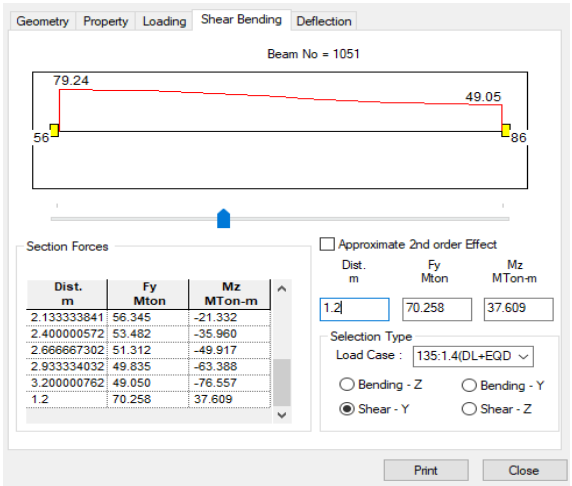
Design of beam : FB-5





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-5

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G9: _FB_5									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1058	86	0.37	-8.4	0	-0	0	-25.6	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Min Fx	1051	56	-0.4	32.4	0	0	0	56.8	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Fy	1051	56	0	79.2	0	0	0	129	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Fy	1058	55	0	-51	0	-0.01	0	39.3	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Max Fz	1051	56	0	33.7	0.31	0	-0.49	60.6	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fz	1058	86	0	-7.2	-0.31	-0	0.49	-25.7	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mx	1051	56	0	79.2	0	0	0	129	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Mx	1058	86	0	-21	0	-0.01	0	-76.6	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Max My	1051	86	0	20.3	0.31	0	0.49	-25.7	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min My	1051	56	0	33.7	0.31	0	-0.49	60.6	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mz	1051	56	0	79.2	0	0	0	129	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Mz	1051	86	0	49.1	0	0	0	-76.6	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4

Max support Top Moment = 90 t-m at the face of the beam

Max span Bottom Moment = -76.6 t-m at the face of the beam

Max Shear = 70.5 t Corresponding Moment = 38 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	600	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.024	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 90.00 t-m
Effective depth ,d = 705.5 mm
Area of steel required ,Ast,required = 3590 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 1491 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 1411 mm^2

Provide 6 nos 20 dia bar althrough at 1st layer
Provide 6 nos 20 dia at 2nd layer
Ast,provided = 3771 mm^2 > 3590 mm^2 ok

Max span Bottom Moment = 76.6 t-m
Effective depth ,d = 705.5 mm
Area of steel required ,Ast,required = 3023 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd 1491 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 1411 mm^2

Provide 6 nos 20 dia bar althrough at 1st layer
Provide 6 nos 20 dia at 2nd layer
Ast,provided = 3771 mm^2 > 3023 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 70.5 t Corresponding Moment, Mu = 38 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 48.5 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 46.5 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 72.6 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 34.9 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 47.5 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 1.6 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.52* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.5* mm^2/m } table 9.6.3.3 SBC 304 : 2018
Provide 4L-12dia stirrups @200 mm c/c

therefore, A_v/s provided = 2.26 mm^2/m ok

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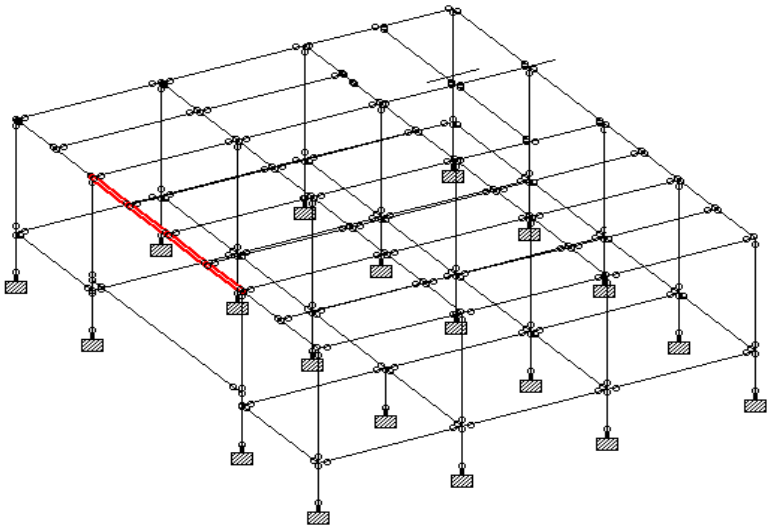
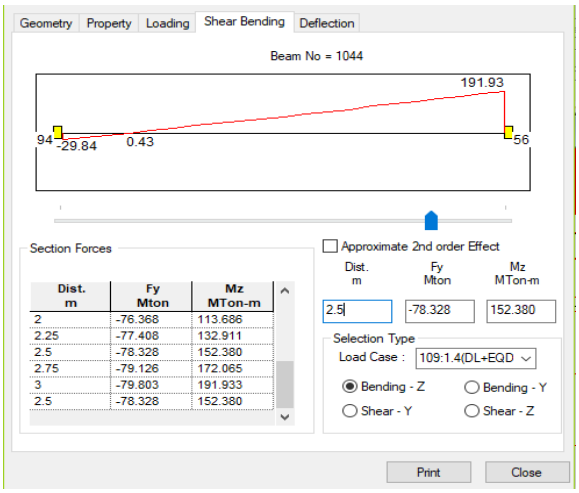
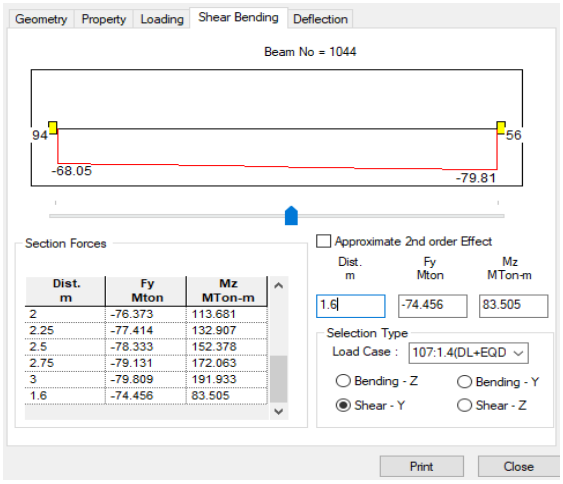
Design of beam : FB-6





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-6

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G10: _FB_6									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1023	43	0.17	55.7	0	0	0	125	122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.
Min Fx	1023	43	-0.2	49.5	0	-0	0	87.7	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Fy	1023	43	0	67.5	0	-0	0	134	103 1.4(DL+EQDL+FL)+1.7(LL+SBT)
Min Fy	1044	56	0	-80	0	0	0	192	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Max Fz	1023	43	-0	50.4	0.25	-0	-0.38	94.8	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fz	1023	43	0	54.7	-0.25	0	0.38	118	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Max Mx	1037	58	0	-35	0	0.01	0	-168	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Mx	1023	43	0	52.4	0	-0	0	108	115 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
Max My	1023	43	0	54.7	-0.25	0	0.38	118	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Min My	1023	43	-0	50.4	0.25	-0	-0.38	94.8	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mz	1044	56	0	-80	0	0	0	192	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)
Min Mz	1037	58	0	-35	0	0.01	0	-168	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4

Max support Top Moment = 153 t-m at the face of the beam

Max span Bottom Moment = -168.0 t-m at the face of the beam

Max Shear = 75 t Corresponding Moment = 84 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete,f_c' =	35	Mpa	
Specified yield strength of main reinforcement,f_y =	420	Mpa	
Yield strength of transverse reinforcement,f_{yt} =	420	Mpa	
Beam width,b =	600	mm	
Beam depth,h =	1200	mm	
Cover =	200	mm	
Dia of stirrups,d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.024	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005] 0.9

Shear,φ 0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 152.50 t-m

Effective depth ,d = 950.5 mm

Area of steel required ,Ast,required = 4495 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 25 dia bar althrough at 1st layer

Provide 6 nos 25 dia at 2nd layer

Ast,provided = 5893 mm^2 > 4495 mm^2 ok

Max span Bottom Moment = 168 t-m

Effective depth ,d = 1100.5 mm

Area of steel required ,Ast,required = 4229 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 25 dia bar althrough at 1st layer

Provide 6 nos 25 dia at 2nd layer

Ast,provided = 5893 mm^2 > 4229 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 75 t Corresponding Moment, Mu = 84 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 72.3 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 72.5 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 113 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 54.3 t

Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 27.7 t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.6 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.52* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.5* mm^2/m } table 9.6.3.3 SBC 304 : 2018

Provide 4L-12dia stirrups @200 mm c/c

therefore, A_v/s provided = 2.26 mm^2/m ok

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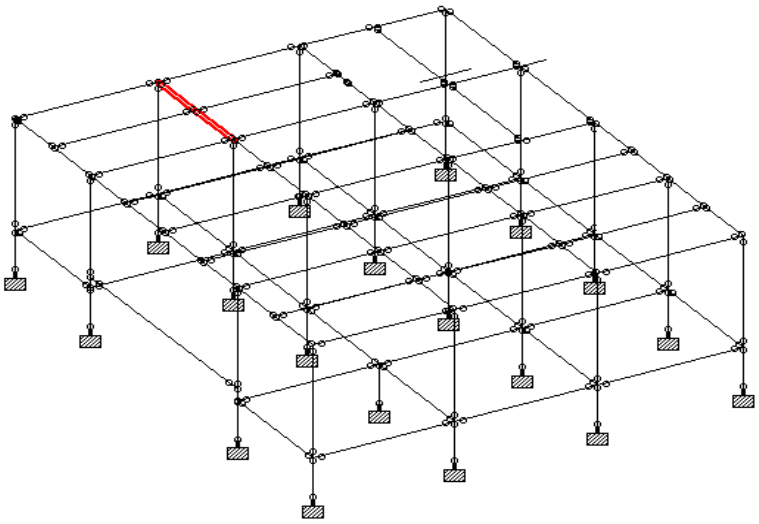
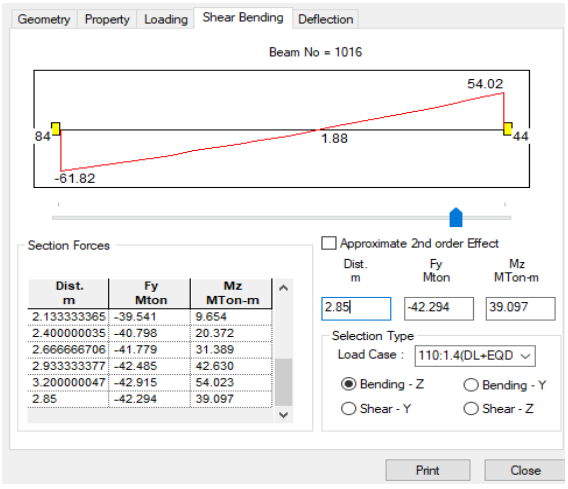
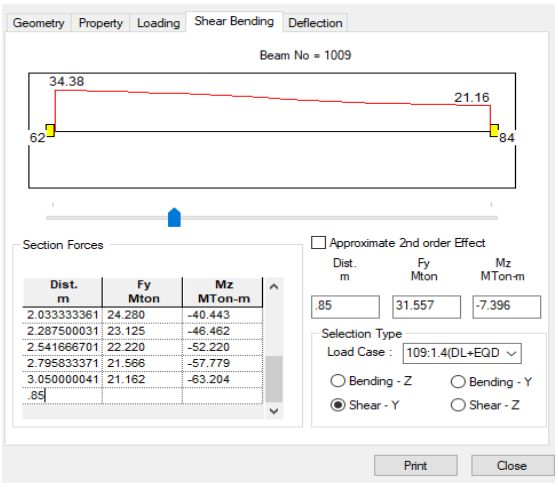
Design of beam : FB-7





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-7

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G10: _FB_7									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1009	62	0.26	23.9	0	0	0	20.2	122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.
Min Fx	1009	62	-0.3	20.9	0	0	0	9.76	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Fy	1006	40	0	141	0	0	0	41.9	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)
Min Fy	1016	44	0	-43	0	0	0	54	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Max Fz	1016	84	0.01	-19	0.06	0	-0.09	-39.2	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Min Fz	1016	84	-0	-20	-0.06	0	0.09	-39.7	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mx	1006	40	0	55.5	0	0	0	24.8	116 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
Min Mx	1016	84	0	-19	0	0	0	-39.2	116 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
Max My	1016	84	-0	-20	-0.06	0	0.09	-39.7	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min My	1016	84	0.01	-19	0.06	0	-0.09	-39.2	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Max Mz	1016	44	0	-43	0	0	0	54	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Min Mz	1016	84	0	-29	0	0	0	-63.2	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)

Max support Top Moment = 39.5 t-m at the face of the beam

Max span Bottom Moment = -63.2 t-m at the face of the beam

Max Shear = 32 t Corresponding Moment = 7.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 39.50 t-m

Effective depth ,d = 705.0 mm

Area of steel required ,Ast,required = 1542 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 4 nos 25 dia bar althrough at 1st layer

Provide 4 nos 20 dia at 2nd layer

Ast,provided = 3221 mm^2 > 1542 mm^2 ok

Max span Bottom Moment = 63.2 t-m

Effective depth ,d = 702.5 mm

Area of steel required ,Ast,required = 2543 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 4 nos 25 dia bar althrough at 1st layer

Provide 4 nos 25 dia at 2nd layer

Ast,provided = 3929 mm^2 > 2543 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 32 t Corresponding Moment, Mu = 7.5 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 37.3 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 30.4 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 49.1 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 22.8 t

Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 12.3 t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.41 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.35* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.33* mm^2/m } table 9.6.3.3 SBC 304 : 2018

Provide 2L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 0.79 mm^2/m ok

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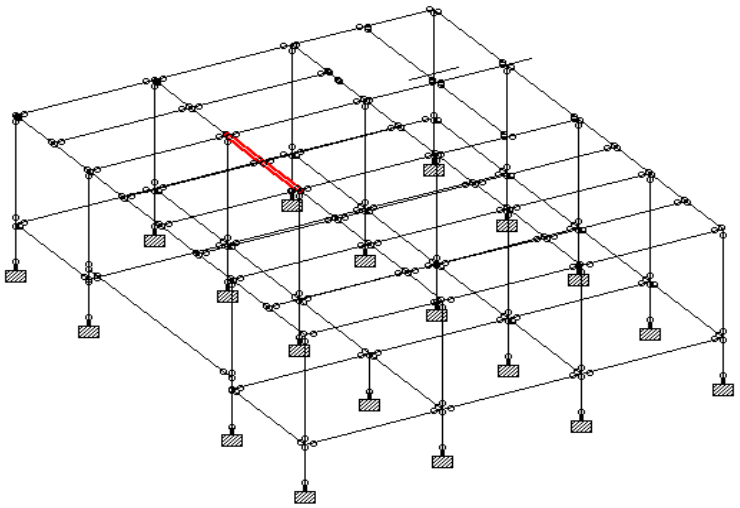
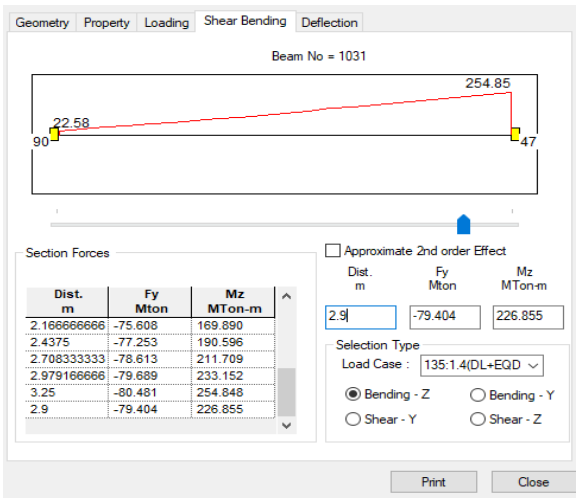
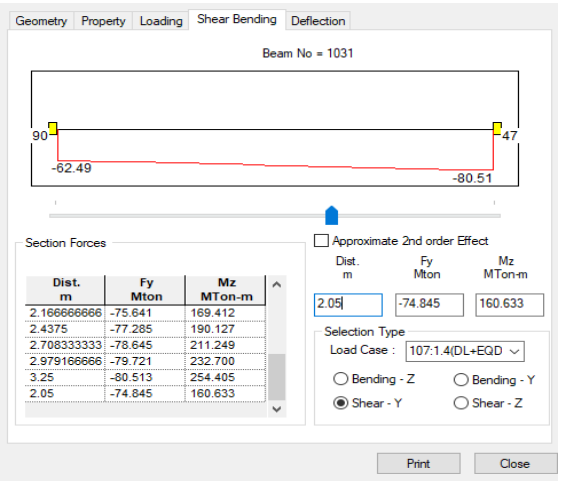
Design of beam : FB-8





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-8

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G11: _FB_8									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1024	44	0.29	17.2	0	0	0	33.6	122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL-
Min Fx	1024	44	-0.3	12	0	0	0	17.6	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL
Max Fy	1024	44	0	20.3	0	0	0	34.1	114 1.2(DL+EQDL)+0.8WL(-Z)
Min Fy	1031	47	0	-81	0	-0	0	254	107 1.4(DL+EQDL+FL+PL)+1.7(LL+E(
Max Fz	1024	44	0	13.4	0.35	0	-0.53	22	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL
Min Fz	1024	44	0	15.8	-0.35	0	0.53	29.1	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL-
Max Mx	1024	44	0	13.2	0	0	0	19	124 0.9(DL+EQDL)+1.6WL(-X)
Min Mx	1031	90	0	-33	0	-0	0	-2	116 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL
Max My	1024	44	0	15.8	-0.35	0	0.53	29.1	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL-
Min My	1024	44	0	13.4	0.35	0	-0.53	22	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL
Max Mz	1031	47	0	-80	0	-0	0	255	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL
Min Mz	1031	90	0	-23	0	-0	0	-12	113 1.2(DL+EQDL)+0.8WL(+Z)

Max support Top Moment = 226 t-m at the face of the beam

Max span Bottom Moment = -12.0 t-m at the face of the beam

Max Shear = 75 t Corresponding Moment = 161 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	700	mm	
Beam depth,h =	1000	mm	
Cover =	200	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.020	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 226.00 t-m

Effective depth ,d = 745.5 mm

Area of steel required ,Ast,required = 9153 mm^2 $As = M_u / \varphi f_y (d - a/2)$

$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 1838 mm^2 cl 9.6.1.2 SBC 304 : 2018

(b) $1.4/f_y$bd 1740 mm^2

Provide 8 nos 32 dia bar althrough at 1st layer

Provide 8 nos 25 dia at 2nd layer

Ast,provided = 10365.1 mm^2 > 9153 mm^2 ok

Max span Bottom Moment = 12 t-m

Effective depth ,d = 927.5 mm

Area of steel required ,Ast,required = 343 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 8 nos 25 dia bar althrough at 1st layer

Ast,provided = 3929 mm^2 > 343 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 75 t Corresponding Moment, Mu = 161 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 64.3 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 68.1 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 111 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 48.3 t

Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 35.7 t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.92 mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.61* mm^2/m table 9.6.3.3 SBC 304 : 2018

(b) $0.35b/f_{yt}$ *0.58* mm^2/m

Provide 4L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 1.57 mm^2/m ok

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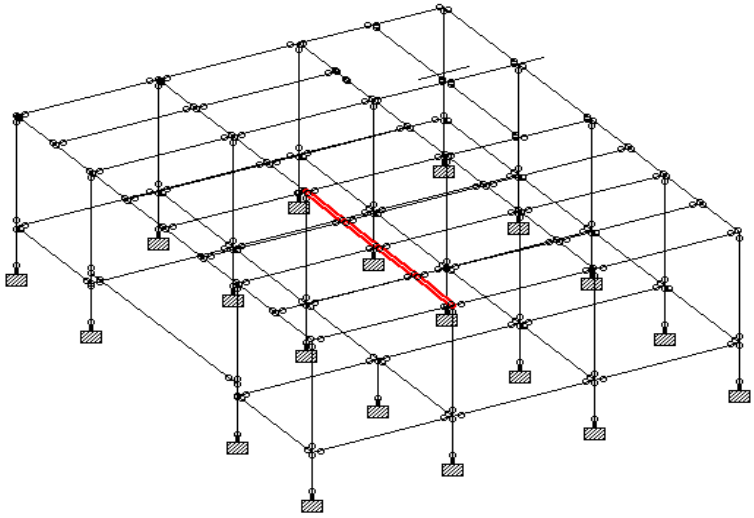
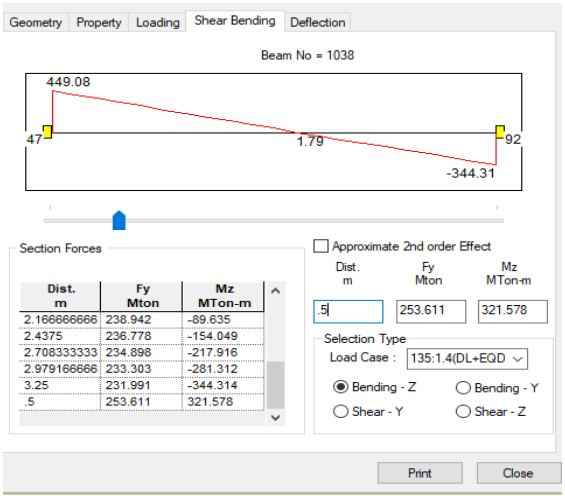
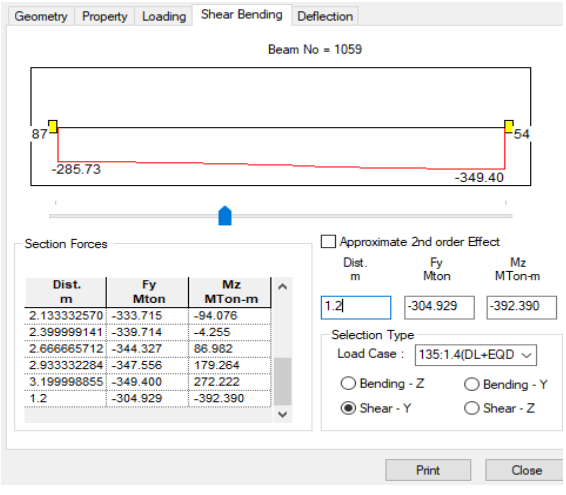
Design of beam : FB-9





Dist. m	Fy Mton	Mz MTon-m
2.166666666	238.942	-89.635
2.4375	236.778	-154.049
2.708333333	234.898	-217.916
2.979166666	233.303	-281.312
3.25	231.991	-344.314
.5	253.611	321.578

Moment at the face of column



Dist. m	Fy Mton	Mz MTon-m
2.133332570	-333.715	-94.076
2.399999141	-339.714	-4.255
2.666665712	-344.327	86.982
2.933332284	-347.556	179.264
3.199998855	-349.400	272.222
1.2	-304.929	-392.390

Shear at the distance d from face of column

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Design of beam = FB-9

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G13: _FB_9									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1059	87	0.78	-126	0	0	0	-302	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Min Fx	1052	57	-0.8	-21	0	0	0	-416	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Fy	1038	47	0	256	0	-0	0	449	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Fy	1059	54	0	-349	0	0	0	272	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Max Fz	1052	57	0	-19	1.06	0	-1.69	-416	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fz	1059	87	0	-124	-1.06	0	1.69	-309	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mx	1059	87	0	-279	0	0	0	-726	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Min Mx	1052	57	0	-13	0	-0	0	-397	116 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
Max My	1052	87	0	-48	1.06	0	1.69	-309	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min My	1052	57	0	-19	1.06	0	-1.69	-416	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mz	1038	47	0	256	0	-0	0	449	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Mz	1045	57	0	174	0	0	0	-899	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4

Max support Top Moment = 322 t-m at the face of the beam

Max span Bottom Moment = -899.4 t-m at the face of the beam

Max Shear = 305 t Corresponding Moment = 393 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	700	mm	
Beam depth,h =	1800	mm	
Cover =	200	mm	
Dia of stirrups, d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.020	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 322.00 t-m
Effective depth ,d = 1572.0 mm
Area of steel required ,Ast,required = 5622 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 3875 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 3668 mm^2

Provide 8 nos 32 dia bar althrough at 1st layer
Provide 8 nos 25 dia at 2nd layer
Ast,provided = 6437 mm^2 > 5622 mm^2 ok

Max span Bottom Moment = 899 t-m
Effective depth ,d = 1690.0 mm
Area of steel required ,Ast,required = 15516 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd 4166 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 3943 mm^2

Provide 8 nos 32 dia althrough at 1st layer
Provide 8 nos 32 dia at 2nd layer
Provide 8 nos 32 dia at 3rd layer
Ast,provided = 19309.7 mm^2 > 15516 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 305 t Corresponding Moment, Mu = 393 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 143 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 136 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 207 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 102 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 271 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 3.74 mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.61* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ *0.58* mm^2/m

Provide 8L-12dia stirrups @200 mm c/c

therefore, A_v/s provided = 4.53 mm^2/m ok

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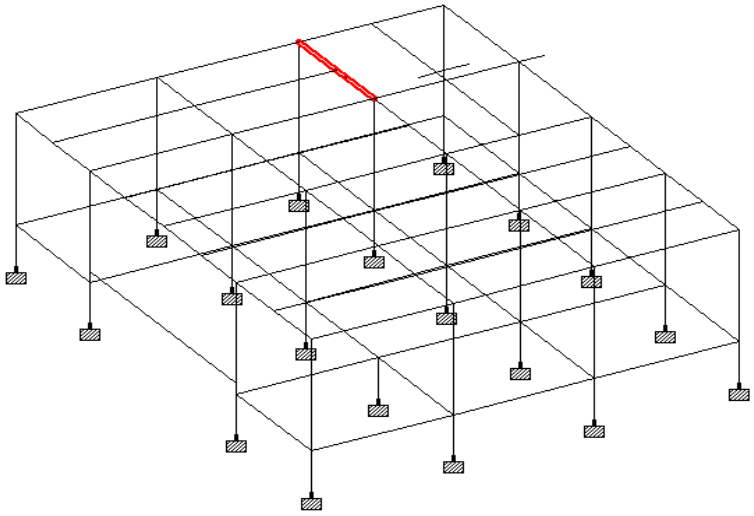
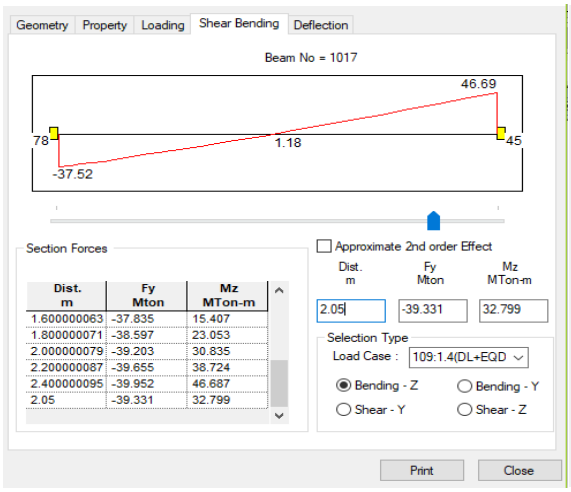
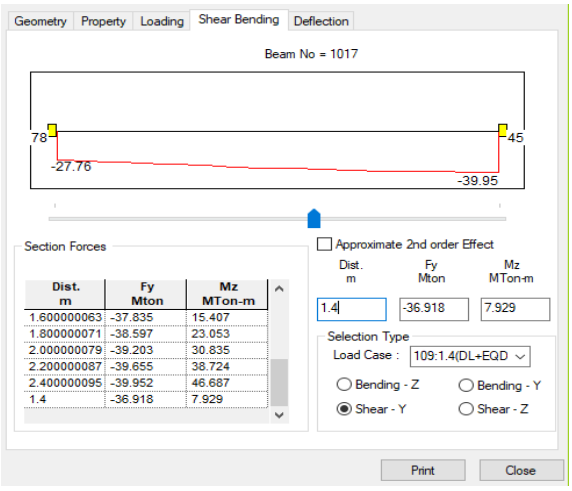
Design of beam : FB-10





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-10

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G14: _FB_10

	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1014	85	0.24	-7.2	0	0	0	-21.2	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Min Fx	1014	85	-0.2	-0.8	0	0	0	-25.5	122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.
Max Fy	1007	41	0	29.2	0	-0	0	28.7	104 1.4(DL+EQDL+FL+TL)+1.7(LL+EQLL)
Min Fy	1017	45	0	-40	0	0	0	46.7	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)
Max Fz	1014	85	-0	-2.8	0.78	0	-0.2	-24.1	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Min Fz	1014	85	0.01	-5.2	-0.78	0	0.2	-22.6	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mx	1014	85	0	-4.7	0	0	0	-39.1	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Min Mx	1007	41	0	27.8	0	-0	0	26.5	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Max My	1014	78	-0	-5.6	0.78	0	0.42	-20.8	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Min My	1014	78	0.01	-8	-0.78	0	-0.42	-17.4	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mz	1017	45	0	-40	0	0	0	46.7	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)
Min Mz	1007	85	0	12.6	0	-0	0	-41.8	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)

Max support Top Moment = 33 t-m at the face of the beam

Max span Bottom Moment = -41.8 t-m at the face of the beam

Max Shear = 37 t Corresponding Moment = 8 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	600	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.024	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 33.00 t-m

Effective depth ,d = 727.5 mm

Area of steel required ,Ast,required = 1224 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 25 dia bar althrough at 1st layer

Ast,provided = 2946 mm^2 > 1224 mm^2 ok

Max span Bottom Moment = 41.8 t-m

Effective depth ,d = 727.5 mm

Area of steel required ,Ast,required = 1559 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 25 dia bar althrough at 1st layer

Ast,provided = 2946 mm^2 > 1559 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 37 t Corresponding Moment, Mu = 8 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 58.2 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 46.3 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 74.9 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 34.7 t

Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 3.01 t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.1 mm^2/m

$A_{v,min}/s >$ max of $\begin{cases} (a)\ 0.062\sqrt{f_c'}b/f_{yt} \\ (b)\ 0.35b/f_{yt} \end{cases}$ *0.52* mm^2/m table 9.6.3.3 SBC 304 : 2018

0.5 mm^2/m

Provide 4L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 1.57 mm^2/m ok

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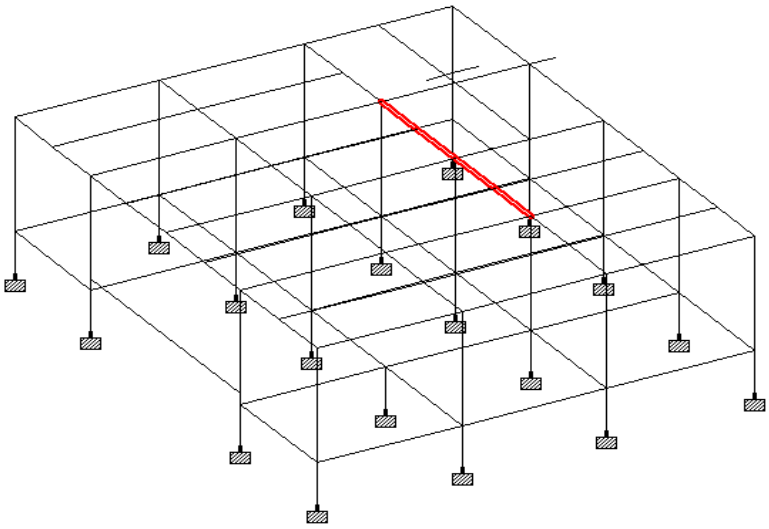
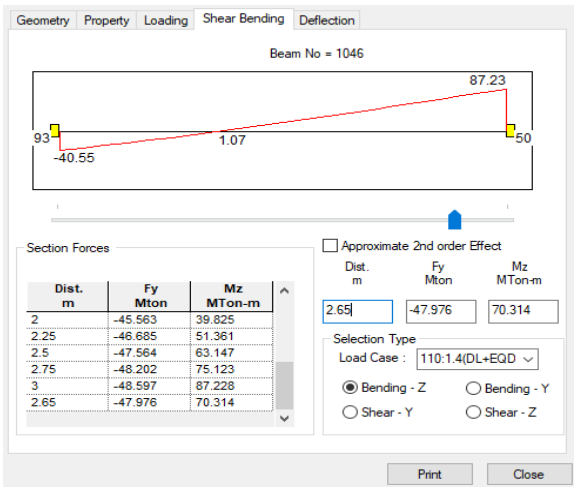
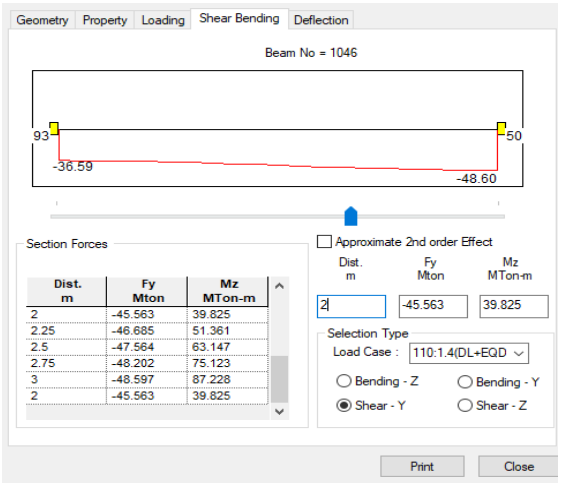
Design of beam : FB-11





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-11

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G15: _FB_11									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1046	93	0.27	-22	0	0	0	-27.7	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Min Fx	1046	93	-0.3	-20	0	0	0	-27.4	122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.
Max Fy	1039	48	0	31.5	0	0	0	39	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Fy	1046	50	0	-49	0	0	0	87.2	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Max Fz	1025	45	0	14.3	0.07	-0	-0.11	19.2	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fz	1025	45	-0	16.1	-0.07	-0	0.11	25	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Max Mx	1032	91	0	-5	0	0	0	-14.9	104 1.4(DL+EQDL+FL+TL)+1.7(LL+EQLL)
Min Mx	1025	45	0	23.2	0	-0	0	34.9	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Max My	1025	45	-0	16.1	-0.07	-0	0.11	25	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Min My	1025	45	0	14.3	0.07	-0	-0.11	19.2	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mz	1046	50	0	-49	0	0	0	87.2	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Min Mz	1046	93	0	-37	0	0	0	-40.8	109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)

Max support Top Moment = 70.5 t-m at the face of the beam

Max span Bottom Moment = -40.8 t-m at the face of the beam

Max Shear = 46 t Corresponding Moment = 40 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	600	mm	
Beam depth,h =	800	mm	
Cover =	200	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.024	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 70.50 t-m

Effective depth ,d = 555.0 mm

Area of steel required ,Ast,required = 3642 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 25 dia bar althrough at 1st layer

Provide 6 nos 20 dia at 2nd layer

Ast,provided = 4832 mm^2 > 3642 mm^2 ok

Max span Bottom Moment = 40.8 t-m

Effective depth ,d = 727.5 mm

Area of steel required ,Ast,required = 1523 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 25 dia bar althrough at 1st layer

Ast,provided = 2946 mm^2 > 1523 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 46 t Corresponding Moment, Mu = 40 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 44.7 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 45.6 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 73.6 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 33.5 t

Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 16.6 t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.55 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.52* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.5* mm^2/m } table 9.6.3.3 SBC 304 : 2018

Provide 4L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 1.57 mm^2/m ok

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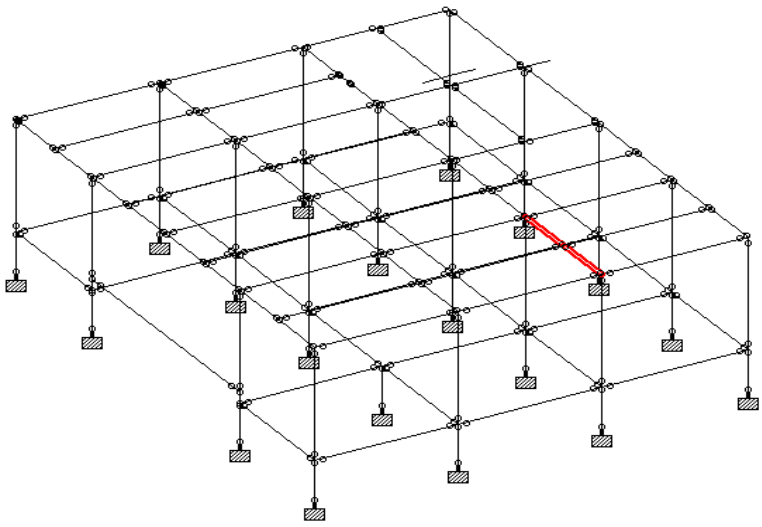
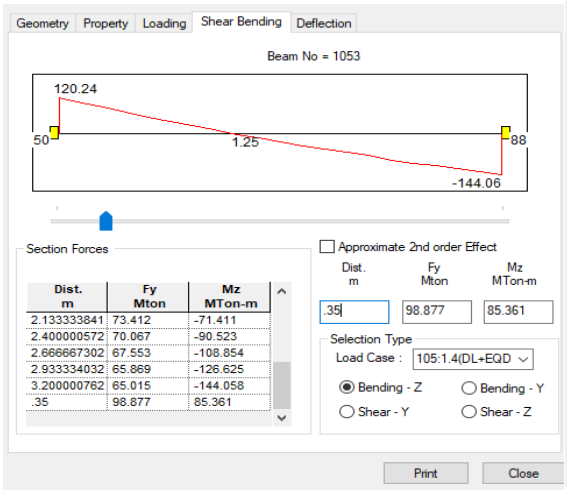
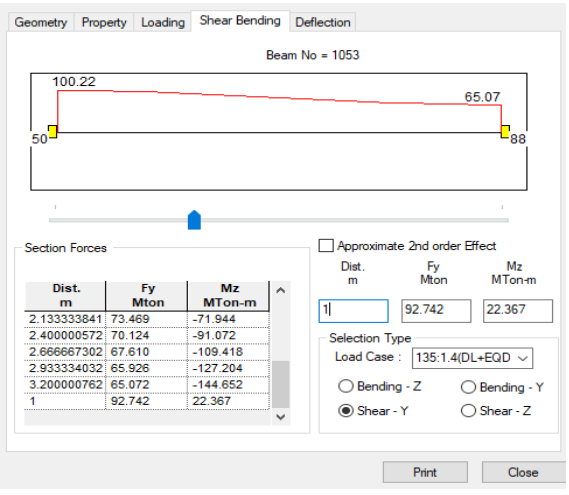
Design of beam : FB-12





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-12

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G15: _FB_12									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1060	88	0.5	-24	0	0	0	-64.5	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Min Fx	1053	50	-0.5	45	0	0	0	53.3	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Fy	1053	50	0	100	0	-0	0	120	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Fy	1060	52	0	-84	0	0	0	69.5	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Max Fz	1053	50	0	46.3	0.33	0	-0.53	56.9	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fz	1060	88	0	-23	-0.33	0	0.53	-64.9	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mx	1060	88	0	-49	0	0	0	-144	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Min Mx	1053	50	0	100	0	-0	0	120	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
Max My	1053	88	0	29.9	0.33	0	0.53	-64.9	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min My	1053	50	0	46.3	0.33	0	-0.53	56.9	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mz	1053	50	0	100	0	-0	0	120	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)
Min Mz	1053	88	0	65.1	0	-0	0	-145	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4

Max support Top Moment = 86 t-m at the face of the beam

Max span Bottom Moment = -144.7 t-m at the face of the beam

Max Shear = 93 t Corresponding Moment = 22.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	600	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.024	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 86.00 t-m
Effective depth ,d = 703.0 mm
Area of steel required ,Ast,required = 3434 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 25 dia bar althrough at 1st layer
Provide 6 nos 20 dia at 2nd layer
Ast,provided = 4832 mm^2 > 3434 mm^2 ok

Max span Bottom Moment = 145 t-m
Effective depth ,d = 697.0 mm
Area of steel required ,Ast,required = 6096 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd 1478 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 1399 mm^2

Provide 6 nos 25 dia bar althrough at 1st layer
Provide 6 nos 32 dia at 2nd layer
Ast,provided = 7774 mm^2 > 6096 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 93 t Corresponding Moment, Mu = 22.5 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 63.8 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 48.1 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 72.4 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 36.1 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 75.9 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 2.57 mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.52* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ *0.5* mm^2/m
Provide 6L-12dia stirrups @200 mm c/c

therefore, A_v/s provided = 3.39 mm^2/m ok

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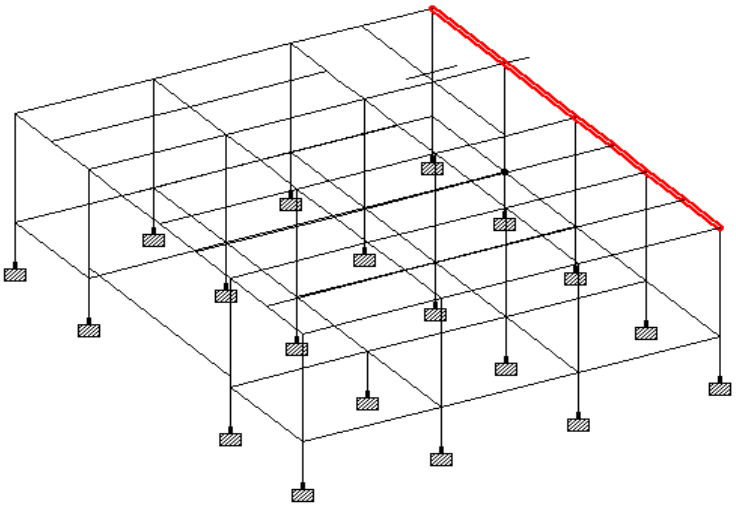
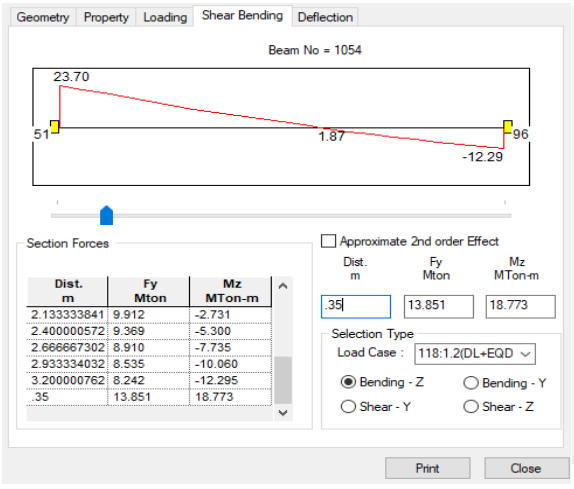
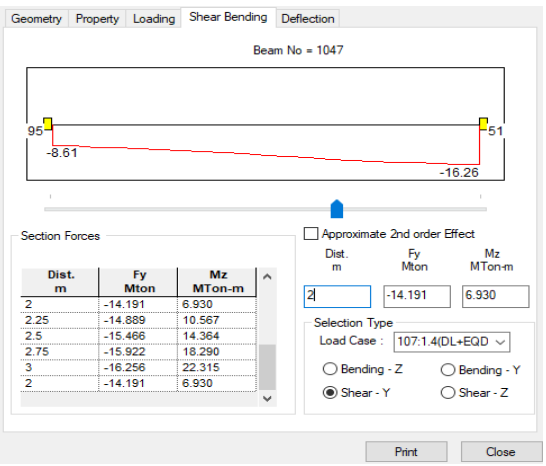
Design of beam : FB-13





Dist. m	Fy Mton	Mz MTon-m
2.133333841	9.912	-2.731
2.400000572	9.369	-5.300
2.666667302	8.910	-7.735
2.933334032	8.535	-10.060
3.200000762	8.242	-12.295
.35	13.851	18.773

Moment at the face of column



Dist. m	Fy Mton	Mz MTon-m
2	-14.191	6.930
2.25	-14.889	10.567
2.5	-15.466	14.364
2.75	-15.922	18.290
3	-16.256	22.315
2	-14.191	6.930

Shear at the distance d from face of column

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Design of beam = FB-13

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G16: _FB_13

	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1061	96	0.15	-5.3	0	0	0	-11.4	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Min Fx	1054	51	-0.1	10	0	0	0	11	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Fy	1054	51	0	15.5	0	0	0	21.2	103 1.4(DL+EQDL+FL)+1.7(LL+SBT)
Min Fy	1047	51	0	-16	0	0	0	22.3	107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
Max Fz	1054	51	0	11.1	0.11	0	-0.17	14.3	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fz	1061	96	0	-4.2	-0.11	0	0.17	-11.7	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mx	1061	96	0	-4.2	0	0	0	-14.9	108 1.4(DL+EQDL+FL+PL)+1.7(LL+SBT)
Min Mx	1054	51	0	15.4	0	0	0	21	108 1.4(DL+EQDL+FL+PL)+1.7(LL+SBT)
Max My	1054	96	0	5.13	0.11	0	0.17	-11.7	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min My	1054	51	0	11.1	0.11	0	-0.17	14.3	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Mz	1054	51	0	14.3	0	0	0	23.7	118 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
Min Mz	1040	95	0	5.57	0	0	0	-15	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)

Max support Top Moment = 19 t-m at the face of the beam

Max span Bottom Moment = -15.0 t-m at the face of the beam

Max Shear = 14.5 t Corresponding Moment = 7 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 19.00 t-m

Effective depth ,d = 730.0 mm

Area of steel required ,Ast,required = 700 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 4 nos 20 dia bar althrough at 1st layer

Ast,provided = 1257 mm^2 > 700 mm^2 ok

Max span Bottom Moment = 15 t-m

Effective depth ,d = 730.0 mm

Area of steel required ,Ast,required = 552 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 4 nos 20 dia bar althrough at 1st layer

Ast,provided = 1257 mm^2 > 552 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 14.5 t Corresponding Moment, Mu = 7 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 30.9 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 29.8 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 50.1 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 22.3 t

Remarks : provide minimum shear reinforcement

$V_s = V_u/\varphi - V_c$ - t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = - mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.35* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.33* mm^2/m } table 9.6.3.3 SBC 304 : 2018

Provide 2L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 0.79 mm^2/m ok

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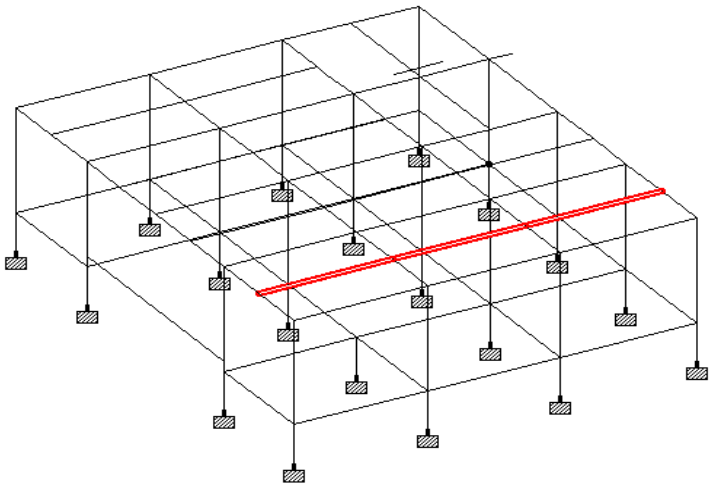
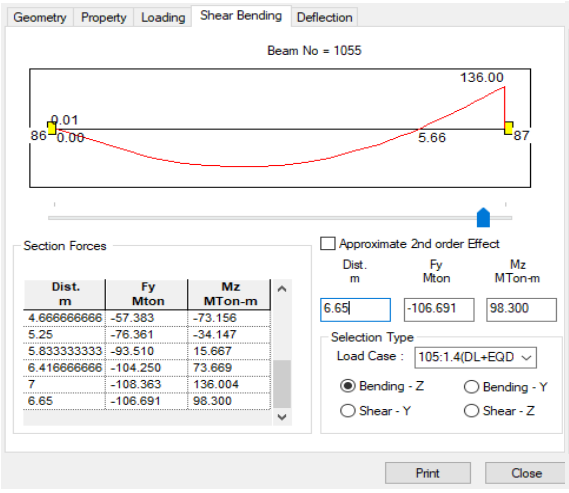
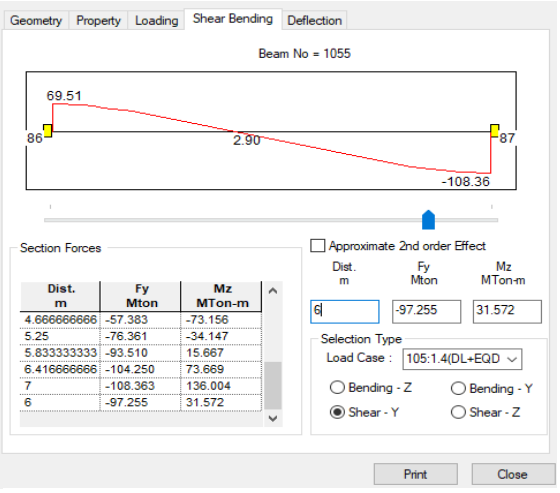
Design of beam : FB-14





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB-14

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G18: _FB_14									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1056	87	0.42	35.4	0	0	0	46.6	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Min Fx	1056	87	-0.4	35.3	0	0	0	46.3	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Max Fy	1056	87	0	97.3	0	-0	0	136	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)
Min Fy	1055	87	0	-108	0	0	0	136	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)
Max Fz	1056	87	0	35.4	0	0	-0	46.5	122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.
Min Fz	1056	87	0	35.4	0	0	0	46.5	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Mx	1055	86	0	69.7	0	0	0	0.01	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Mx	1056	87	0	97	0	-0	0	135	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Max My	1057	88	0	19.8	0	0	0	37.3	122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.
Min My	1057	88	0	19.8	0	0	-0	37.4	121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0
Max Mz	1055	87	0	-108	0	0	0	136	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)
Min Mz	1057	96	0.08	-7	0	0	0	0	128 0.9(DL+EQDL)+1.0SL(-X)

Max support Top Moment = 98.5 t-m at the face of the beam

Max span Bottom Moment = -123.3 t-m at the face of the beam

Max Shear = 97.5 t Corresponding Moment = 32 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	500	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.028	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 98.50 t-m
Effective depth ,d = 700.5 mm
Area of steel required ,Ast,required = 4051 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 6 nos 25 dia bar althrough at 1st layer
Provide 6 nos 25 dia at 2nd layer
Ast,provided = 5893 mm^2 > 4051 mm^2 ok

Max span Bottom Moment = 123 t-m
Effective depth ,d = 725.5 mm
Area of steel required ,Ast,required = 5201 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater (a) $0.25\sqrt{f_c'}/f_y$ bd = 1233 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd = 1168 mm^2

Provide 6 nos 25 dia bar althrough at 1st layer

Ast,provided = 2946 mm^2 > 5201 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 97.5 t Corresponding Moment, Mu = 32 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ = 45.4 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ = 39.3 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ = 62.2 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 29.5 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ = 90.7 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 2.98 mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.44* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ = *0.42* mm^2/m
Provide 6L-12dia stirrups @200 mm c/c

therefore, A_v/s provided = 3.39 mm^2/m ok

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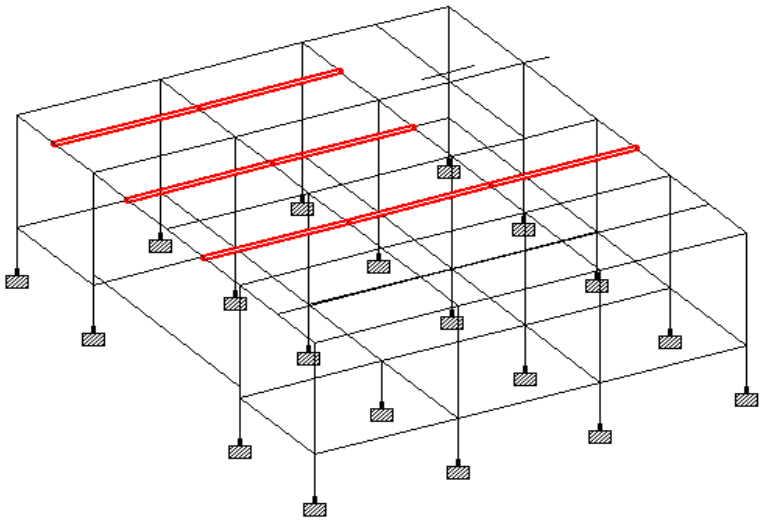
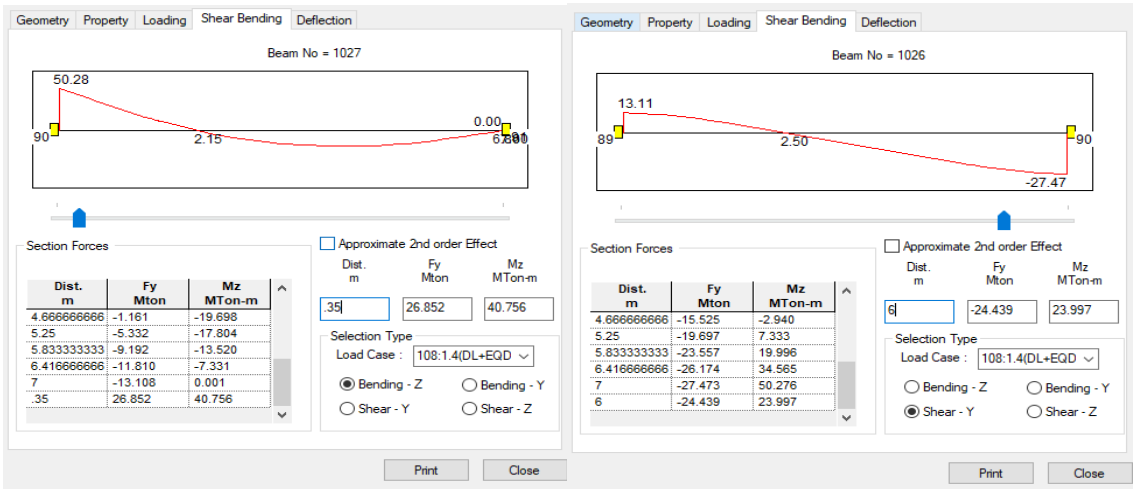
Design of beam : FB-15





Moment at the face of column

Shear at the distance d from face of column

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Design of beam = FB-15

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G19: _FB_15									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1011	84	0.56	16.4	0.01	0	-0.02	21.1	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fx	1011	84	-0.6	16.5	-0.01	0	0.02	21.4	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Max Fy	1043	93	0	27.6	0	0	0	48.6	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Fy	1026	90	0	-27	0	0	0	50.3	108 1.4(DL+EQDL+FL+PL)+1.7(LL+SBT)
Max Fz	1011	84	0.56	16.4	0.01	0	-0.02	21.1	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min Fz	1011	84	-0.6	16.5	-0.01	0	0.02	21.4	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Max Mx	1042	92	0	14.6	0	0	0	8.44	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Min Mx	1041	94	0	19.1	0	-0	0	0.01	135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
Max My	1011	85	0.56	-10	0.01	0	0.05	0	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0
Min My	1011	85	-0.6	-10	-0.01	0	-0.05	0	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.
Max Mz	1027	90	0	27.5	0	0	0	50.3	108 1.4(DL+EQDL+FL+PL)+1.7(LL+SBT)
Min Mz	1026	89	-0.1	8.57	-0	0	0	0	119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0

Max support Top Moment = 41 t-m at the face of the beam

Max span Bottom Moment = -37.1 t-m at the face of the beam

Max Shear = 25 t Corresponding Moment = 23 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 41.00 t-m

Effective depth ,d = 727.5 mm

Area of steel required ,Ast,required = 1549 mm^2 $As = M_u / \varphi f_y (d - a/2)$

$A_{s,min}$=greater of : (a) $0.25\sqrt{f_c'}/f_y$ bd 1025 mm^2 cl 9.6.1.2 SBC 304 : 2018

(b) $1.4/f_y$bd 970 mm^2

Provide 4 nos 25 dia bar althrough at 1st layer

Ast,provided = 1964 mm^2 > 1549 mm^2 ok

Max span Bottom Moment = 37.1 t-m

Effective depth ,d = 727.5 mm

Area of steel required ,Ast,required = 1396 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 4 nos 25 dia bar althrough at 1st layer

Ast,provided = 1964 mm^2 > 1396 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 25 t Corresponding Moment, Mu = 23 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 30.2 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 30.9 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 49.9 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 22.6 t

Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 3.15 t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.1 mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *0.35* mm^2/m ; (b) $0.35b/f_{yt}$ = *0.33* mm^2/m } table 9.6.3.3 SBC 304 : 2018

Provide 4L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 1.57 mm^2/m ok

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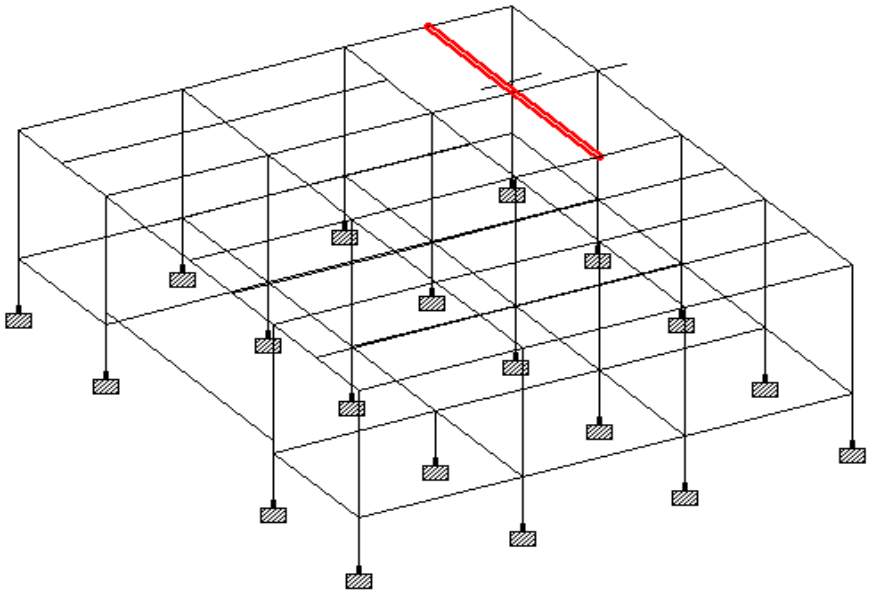
Design of beam : FB-16



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Design of beam = FB-16

Refer staad file : Relining Area,Maaden, R0

Beam End Forces : G20: _FB_16

	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	1012	65	0	47.4	0	0	0	0	101 1.4(DL+EQDL+FL)
Min Fx	1012	65	0	47.4	0	0	0	0	101 1.4(DL+EQDL+FL)
Max Fy	1012	65	0	54.3	0	-0	0	0	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SE
Min Fy	1028	70	0	-52	0	0	0	0.05	105 1.4(DL+EQDL+FL+TL)+1.7(LL+SE
Max Fz	1028	68	0	29	0	-0	0	-37.7	116 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL
Min Fz	1012	65	0	43	0	0	0	0	116 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL
Max Mx	1028	68	0	39.2	0	0	0	-39.2	107 1.4(DL+EQDL+FL+PL)+1.7(LL+E(
Min Mx	1012	65	0	54.1	0	-0	0	0	107 1.4(DL+EQDL+FL+PL)+1.7(LL+E(
Max My	1028	68	0	20.7	0	0	0	-25.6	127 0.9(DL+EQDL)+1.0SL(+X)
Min My	1028	68	0	29	0	-0	0	-37.5	120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL-
Max Mz	1028	70	0	-52	0	0	0	0.05	110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL
Min Mz	1028	68	0	31.7	0	0	0	-42.4	101 1.4(DL+EQDL+FL)

Max support Top Moment = 0.05 t-m at the face of the beam

Max span Bottom Moment = -101.8 t-m at the face of the beam

Max Shear = 54.3 t Corresponding Moment = 0 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	3110	mm	
Beam depth,h =	700	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.005	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 0.05 t-m

Effective depth ,d = 630.0 mm

Area of steel required ,Ast,required = 2 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 22 nos 20 dia bar althrough at 1st layer

Ast,provided = 6914 mm^2 > 2.1 mm^2 ok

Max span Bottom Moment = 102 t-m

Effective depth ,d = 630.0 mm

Area of steel required ,Ast,required = 4341 mm^2 $As = M_u / \varphi f_y (d - a/2)$

Provide 22 nos 20 dia bar althrough at 1st layer

Ast,provided = 6914 mm^2 > 4341 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 54.3 t Corresponding Moment, Mu = 0 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ - t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 197 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 336 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 148 t

Remarks : provide minimum shear reinforcement

$V_s = V_u/\varphi - V_c$ - t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = - mm^2/m

$A_{v,min}/s >$ max of { (a) $0.062\sqrt{f_c'}b/f_{yt}$ = *2.72* mm^2/m ; (b) $0.35b/f_{yt}$ = *2.59* mm^2/m } table 9.6.3.3 SBC 304 : 2018

Provide 8L-10dia stirrups @200 mm c/c

therefore, A_v/s provided = 3.14 mm^2/m ok

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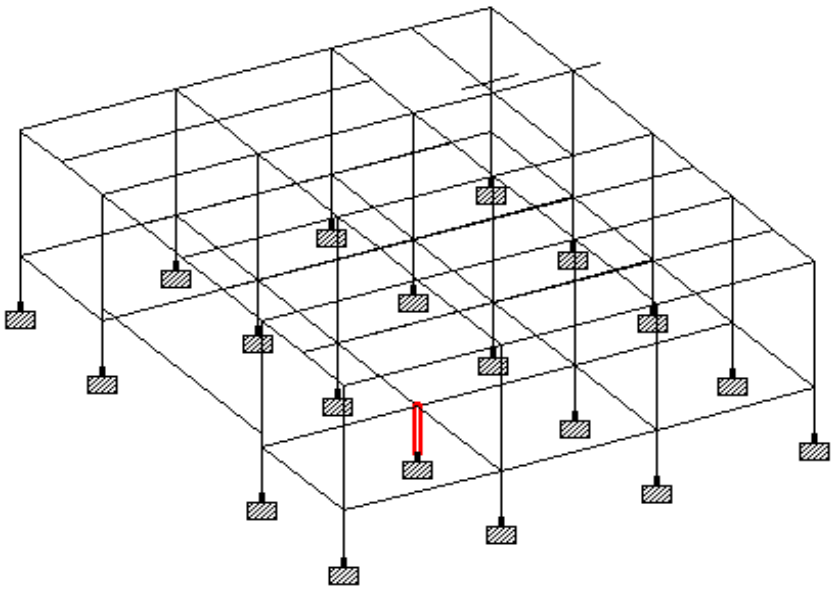
7.4. Design of Columns

Design of column C1



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P_n : Nominal axial load P_u : Factored design axial load Use interaction diagram R35-420.7

M_n : Nominal moment M_u : Factored design moment ACI DESIGN HANDBOOK-SP-17M(09)

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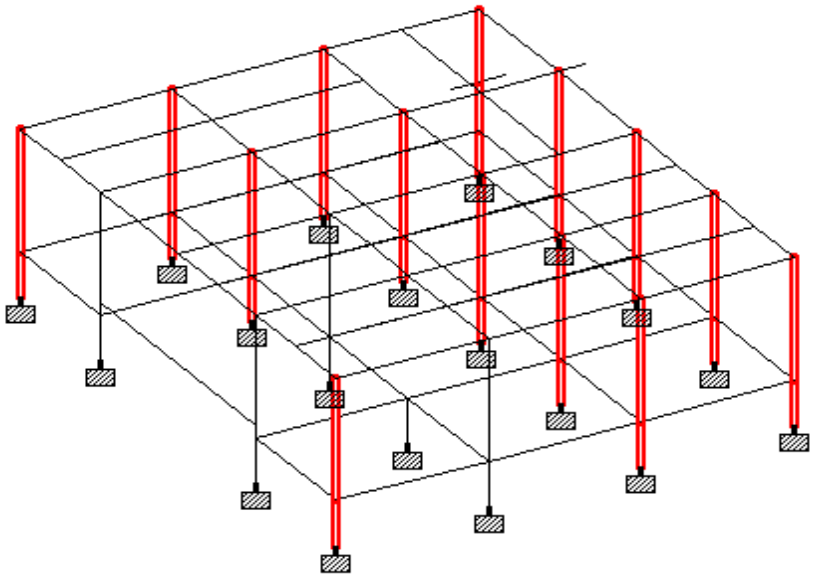
Design of column C2



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Design of column C2

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.90

% of reinforcement, p_t = 1.1 % i.e. ρ_g = 0.011

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 700 mm

Depth parallel to Z-Z axis, b = 700 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.79

Y_b = (b-d')/b = 0.79

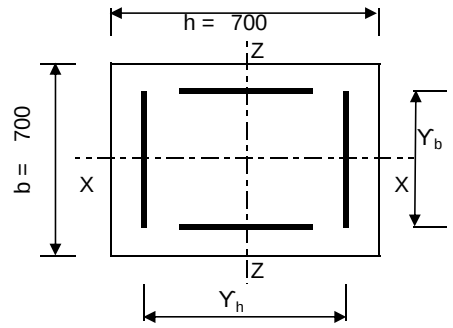
Unsupported length of column parallel to X-X axis, l_{ux} = 5.45 m kl_{ux}/b = 7.786 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 5.45 m kl_{uz}/h = 7.786

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 13501.8 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
141	3791.1	115.2	115.4	4212	127.9	128.2	0.246	0.120	1441	0.09	0.120	1441	0.09		0.04	safe
115	1982.7	53.0	32.6	2203	58.8	36.2	0.128	0.100	1201	0.05	0.100	1201	0.03		0.01	safe
116	1302.3	0.9	22.5	1447	1.0	25.0	0.084	0.080	960.4	0.00	0.080	960.4	0.03		0	safe
135	3788.4	115.4	115.6	4209	128.2	128.5	0.245	0.120	1441	0.09	0.120	1441	0.09		0.04	safe
107	1543.0	153.2	270.3	1714	170.2	300.3	0.1	0.080	960.4	0.18	0.080	960.4	0.31		0.22	safe
128	325.6	15.6	38.8	362	17.4	43.1	0.021	0.060	720.3	0.02	0.060	720.3	0.06	1.61	0.01	safe
109	675.5	25.5	47.8	751	28.3	53.1	0.044	0.060	720.3	0.04	0.060	720.3	0.07		0.02	safe
141	1577.2	89.8	433.3	1752	99.8	481.5	0.102	0.080	960.4	0.10	0.080	960.4	0.50		0.36	safe
107	1440.1	153.2	687.0	1600	170.2	763.3	0.093	0.080	960.4	0.18	0.080	960.4	0.79		0.75	safe
116	1214.1	0.9	17.0	1349	1.0	18.9	0.079	0.080	960.4	0.00	0.080	960.4	0.02		0	safe
108	2916.7	73.8	331.9	3241	82.0	368.8	0.189	0.110	1321	0.06	0.110	1321	0.28		0.14	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
5390	12	25	+	0	0	5893	1.2	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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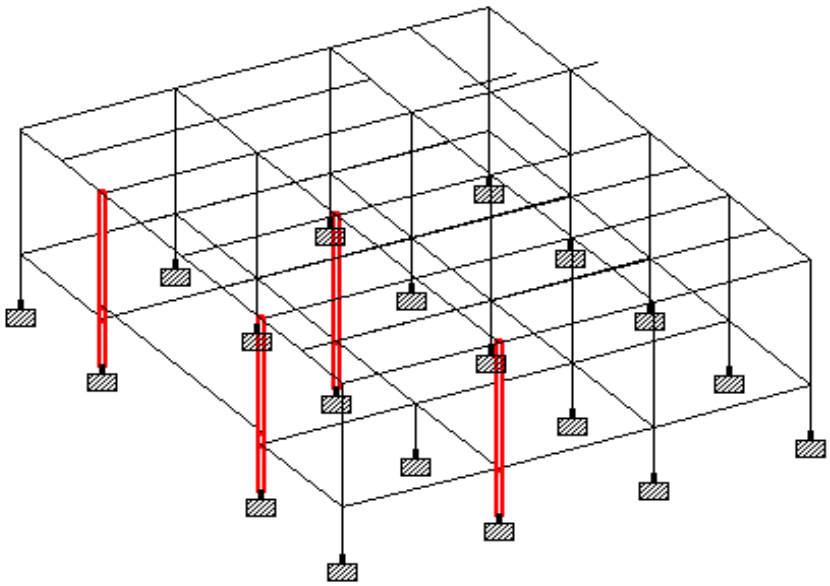
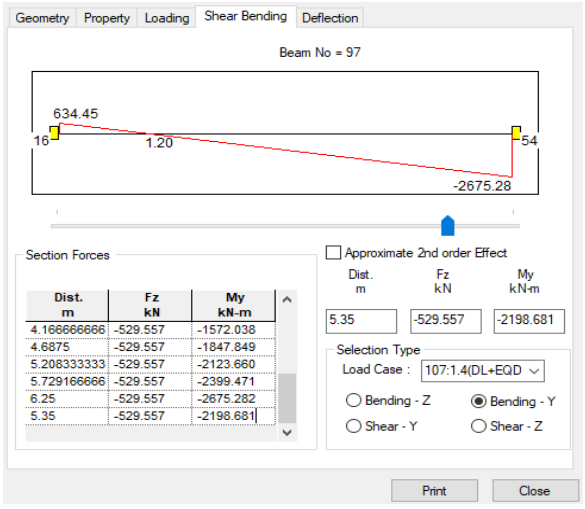
Design of column C3





Design Moment at center of beam

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Design of column C3

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.90

% of reinforcement, p_t = 2.2 % i.e. ρ_g = 0.022

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 700 mm

Depth parallel to Z-Z axis, b = 1000 mm

Clear Cover = 50 mm

d' = 76 mm

Y_h = (h-d')/h = 0.78

Y_b = (b-d')/b = 0.85

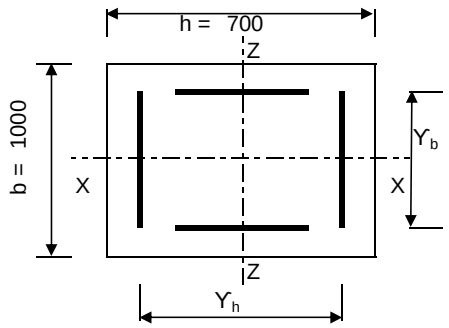
Unsupported length of column parallel to X-X axis, l_{ux} = 5.45 m kl_{ux}/b = 5.45 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 6.45 m kl_{uz}/h = 9.214

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 21683.9 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx}= M_{ux}/\varphi$	$M_{nz}= M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
135	4959.9	0.1	511.6	5511	0.1	568.5	0.225	0.160	3920	0.00	0.160	2744	0.21	1.61	0.08	safe
126	596.9	52.8	213.8	663	58.7	237.6	0.027	0.160	3920	0.01	0.160	2744	0.09		0.02	safe
115	2891.2	185.2	265.8	3212	205.8	295.3	0.131	0.160	3920	0.05	0.160	2744	0.11		0.04	safe
135	3220.6	188.6	426.8	3578	209.6	474.2	0.146	0.160	3920	0.05	0.160	2744	0.17		0.07	safe
135	4800.1	398.8	567.6	5333	443.1	630.6	0.218	0.160	3920	0.11	0.160	2744	0.23		0.12	safe
107	4529.2	532.4	640.8	5032	591.6	712.0	0.205	0.160	3920	0.15	0.160	2744	0.26		0.16	safe
109	3611.8	94.5	280.6	4013	105.0	311.8	0.164	0.160	3920	0.03	0.160	2744	0.11		0.03	safe
110	3668.1	145.7	194.1	4076	161.9	215.6	0.166	0.160	3920	0.04	0.160	2744	0.08		0.02	safe
135	4653.1	398.8	1924.8	5170	443.1	2138.6	0.211	0.160	3920	0.11	0.160	2744	0.78		0.7	safe
107	4382.2	532.4	2200.0	4869	591.6	2444.4	0.199	0.160	3920	0.15	0.160	2744	0.89		0.88	safe
135	3087.7	188.6	639.1	3431	209.6	710.1	0.14	0.160	3920	0.05	0.160	2744	0.26		0.12	safe
115	2765.2	185.2	891.8	3072	205.8	990.8	0.125	0.160	3920	0.05	0.160	2744	0.36		0.2	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
15400	20	32	+	0	0	16092	2.3	10	480	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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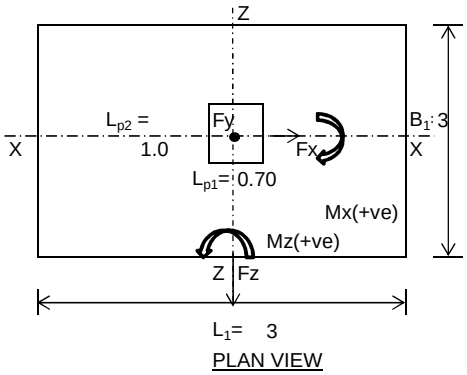
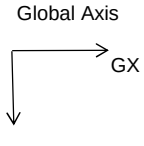
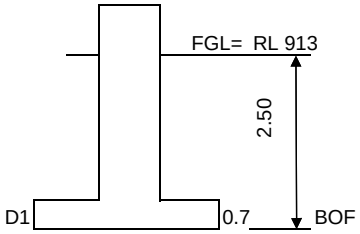
7.4 Design of Foundations

Foundation MKD. **F1**



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

EL of Finish Ground Level ,FGL =	RL	913.0	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		750	kN/m^2

Material Properties :

Unit Weight of Soil =		18.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	3	m
Footing dimension along Z direction,B_1	=	3	m
Depth, D_1	=	0.7	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap =	0.7	x	1	=	0.7	m^2
Area of Footing , A =	3.00	x	3.00	=	9	m^2
Zz-z =	3	x	3.00 x 3.00 /	6	=	4.5	m^3
Zx-x =	3	x	3 x 3 /	6	=	4.5	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	158	kN
Wt of soil above ftg =	269	kN
Total weight,W=	426	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.7	m
Pedestal dimension along Z direction,Lp2=	1	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 18 = 795 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1 STAAD node35	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:								Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis									
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zzz+Mx/Zxx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	KN	KN	KN	KN-m	KN-m	KN-m	Kn/m^2	Kn/m^2		Kn/m^2		Kn-m	Kn-m			Kn-m	Kn-m			Kn	Kn	Kn	Kn		
201	-4.93	-1580	-2.4	-118	0	14.3	205.1	795.0	safe	146.1	ok	14.3	2370	165	ok	118.4	2370	20	ok	4.9	2.4	5.5	1580	268	ok
202	-4.69	-3534	0.23	-315	0	37	470.9	795.0	safe	314.5	ok	37.0	5301	143	ok	314.9	5301	16.8	ok	4.7	0.2	4.7	3534	6146	ok
203	-7.45	-3499	0.04	-308	0	45.4	467.3	795.0	safe	310.4	ok	45.4	5249	116	ok	307.5	5249	17.1	ok	7.5	0.0	7.5	3499		ok
204	-4.75	-3046	-0.4	-266	0	31.3	404.5	795.0	safe	272.4	ok	31.3	4568	146	ok	265.8	4568	17.2	ok	4.8	0.4	4.8	3046	2921	ok
205	-6.82	-3019	-0.6	-260	0	37.6	401.7	795.0	safe	269.3	ok	37.6	4529	120	ok	260.3	4529	17.4	ok	6.8	0.6	6.8	3019	2168	ok
206	-10.4	-1587	-6.2	-139	0	31.2	214.2	795.0	safe	138.6	ok	31.2	2381	76.3	ok	138.8	2381	17.2	ok	10.4	6.2	12.1	1587	103	ok
207	-10.2	-3541	-3.6	-335	0	53.8	480.0	795.0	safe	307.1	ok	53.8	5312	98.7	ok	335.3	5312	15.9	ok	10.2	3.6	10.8	3541	397	ok
208	-12.9	-3506	-3.8	-328	0	62.3	476.4	795.0	safe	302.9	ok	62.3	5260	84.4	ok	327.9	5260	16.1	ok	12.9	3.8	13.5	3506	373	ok
209	-10.2	-3053	-4.2	-286	0	48.2	413.5	795.0	safe	265.0	ok	48.2	4579	95.1	ok	286.1	4579	16	ok	10.2	4.2	11.1	3053	290	ok
210	-12.3	-3027	-4.4	-281	0	54.5	410.8	795.0	safe	261.9	ok	54.5	4540	83.3	ok	280.6	4540	16.2	ok	12.3	4.4	13.0	3027	278	ok
211	-43.6	-1587	-1.4	-115	0	125	229.7	795.0	safe	122.9	ok	125.0	2380	19	ok	115.3	2380	20.6	ok	43.6	1.4	43.7	1587	455	ok
212	31.87	-1569	-3.7	-122	0	-95	222.7	795.0	safe	126.1	ok	95.0	2354	24.8	ok	122.2	2354	19.3	ok	31.9	3.7	32.1	1569	171	ok
213	-3.58	-1617	-36	-286	0	10.1	245.5	795.0	safe	113.9	ok	10.1	2426	239	ok	285.9	2426	8.49	ok	3.6	36.3	36.5	1617	17.8	ok
214	-7.4	-1541	36.1	62.8	0	21.9	190.1	795.0	safe	152.4	ok	21.9	2311	105	ok	62.8	2311	36.8	ok	7.4	36.1	36.9	1541	17.1	ok
215	-40.7	-1608	-16	-192	0	123	248.7	795.0	safe	108.7	ok	122.9	2412	19.6	ok	192.1	2412	12.6	ok	40.7	16.2	43.8	1608	39.6	ok
216	30.8	-1552	11.5	-45	0	-94	203.4	795.0	safe	141.7	ok	94.2	2328	24.7	ok	44.7	2328	52.1	ok	30.8	11.5	32.9	1552	53.9	ok
217	-21.5	-1622	-32	-272	0	65.5	255.1	795.0	safe	105.3	ok	65.5	2432	37.2	ok	271.6	2432	8.96	ok	21.5	32.3	38.8	1622	20.1	ok
218	11.64	-1538	27.6	34.8	0	-37	186.9	795.0	safe	155.1	ok	36.8	2308	62.7	ok	34.8	2308	66.4	ok	11.6	27.6	30.0	1538	22.3	ok
219	-33.8	-3051	0.29	-264	0	114	423.1	795.0	safe	255.1	ok	114.2	4577	40.1	ok	263.7	4577	17.4	ok	33.8	0.3	33.8	3051	4166	ok
220	22.88	-3038	-1.4	-269	0	-51	408.7	795.0	safe	266.6	ok	50.8	4558	89.8	ok	268.9	4558	17	ok	22.9	1.4	22.9	3038	864	ok
221	-3.71	-3074	-26	-392	0	28.1	434.9	795.0	safe	248.4	ok	28.1	4612	164	ok	391.7	4612	11.8	ok	3.7	25.9	26.2	3074	47.5	ok
222	-6.57	-3017	28.4	-130	0	36.9	372.4	795.0	safe	298.2	ok	36.9	4526	123	ok	130.1	4526	34.8	ok	6.6	28.4	29.2	3017	42.5	ok
223	-31.5	-3067	-11	-321	0	113	437.3	795.0	safe	244.4	ok	112.6	4601	40.9	ok	321.3	4601	14.3	ok	31.5	10.8	33.3	3067	113	ok
224	22.08	-3026	9.98	-211	0	-50	394.2	795.0	safe	278.3	ok	50.2	4539	90.4	ok	210.7	4539	21.5	ok	22.1	10.0	24.2	3026	121	ok
225	-17.2	-3078	-23	-381	0	69.6	442.1	795.0	safe	241.9	ok	69.6	4617	66.4	ok	380.9	4617	12.1	ok	17.2	22.9	28.6	3078	53.8	ok

226	7.709	-3015	22	-151	0	-7.2	370.3		795.0	safe	299.9	ok	7.2	4523	632	ok	151.2	4523	29.9	ok	7.7	22.0	23.3	3015	54.7	ok
227	-38.4	-1107	1.77	-53	0	109	159.2		795.0	safe	87.0	ok	109.4	1661	15.2	ok	53.1	1661	31.3	ok	38.4	1.8	38.5	1107	251	ok
228	37.07	-1090	-0.5	-60	0	-111	159.1		795.0	safe	83.2	ok	110.7	1635	14.8	ok	60.0	1635	27.2	ok	37.1	0.5	37.1	1090	869	ok
229	1.616	-1138	-33	-224	0	-5.6	177.4		795.0	safe	75.5	ok	5.6	1707	307	ok	223.7	1707	7.64	ok	1.6	33.2	33.2	1138	13.7	ok
230	-2.21	-1061	39.3	125	0	6.22	147.2		795.0	safe	88.8	ok	6.2	1592	256	ok	125.1	1592	12.7	ok	2.2	39.3	39.3	1061	10.8	ok
231	-35.5	-1129	-13	-130	0	107	178.1		795.0	safe	72.8	ok	107.2	1693	15.8	ok	129.9	1693	13	ok	35.5	13.1	37.8	1129	34.5	ok
232	35.99	-1073	14.7	17.5	0	-110	147.6		795.0	safe	91.0	ok	109.9	1610	14.7	ok	17.5	1610	91.8	ok	36.0	14.7	38.9	1073	29.3	ok
233	-16.3	-1142	-29	-209	0	49.8	184.6		795.0	safe	69.4	ok	49.8	1714	34.4	ok	209.4	1714	8.19	ok	16.3	29.2	33.4	1142	15.7	ok
234	16.84	-1059	30.8	97	0	-53	151.0		795.0	safe	84.5	ok	52.5	1589	30.3	ok	97.0	1589	16.4	ok	16.8	30.8	35.1	1059	13.8	ok
235	-0.02	-1689	2.93	-120	0	-0.7	214.5		795.0	safe	160.9	ok	0.7	2534	3619	ok	119.8	2534	21.2	ok	0.0	2.9	2.9	1689	230	ok
236	-4.64	-3536	0.22	-315	0	36.8	471.2		795.0	safe	314.7	ok	36.8	5304	144	ok	315.4	5304	16.8	ok	4.6	0.2	4.6	3536	6458	ok
237	-4.65	-3540	0.27	-316	0	36.8	471.7		795.0	safe	314.9	ok	36.8	5309	144	ok	316.0	5309	16.8	ok	4.6	0.3	4.7	3540	5343	ok
238	-4.66	-3543	0.31	-317	0	36.9	472.3		795.0	safe	315.2	ok	36.9	5315	144	ok	316.7	5315	16.8	ok	4.7	0.3	4.7	3543	4557	ok
239	-4.66	-3545	0.34	-317	0	36.9	472.6		795.0	safe	315.3	ok	36.9	5317	144	ok	317.1	5317	16.8	ok	4.7	0.3	4.7	3545	4208	ok
240	-4.65	-3540	0.28	-316	0	36.8	471.9		795.0	safe	315.0	ok	36.8	5311	144	ok	316.2	5311	16.8	ok	4.7	0.3	4.7	3540	5094	ok
241	-4.64	-3536	0.22	-315	0	36.8	471.2		795.0	safe	314.7	ok	36.8	5304	144	ok	315.3	5304	16.8	ok	4.6	0.2	4.6	3536	6458	ok
242	-4.66	-3536	0.22	-315	0	36.8	471.1		795.0	safe	314.6	ok	36.8	5303	144	ok	315.3	5303	16.8	ok	4.7	0.2	4.7	3536	6547	ok
243	-4.67	-3535	0.21	-315	0	36.9	471.1		795.0	safe	314.6	ok	36.9	5303	144	ok	315.3	5303	16.8	ok	4.7	0.2	4.7	3535	6639	ok
244	-4.68	-3535	0.21	-315	0	36.9	471.1		795.0	safe	314.6	ok	36.9	5303	144	ok	315.3	5303	16.8	ok	4.7	0.2	4.7	3535	6671	ok
245	-4.66	-3542	0.3	-316	0	36.8	472.1		795.0	safe	315.1	ok	36.8	5313	144	ok	316.5	5313	16.8	ok	4.7	0.3	4.7	3542	4786	ok
246	-4.66	-3543	0.31	-317	0	36.9	472.3		795.0	safe	315.1	ok	36.9	5314	144	ok	316.7	5314	16.8	ok	4.7	0.3	4.7	3543	4572	ok

Design pressure for Strength Calculation

Foundation Mkd. F1 STAAD node35	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment						
	Fx	Fy	Fz	Mx	My	Mz	P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
L/C	KN	KN	KN	KN-m	KN-m	KN-m	KN/m^2			
101	-6.91	-1616	-3.3	-166	0	20.1	220.9	212	147.2	138.3
102	-6.43	-4940	1.07	-500	-0	58.3	673.1	647	450.7	424.8
103	-11.1	-4880	0.75	-488	-0	72.7	666.9	635	450.1	417.8
104	-6.49	-4938	1.09	-500	-0	58.5	672.8	647	450.6	424.6
105	-11.2	-4878	0.78	-487	-0	72.9	666.6	634	450.0	417.6
106	-14.6	-1626	-8.6	-194	0	43.7	233.6	214	147.2	127.8
107	-14.1	-4950	-4.2	-529	-0	81.9	685.8	649	450.7	414.3
108	-18.8	-4891	-4.6	-516	-0	96.3	679.6	637	450.1	407.3
109	-14.1	-4948	-4.2	-528	-0	82.1	685.5	649	450.6	414.2
110	-18.8	-4889	-4.5	-516	-0	96.5	679.3	636	450.0	407.2
111	-30.4	-1355	2.37	-110	0	85.8	194.1	156	145.3	107.1
112	29.97	-1341	0.56	-115	0	-90	194.7	155	143.4	103.4
113	1.608	-1379	-26	-246	0	-6.1	209.4	207	99.9	97.2

114	-1.45	-1318	32.4	32.6	0	3.34	154.5	153	140.0	138.5
115	-61.3	-2338	4.43	-206	0	186	346.8	264	255.3	172.8
116	59.56	-2310	0.8	-217	0	-166	341.9	268	245.5	171.5
117	2.833	-2387	-51	-479	0	1.82	372.1	371	159.2	158.4
118	-3.29	-2264	64.5	79.2	0	20.7	273.8	265	238.6	229.5
119	-53.1	-2382	-19	-329	-0	172	375.9	300	229.9	153.5
120	49.02	-2302	20.9	-118	0	-138	312.8	251	260.4	199
121	-25.7	-2401	-42	-442	0	89.9	385.1	345	188.6	148.7
122	21.66	-2283	43.8	-4.4	0	-56	267.1	242	265.2	240.2
123	-61.5	-1023	2.74	-79	0	175	170.2	92.4	135.0	57.2
124	59.29	-995	-0.9	-90	0	-177	170.0	91.4	129.8	51.2
125	2.56	-1072	-53	-352	0	-8.7	199.4	196	42.8	38.9
126	-3.56	-949	62.8	206	0	10.1	153.5	149	62.1	57.6
127	-50.6	-1052	-19	-190	0	153	193.1	125	108.8	40.8
128	51.43	-973	21	21	0	-157	147.7	77.9	138.4	68.5
129	-23.3	-1072	-42	-303	0	71	202.3	171	67.5	36
130	24.07	-953	44	135	0	-75	152.5	119	92.7	59.3
131	-0.1	-1478	3.09	-136	0	-0.7	194.7	194	134.1	133.8
132	-6.41	-4941	1.08	-501	-0	58.3	673.2	647	450.7	424.8
133	-6.42	-4947	1.16	-502	-0	58.3	674.1	648	451.1	425.2
134	-6.44	-4953	1.24	-503	-0	58.4	675.1	649	451.6	425.6
135	-6.44	-4956	1.28	-504	-0	58.4	675.6	650	451.8	425.9
136	-6.42	-4948	1.18	-502	-0	58.3	674.4	649	451.3	425.3
137	-6.41	-4940	1.08	-501	-0	58.2	673.1	647	450.7	424.8
138	-6.43	-4940	1.07	-500	-0	58.3	673.1	647	450.7	424.8
139	-6.46	-4940	1.07	-500	-0	58.4	673.1	647	450.7	424.7
140	-6.48	-4940	1.07	-500	-0	58.5	673.1	647	450.7	424.7
141	-6.43	-4951	1.21	-503	-0	58.3	674.8	649	451.4	425.5
142	-6.44	-4953	1.24	-503	-0	58.3	675.1	649	451.6	425.6

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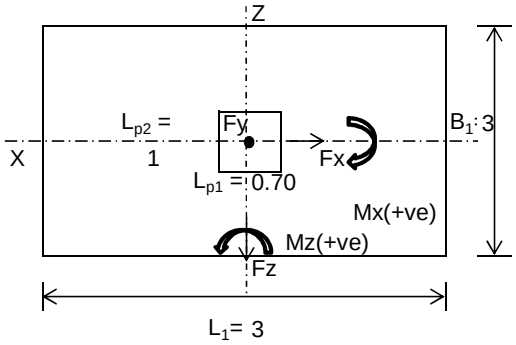


PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	20	157.4	150	20	202.3	150

Footing Dimension :

Footing dimension along X direction, L_1	=	3	m
Footing dimension along Z direction, B_1	=	3	m
Depth, D_1	=	0.7	m
Clear cover	=	75	mm

Column Dimension :

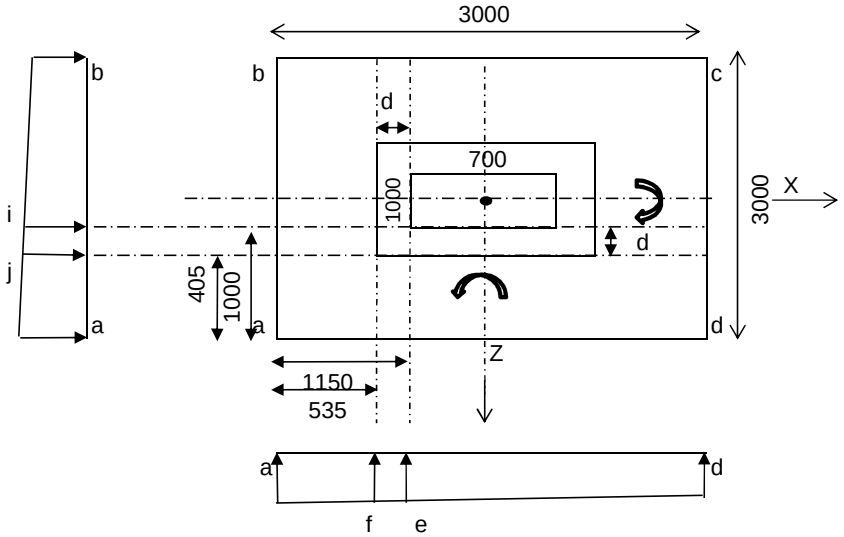
Pedestal dimension along X direction, Lp1	=	0.7	m
Pedestal dimension along Z direction, Lp2	=	1	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 685.8 KN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*685.8]*3*1.15*1.15	=	1360.5	KN-m
Shear force at Face "f"	= 685.8*0.535*3	=	1100.7	KN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*685.8]*3*1*1	=	1028.7	KN-m
Shear force at Face "j"	= 685.8*0.405*3	=	833.2	KN

Design for Bottom Reinforcement :

In X Direction (Along Length 3000 mm)

Cantilever Length		=	1150	mm
Dia of Bar		d_b =	20	mm
Effective Depth		d =	615	mm
Bending Moment at "e"		M_u =	1360.5	KN-m
$C = 0.85f_c'ba$ and $T = A_sf_y$	Therefore 'a' = $f_y/0.85f_c'b * A_s$	a =	0.0047	$*A_s$
$\varphi M_n \geq M_u = \varphi A_sf_y(d-a/2)$	0.9*420*(615-0.0047As/2)As=1360.5*10^6	A_s =	5989.4	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*3000*700	$A_{s,min}$ =	3780	mm^2
Spacing required	3000*22/28*20^2/5989.4	S_{reqd} =	157.4	mm c/c
Spacing Provided		S_{prov} =	150	mm c/c
A_s Provided	3000*22/28*20^2/150	$A_{s,pro}$ =	6285.8	mm^2
$a = A_sf_y/0.85f'_cb$	6285.8*420/(0.85*35*3000)	a =	29.59	mm
$c = a/\beta_1$	29.59/0.8	c =	36.99	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/36.99*(615-36.99)	ε_t =	0.047	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	1100.7	kn
Moment at "f"	= [0.5*685.8]*3*0.535*0.535	M_u =	294.4	kn-m
ρ_w provided	6285.8/3000/615	$\rho_{w,pro}$ =	0.0034	%
Effective Shear Area	=615*3000/10^6	bd =	1.845	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$ =	1992.1	KN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	1853.3	KN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	3165.4	KN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	6793.0	KN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 3000 mm)

Cantilever Length		=	1000	mm
Dia of Bar		d_b =	20	mm
Effective Depth		d =	595	mm
Bending Moment at "i"		M_u =	1028.7	KN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0047	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(595-0.0047As/2)As=1028.7*10^6	A_s =	4659.6	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*3000*700	$A_{s,min}$ =	3780	mm^2
Spacing required	3000*22/28*20^2/4659.6	S_{reqd} =	202.35	mm c/c
Spacing Provided		S_{prov} =	150	mm c/c
A_s Provided	3000*22/28*20^2/150	$A_{s,pro}$ =	6285.8	mm^2
$a = A_{st} f_y/0.85f'_c b$	6285.8*420/(0.85*35*3000)	a =	29.58	mm
$c = a/\beta_1$	29.58/0.8	c =	36.98	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/36.98*(595-36.98)	ε_t =	0.045	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	833.2	KN
Moment at "j"	= [0.5*685.8]*3*0.405*0.405	M_u =	168.7	KN-m
ρ_w provided	= 6285.8/3000/595	$\rho_{w,pro}$ =	0.0035	%
Effective Shear Area	=595*3000/10^6	bd =	1.785	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	2003.6	KN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	1796.5	KN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	3062.5	KN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	6574.7	KN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 0.70

Average soil pressure		=	685.80	Kn/sqm
Shear force,V_u	= 685.8*[3*3-(0.7+0.615)*(1+0.595)]	=	4733.79	KN
Shear peremeter,b_0	= 2*[(0.7+0.615)+(1+0.595)]	=	5.82	m
$V_c = 0.17(1+2/\beta)\sqrt{f_c'}b_0d$	= 0.66 $\sqrt{f_c'}b_0d$			
$V_c = 0.083(\alpha_s d/b_0 + 2)\sqrt{f_c'}b_0d$	= 0.43 $\sqrt{f_c'}b_0d$	For Edge column α_s	30	
$V_c = 0.33\sqrt{f_c'}b_0d$	= 0.33 $\sqrt{f_c'}b_0d$			
φV_c	= 0.75*0.33*SQRT(35)*5820*615/1000	=	5240.9	KN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

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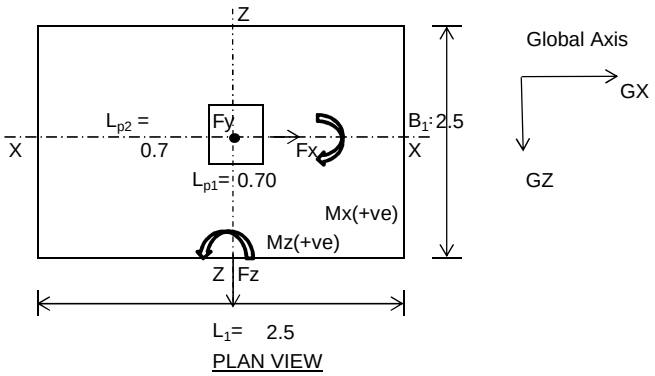
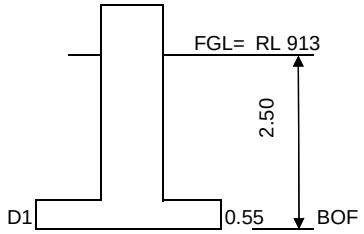
Foundation MKD. **F2**



PLAN VIEW



SECTIONAL VIEW

Foundation Level Details :

EL of Finish Ground Level ,FGL =	RL	913.0	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		750	kN/m^2

Material Properties :

Unit Weight of Soil =		18.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2.5	m
Footing dimension along Z direction,B_1	=	2.5	m
Depth, D_1	=	0.55	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap = 0.7 x 0.7	=	0.49	m^2
Area of Footing , A = 2.50 x 2.50	=	6.25	m^2
Zz-z = 2.5 x 2.50 x 2.50 / 6	=	2.6	m^3
Zx-x = 2.5 x 2.5 x 2.5 / 6	=	2.6	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	85.9	kN
Wt of soil above ftg =	202	kN
Total weight,W=	288	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.7	m
Pedestal dimension along Z direction,Lp2=	0.7	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 18 = 795 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F2 STAAD node31	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:									Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis										
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zzz+Mx/Zxx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	KN	KN	KN	KN-m	KN-m	KN-m	Kn/m^2	Kn/m^2		Kn/m^2		Kn-m	Kn-m			Kn-m	Kn-m			Kn	Kn	Kn	Kn		
201	9.362	-1435	-3.1	-4.9	0	-2.2	232.4	795.0	safe	226.9	ok	2.2	1793	800	ok	4.9	1793	367	ok	9.4	3.1	9.9	1435	185	ok
202	42.22	-2605	-40	-31	0	-3.1	429.8	795.0	safe	403.9	ok	3.1	3256	1042	ok	30.6	3256	107	ok	42.2	39.5	57.8	2605	26.4	ok
203	39.9	-2509	-39	-30	0	2.31	413.9	795.0	safe	389.0	ok	2.3	3136	1359	ok	30.1	3136	104	ok	39.9	39.2	55.9	2509		ok
204	34.01	-2312	-30	-24	0	-2.9	380.4	795.0	safe	359.6	ok	2.9	2890	995	ok	24.2	2890	120	ok	34.0	30.4	45.6	2312	30.4	ok
205	32.26	-2240	-30	-24	0	1.17	368.1	795.0	safe	348.9	ok	1.2	2800	2393	ok	23.8	2800	118	ok	32.3	30.2	44.2	2240	29.7	ok
206	6.886	-1433	-5	-9.8	0	6.4	235.6	795.0	safe	223.2	ok	6.4	1792	280	ok	9.8	1792	183	ok	6.9	5.0	8.5	1433	114	ok
207	39.74	-2603	-41	-35	0	5.51	432.3	795.0	safe	400.8	ok	5.5	3254	590	ok	35.5	3254	91.8	ok	39.7	41.5	57.4	2603	25.1	ok
208	37.42	-2507	-41	-35	0	10.9	418.8	795.0	safe	383.6	ok	10.9	3134	286	ok	35.0	3134	89.6	ok	37.4	41.1	55.6	2507	24.4	ok
209	31.53	-2311	-32	-29	0	5.73	383.1	795.0	safe	356.4	ok	5.7	2889	504	ok	29.0	2889	99.5	ok	31.5	32.4	45.2	2311	28.6	ok
210	29.79	-2239	-32	-29	0	9.81	373.0	795.0	safe	343.5	ok	9.8	2798	285	ok	28.7	2798	97.5	ok	29.8	32.1	43.8	2239	27.9	ok
211	-17.2	-1417	-4.6	-8.7	0	74.4	258.7	795.0	safe	194.9	ok	74.4	1772	23.8	ok	8.7	1772	204	ok	17.2	4.6	17.8	1417	123	ok
212	36.89	-1435	-1.9	-1.9	0	-80	261.1	795.0	safe	198.3	ok	79.9	1794	22.5	ok	1.9	1794	957	ok	36.9	1.9	36.9	1435	301	ok
213	10.04	-1435	-29	-67	0	-4.1	257.1	795.0	safe	202.3	ok	4.1	1794	442	ok	67.3	1794	26.7	ok	10.0	29.1	30.8	1435	19.7	ok
214	8.229	-1408	28.3	64.5	0	0.66	250.4	795.0	safe	200.4	ok	0.7	1760	2655	ok	64.5	1760	27.3	ok	8.2	28.3	29.5	1408	19.9	ok
215	-9.2	-1430	-11	-26	0	60.2	261.8	795.0	safe	195.8	ok	60.2	1787	29.7	ok	25.7	1787	69.5	ok	9.2	11.4	14.7	1430	50.2	ok
216	27.93	-1440	5.2	16	0	-65	261.4	795.0	safe	199.5	ok	64.7	1800	27.8	ok	16.0	1800	113	ok	27.9	5.2	28.4	1440	111	ok
217	1.093	-1437	-24	-56	0	26.5	261.7	795.0	safe	198.2	ok	26.5	1796	67.7	ok	56.2	1796	32	ok	1.1	23.9	23.9	1437	24.1	ok
218	17.63	-1433	17.7	46.4	0	-31	259.0	795.0	safe	199.5	ok	31.0	1791	57.7	ok	46.4	1791	38.6	ok	17.6	17.7	25.0	1433	32.4	ok
219	14.11	-2298	-32	-27	0	54.6	399.1	795.0	safe	336.4	ok	54.6	2873	52.7	ok	27.1	2873	106	ok	14.1	31.6	34.6	2298	29.1	ok
220	54.66	-2312	-30	-22	0	-61	401.8	795.0	safe	338.0	ok	61.2	2889	47.2	ok	21.9	2889	132	ok	54.7	29.6	62.2	2312	31.3	ok
221	34.52	-2312	-50	-71	0	-4.3	398.9	795.0	safe	341.0	ok	4.3	2890	674	ok	71.0	2890	40.7	ok	34.5	50.0	60.8	2312	18.5	ok
222	33.17	-2292	-6.9	27.8	0	-0.7	377.7	795.0	safe	355.7	ok	0.7	2864	3840	ok	27.8	2864	103	ok	33.2	6.9	33.9	2292	133	ok
223	20.09	-2307	-37	-40	0	43.9	401.4	795.0	safe	337.1	ok	43.9	2884	65.7	ok	39.8	2884	72.4	ok	20.1	36.7	41.9	2307	25.1	ok
224	47.94	-2315	-24	-8.6	0	-50	392.9	795.0	safe	348.1	ok	49.7	2894	58.2	ok	8.6	2894	337	ok	47.9	24.3	53.7	2315	38.2	ok
225	27.82	-2313	-46	-63	0	18.7	401.4	795.0	safe	338.9	ok	18.7	2891	155	ok	62.7	2891	46.1	ok	27.8	46.1	53.8	2313	20.1	ok

226	40.22	-2310	-15	14.3	0	-25	384.5		795.0	safe	354.7	ok	24.5	2887	118	ok	14.3	2887	202	ok	40.2	14.9	42.9	2310	62	ok
227	-19.8	-956	-2.2	-3.9	0	70.5	181.6		795.0	safe	124.5	ok	70.5	1195	17	ok	3.9	1195	308	ok	19.8	2.2	20.0	956	173	ok
228	34.23	-974	0.5	2.95	0	-84	189.2		795.0	safe	122.6	ok	83.9	1218	14.5	ok	3.0	1218	413	ok	34.2	0.5	34.2	974	776	ok
229	7.377	-974	-27	-63	0	-8	183.0		795.0	safe	128.9	ok	8.0	1218	152	ok	62.5	1218	19.5	ok	7.4	26.7	27.7	974	14.6	ok
230	5.571	-947	30.7	69.3	0	-3.3	179.5		795.0	safe	123.8	ok	3.3	1184	361	ok	69.3	1184	17.1	ok	5.6	30.7	31.2	947	12.3	ok
231	-11.9	-969	-9	-21	0	56.2	184.7		795.0	safe	125.4	ok	56.2	1211	21.5	ok	20.9	1211	57.9	ok	11.9	9.0	14.9	969	43.1	ok
232	25.27	-979	7.61	20.8	0	-69	191.0		795.0	safe	122.3	ok	68.6	1224	17.8	ok	20.8	1224	58.9	ok	25.3	7.6	26.4	979	51.5	ok
233	-1.57	-976	-21	-51	0	22.6	184.6		795.0	safe	127.8	ok	22.6	1220	54	ok	51.4	1220	23.7	ok	1.6	21.5	21.5	976	18.2	ok
234	14.97	-972	20.1	51.3	0	-35	188.6		795.0	safe	122.4	ok	35.0	1215	34.7	ok	51.3	1215	23.7	ok	15.0	20.1	25.0	972	19.4	ok
235	14.65	-1426	-1.1	0.23	0	-12	232.8		795.0	safe	223.7	ok	11.6	1783	154	ok	0.2	1783	7751	ok	14.6	1.1	14.7	1426	526	ok
236	42.24	-2604	-40	-31	0	-3.2	429.6		795.0	safe	403.6	ok	3.2	3255	1028	ok	30.6	3255	106	ok	42.2	39.6	57.9	2604	26.3	ok
237	42.35	-2604	-40	-31	0	-3.2	429.7		795.0	safe	403.7	ok	3.2	3255	1011	ok	30.6	3255	106	ok	42.4	39.6	58.0	2604	26.3	ok
238	42.46	-2605	-40	-31	0	-3.3	429.9		795.0	safe	403.8	ok	3.3	3256	994	ok	30.6	3256	106	ok	42.5	39.6	58.1	2605	26.3	ok
239	42.51	-2606	-40	-31	0	-3.3	430.0		795.0	safe	403.9	ok	3.3	3257	985	ok	30.6	3257	106	ok	42.5	39.6	58.1	2606	26.3	ok
240	42.35	-2616	-40	-31	0	-3.2	431.7		795.0	safe	405.7	ok	3.2	3270	1022	ok	30.7	3270	107	ok	42.4	39.6	58.0	2616	26.4	ok
241	42.19	-2627	-40	-31	0	-3.1	433.4		795.0	safe	407.4	ok	3.1	3284	1060	ok	30.7	3284	107	ok	42.2	39.6	57.9	2627	26.5	ok
242	42.23	-2616	-40	-31	0	-3.1	431.5		795.0	safe	405.6	ok	3.1	3269	1041	ok	30.7	3269	107	ok	42.2	39.6	57.9	2616	26.4	ok
243	42.27	-2604	-40	-31	0	-3.2	429.6		795.0	safe	403.7	ok	3.2	3255	1023	ok	30.7	3255	106	ok	42.3	39.6	57.9	2604	26.3	ok
244	42.29	-2597	-40	-31	0	-3.2	428.6		795.0	safe	402.6	ok	3.2	3246	1013	ok	30.6	3246	106	ok	42.3	39.6	57.9	2597	26.2	ok
245	42.42	-2605	-40	-31	0	-3.3	429.8		795.0	safe	403.8	ok	3.3	3256	1001	ok	30.6	3256	106	ok	42.4	39.6	58.0	2605	26.3	ok
246	42.45	-2605	-40	-31	0	-3.3	429.9		795.0	safe	403.8	ok	3.3	3256	995	ok	30.6	3256	106	ok	42.5	39.6	58.1	2605	26.3	ok

Design pressure for Strength Calculation

Foundation Mkd. F2 STAAD node31	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment			P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
	Fx	Fy	Fz	Mx	My	Mz				
L/C	KN	KN	KN	KN-m	KN-m	KN-m	KN/m^2			
101	13.11	-1605	-4.3	-6.8	0	-3.1	260.8	258	255.5	253.1
102	68.99	-3592	-66	-51	0	-4.7	596.1	593	557.2	553.6
103	65.05	-3429	-66	-50	0	4.55	569.5	566	531.3	527.8
104	68.97	-3594	-66	-51	0	-4.6	596.3	593	557.5	553.9
105	65.02	-3431	-66	-50	0	4.59	569.8	566	531.6	528.1
106	9.641	-1603	-7.1	-14	0	8.95	265.3	258	254.8	247.9
107	65.52	-3590	-69	-57	0	7.41	599.4	594	555.3	549.6
108	61.58	-3427	-69	-57	0	16.6	576.5	564	533.0	520.2
109	65.5	-3592	-69	-57	0	7.45	599.7	594	555.6	549.9
110	61.56	-3429	-68	-57	0	16.7	576.8	564	533.3	520.5
111	-7.83	-1358	-2.6	-3.2	0	49	237.3	200	234.8	197.2
112	35.43	-1372	-0.4	2.28	0	-75	249.0	192	247.3	190
113	13.95	-1372	-22	-50	0	-14	244.1	234	205.6	195

114	12.5	-1350	23.8	55.4	0	-10	241.2	234	198.7	191
115	-12.6	-1928	-22	-19	0	110	358.1	274	343.4	259
116	73.88	-1956	-18	-8.2	0	-137	368.9	264	362.6	257.3
117	30.92	-1957	-61	-113	0	-16	362.5	351	275.8	263.7
118	28.03	-1914	30.7	98	0	-8.2	347.0	341	271.7	265.5
119	2.416	-1950	-32	-45	0	80.4	360.3	299	325.6	263.9
120	55.46	-1965	-8.8	14.4	0	-98	357.6	282	346.6	271.3
121	17.13	-1961	-50	-89	0	32.3	360.3	335	292.1	267.3
122	40.75	-1955	9.06	57.9	0	-50	354.2	316	309.7	271.4
123	-32.4	-1001	-3.5	-6.2	0	113	206.1	119	201.3	114.3
124	54.09	-1029	0.87	4.73	0	-134	217.8	115	214.2	111.6
125	11.13	-1030	-43	-100	0	-12	207.9	199	131.0	121.7
126	8.243	-986	49.3	111	0	-4.6	202.2	199	117.1	113.5
127	-16.5	-1021	-13	-30	0	79.9	205.6	144	182.7	121.3
128	36.58	-1036	10.8	29.7	0	-98	215.0	139	192.2	116.6
129	-1.76	-1032	-31	-73	0	31.9	205.5	181	149.2	124.7
130	21.87	-1026	28.6	73.2	0	-50	211.6	173	155.4	116.7
131	16.5	-1359	-1.3	0.19	0	-13	222.6	213	222.4	212.3
132	69.01	-3593	-66	-51	0	-4.7	596.1	593	557.2	553.6
133	69.19	-3594	-66	-51	0	-4.8	596.3	593	557.4	553.8
134	69.37	-3595	-66	-51	0	-4.9	596.5	593	557.7	553.9
135	69.47	-3596	-66	-51	0	-4.9	596.7	593	557.8	554
136	69.19	-3614	-66	-51	0	-4.8	599.6	596	560.7	557
137	68.91	-3632	-66	-51	0	-4.6	602.5	599	563.5	560
138	68.98	-3613	-66	-51	0	-4.7	599.4	596	560.4	556.9
139	69.05	-3593	-66	-51	0	-4.7	596.2	593	557.3	553.6
140	69.09	-3581	-66	-51	0	-4.8	594.3	591	555.4	551.8
141	69.3	-3594	-66	-51	0	-4.9	596.4	593	557.6	553.8
142	69.36	-3595	-66	-51	0	-4.9	596.5	593	557.7	553.9

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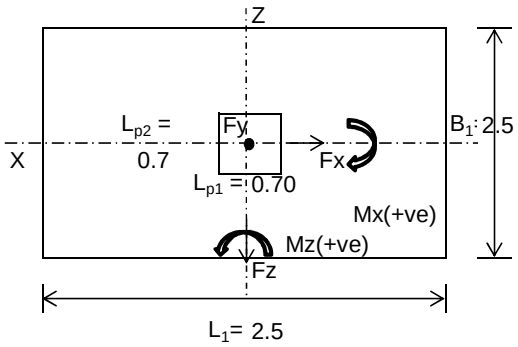


PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	20	221.6	200	20	211.6	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2.5	m
Footing dimension along Z direction, B_1	=	2.5	m
Depth, D_1	=	0.55	m
Clear cover	=	75	mm

Column Dimension :

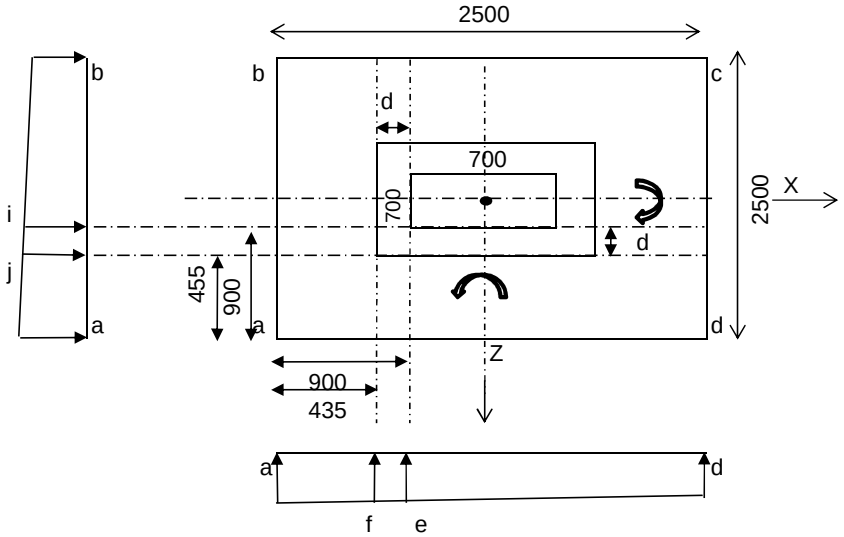
Pedestal dimension along X direction,Lp1	=	0.7	m
Pedestal dimension along Z direction,Lp2	=	0.7	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 602.5 KN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*602.5]*2.5*0.9*0.9	=	610.0	KN-m
Shear force at Face "f"	= 602.5*0.435*2.5	=	655.2	KN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*602.5]*2.5*0.9*0.9	=	610.0	KN-m
Shear force at Face "j"	= 602.5*0.455*2.5	=	685.3	KN

Design for Bottom Reinforcement :

In X Direction (Along Length 2500 mm)

Cantilever Length		=	900	mm
Dia of Bar		d_b =	20	mm
Effective Depth		d =	465	mm
Bending Moment at "e"		M_u =	610.0	KN-m
C = $0.85f_c'ba$ and T = A_sf_y	Therefore 'a' = $f_y/0.85f_c'b$ * A_s	a =	0.0056	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(465-0.0056As/2)As=610*10^6	A_s =	3546.2	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*2500*550	$A_{s,min}$ =	2475	mm^2
Spacing required	2500*22/28*20^2/3546.2	S_{reqd} =	221.6	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2500*22/28*20^2/200	$A_{s,pro}$ =	3928.6	mm^2
$a = A_sf_y/0.85f'_cb$	3928.6*420/(0.85*35*2500)	a =	22.19	mm
$c = a/\beta_1$	22.19/0.8	c =	27.74	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/27.74*(465-27.74)	ε_t =	0.047	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	655.2	kn
Moment at "f"	= [0.5*602.5]*2.5*0.435*0.435	M_u =	142.5	kn-m
ρ_w provided	3928.6/2500/465	$\rho_{w,pro}$ =	0.0034	%
Effective Shear Area	=465*2500/10^6	bd =	1.163	m^2

V_c = lesser of

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$ =	1243.2	KN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	1167.2	KN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	1994.5	KN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	4279.7	KN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 2500 mm)

Cantilever Length		=	900	mm
Dia of Bar		d_b =	20	mm
Effective Depth		d =	445	mm
Bending Moment at "i"		M_u =	610.0	KN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0056	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(445-0.0056As/2)As=610.04*10^6	A_s =	3713.4	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2500*550	$A_{s,min}$ =	2475	mm^2
Spacing required	2500*22/28*20^2/3713.4	S_{reqd} =	211.59	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2500*22/28*20^2/200	$A_{s,pro}$ =	3928.6	mm^2
a = $A_{st}f_y/0.85f'_cb$	3928.6*420/(0.85*35*2500)	a =	22.19	mm
c = a/β_1	22.19/0.8	c =	27.74	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/27.74*(445-27.74)	ε_t =	0.045	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	685.3	KN
Moment at "j"	= [0.5*602.5]*2.5*0.455*0.455	M_u =	155.9	KN-m
ρ_w provided	= 3928.6/2500/445	$\rho_{w,pro}$ =	0.0035	%
Effective Shear Area	=445*2500/10^6	bd =	1.113	m^2
V_c = lesser of				
a) (0.16λ$\sqrt{f_c'}$ + 17$\rho_w V_u d/M_u$)bd		$V_{c,a}$ =	1183.7	KN
b) (0.16λ$\sqrt{f_c'}$ + 17ρ_w)bd		$V_{c,b}$ =	1119.8	KN
c) 0.29λ$\sqrt{f_c'}$bd		$V_{c,c}$ =	1908.7	KN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	4097.8	KN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	602.50	Kn/sqm
Shear force,V_u	= 602.5*[2.5*2.5-(0.7+0.465)*(0.7+0.445)]	=	2961.94	KN
Shear peremeter,b_0	= 2*[(0.7+0.465)+(0.7+0.445)]	=	4.62	m

V_c =0.17(1+2/β)$\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$

V_c = 0.083($\alpha_s d/b_0$ + 2)$\sqrt{f_c'}b_0d$ = 0.42 $\sqrt{f_c'}b_0d$ — For Edge column, α_s = 30

V_c = 0.33$\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$

φV_c	= 0.75*0.33*SQRT(35)*4620*465/1000	=	3145.6	KN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

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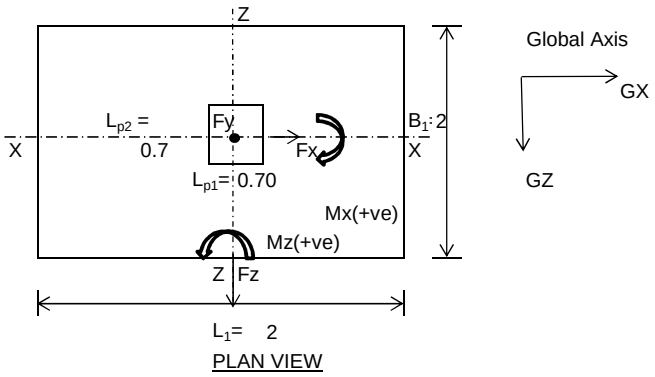
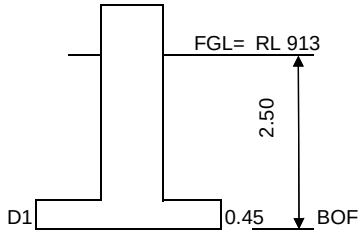
Foundation MKD. **F3**



PLAN VIEW



SECTIONAL VIEW

Foundation Level Details :

EL of Finish Ground Level ,FGL =	RL	913.0	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		750	kN/m^2

Material Properties :

Unit Weight of Soil =		18.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2	m
Footing dimension along Z direction,B_1	=	2	m
Depth, D_1	=	0.45	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap = 0.7 x 0.7	=	0.49	m^2
Area of Footing , A = 2.00 x 2.00	=	4	m^2
Zz-z = 2 x 2.00 x 2.00 / 6	=	1.33	m^3
Zx-x = 2 x 2 x 2 / 6	=	1.33	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	45	kN
Wt of soil above ftg =	130	kN
Total weight,W=	175	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.7	m
Pedestal dimension along Z direction,Lp2=	0.7	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 18 = 795 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F3 STAAD node33	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:								Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis									
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zzz+Mx/Zxx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	KN	KN	KN	KN-m	KN-m	KN-m	Kn/m^2	Kn/m^2		Kn/m^2		Kn-m	Kn-m			Kn-m	Kn-m			Kn	Kn	Kn	Kn		
201	-42.2	-831	-25	-22	0	31.2	247.7	795.0	safe	167.8	ok	31.2	831	26.7	ok	22.0	831	37.7	ok	42.2	25.1	49.1	831	13.3	ok
202	-38.7	-1419	3.08	-24	-0	53.6	413.3	795.0	safe	296.3	ok	53.6	1419	26.5	ok	24.4	1419	58.2	ok	38.7	3.1	38.8	1419	184	ok
203	-41	-1417	3.04	-24	-0	59.8	417.0	795.0	safe	291.4	ok	59.8	1417	23.7	ok	24.0	1417	59.1	ok	41.0	3.0	41.2	1417		ok
204	-39.6	-1272	-4	-24	-0	48	371.9	795.0	safe	264.2	ok	48.0	1272	26.5	ok	23.8	1272	53.5	ok	39.6	4.0	39.8	1272	128	ok
205	-41.3	-1270	-4	-23	-0	52.6	374.7	795.0	safe	260.5	ok	52.6	1270	24.1	ok	23.5	1270	54.1	ok	41.3	4.0	41.5	1270	127	ok
206	-46.9	-833	-26	-27	0	43.5	260.7	795.0	safe	155.7	ok	43.5	833	19.1	ok	26.5	833	31.4	ok	46.9	26.4	53.8	833	12.6	ok
207	-43.4	-1421	1.78	-29	-0	65.9	426.4	795.0	safe	284.2	ok	65.9	1421	21.6	ok	28.9	1421	49.2	ok	43.4	1.8	43.4	1421	320	ok
208	-45.7	-1418	1.74	-28	-0	72.1	430.0	795.0	safe	279.2	ok	72.1	1418	19.7	ok	28.4	1418	49.9	ok	45.7	1.7	45.7	1418	326	ok
209	-44.2	-1274	-5.3	-28	-0	60.3	385.0	795.0	safe	252.0	ok	60.3	1274	21.1	ok	28.3	1274	45	ok	44.2	5.3	44.5	1274	96.8	ok
210	-46	-1272	-5.3	-28	-0	65	387.7	795.0	safe	248.3	ok	65.0	1272	19.6	ok	28.0	1272	45.5	ok	46.0	5.3	46.3	1272	96.2	ok
211	-72.8	-825	-26	-26	0	110	308.0	795.0	safe	104.8	ok	110.0	825	7.51	ok	25.5	825	32.3	ok	72.8	26.1	77.3	825	12.7	ok
212	-9.09	-825	-24	-19	0	-51	259.2	795.0	safe	153.5	ok	51.2	825	16.1	ok	19.3	825	42.8	ok	9.1	24.3	26.0	825	13.6	ok
213	-41.1	-850	-42	-79	0	28.1	292.6	795.0	safe	132.6	ok	28.1	850	30.2	ok	78.5	850	10.8	ok	41.1	42.4	59.0	850	8.02	ok
214	-44.3	-801	-8.1	42.1	0	36.7	259.5	795.0	safe	141.3	ok	36.7	801	21.8	ok	42.1	801	19	ok	44.3	8.1	45.0	801	39.8	ok
215	-72.4	-838	-31	-41	0	110	323.2	795.0	safe	95.8	ok	110.5	838	7.59	ok	41.1	838	20.4	ok	72.4	30.6	78.6	838	11	ok
216	-12	-824	-20	-3	0	-48	244.3	795.0	safe	167.7	ok	48.1	824	17.1	ok	3.0	824	279	ok	12.0	19.6	22.9	824	16.9	ok
217	-56.3	-851	-39	-69	0	68.6	316.1	795.0	safe	109.5	ok	68.6	851	12.4	ok	69.2	851	12.3	ok	56.3	39.2	68.6	851	8.69	ok
218	-28.1	-811	-11	25.1	0	-6.2	226.2	795.0	safe	179.2	ok	6.2	811	131	ok	25.1	811	32.3	ok	28.1	10.9	30.2	811	29.6	ok
219	-62.5	-1268	-4.7	-26	-0	107	417.2	795.0	safe	217.1	ok	107.0	1268	11.9	ok	26.4	1268	48	ok	62.5	4.7	62.7	1268	108	ok
220	-14.7	-1268	-3.4	-22	-0	-14	343.8	795.0	safe	290.4	ok	13.9	1268	91.5	ok	21.7	1268	58.3	ok	14.7	3.4	15.1	1268	150	ok
221	-38.7	-1287	-17	-66	-0	45.6	405.7	795.0	safe	237.9	ok	45.6	1287	28.2	ok	66.2	1287	19.5	ok	38.7	17.0	42.2	1287	30.4	ok
222	-41.1	-1250	8.81	24.3	-0	52.1	369.9	795.0	safe	255.4	ok	52.1	1250	24	ok	24.3	1250	51.5	ok	41.1	8.8	42.1	1250	56.8	ok
223	-62.2	-1278	-8.1	-38	-0	107	428.6	795.0	safe	210.3	ok	107.4	1278	11.9	ok	38.1	1278	33.5	ok	62.2	8.1	62.8	1278	63.1	ok
224	-16.9	-1267	0.19	-9.5	0	-12	332.7	795.0	safe	301.1	ok	11.6	1267	110	ok	9.5	1267	134	ok	16.9	0.2	16.9	1267	2740	ok
225	-50.1	-1288	-15	-59	-0	76	423.3	795.0	safe	220.6	ok	76.0	1288	17	ok	59.2	1288	21.8	ok	50.1	14.6	52.2	1288	35.4	ok

226	-29	-1257	6.65	11.6	0	19.9	338.0		795.0	safe	290.8	ok	19.9	1257	63.2	ok	11.6	1257	109	ok	29.0	6.6	29.7	1257	75.7	ok
227	-53.2	-562	-15	-14	0	90.2	218.7		795.0	safe	62.2	ok	90.2	562	6.23	ok	14.1	562	39.8	ok	53.2	15.3	55.3	562	14.7	ok
228	10.54	-561	-14	-7.9	0	-71	199.5		795.0	safe	81.3	ok	71.0	561	7.91	ok	7.9	561	71.4	ok	10.5	13.6	17.2	561	16.6	ok
229	-21.4	-586	-32	-67	0	8.36	203.2		795.0	safe	90.0	ok	8.4	586	70.1	ok	67.1	586	8.74	ok	21.4	31.7	38.2	586	7.41	ok
230	-24.7	-538	2.7	53.5	0	17	187.3		795.0	safe	81.6	ok	17.0	538	31.7	ok	53.5	538	10.1	ok	24.7	2.7	24.8	538	79.7	ok
231	-52.8	-574	-20	-30	0	90.7	233.9		795.0	safe	53.2	ok	90.7	574	6.33	ok	29.7	574	19.3	ok	52.8	19.8	56.4	574	11.6	ok
232	7.662	-560	-8.8	8.48	0	-68	197.3		795.0	safe	82.8	ok	67.9	560	8.25	ok	8.5	560	66.1	ok	7.7	8.8	11.7	560	25.5	ok
233	-36.7	-587	-28	-58	0	48.8	226.8		795.0	safe	66.9	ok	48.8	587	12	ok	57.8	587	10.2	ok	36.7	28.5	46.4	587	8.26	ok
234	-8.5	-547	-0.2	36.5	0	-26	183.6		795.0	safe	89.9	ok	26.0	547	21.1	ok	36.5	547	15	ok	8.5	0.2	8.5	547	1182	ok
235	-38.1	-832	-24	-17	0	20.3	236.3		795.0	safe	179.8	ok	20.3	832	41.1	ok	17.4	832	47.9	ok	38.1	23.6	44.8	832	14.1	ok
236	-38.7	-1420	3.08	-24	-0	53.5	413.4		795.0	safe	296.6	ok	53.5	1420	26.6	ok	24.4	1420	58.2	ok	38.7	3.1	38.8	1420	184	ok
237	-38.7	-1420	3.09	-24	-0	53.5	413.4		795.0	safe	296.6	ok	53.5	1420	26.5	ok	24.4	1420	58.2	ok	38.7	3.1	38.8	1420	184	ok
238	-38.7	-1420	3.11	-24	-0	53.5	413.5		795.0	safe	296.6	ok	53.5	1420	26.5	ok	24.4	1420	58.3	ok	38.7	3.1	38.8	1420	183	ok
239	-38.7	-1420	3.11	-24	-0	53.5	413.5		795.0	safe	296.6	ok	53.5	1420	26.5	ok	24.4	1420	58.3	ok	38.7	3.1	38.8	1420	183	ok
240	-38.7	-1420	3.1	-24	-0	53.5	413.5		795.0	safe	296.6	ok	53.5	1420	26.5	ok	24.4	1420	58.2	ok	38.7	3.1	38.8	1420	183	ok
241	-38.7	-1420	3.09	-24	-0	53.5	413.5		795.0	safe	296.6	ok	53.5	1420	26.6	ok	24.4	1420	58.1	ok	38.7	3.1	38.8	1420	184	ok
242	-38.7	-1420	3.09	-24	-0	53.5	413.5		795.0	safe	296.6	ok	53.5	1420	26.5	ok	24.4	1420	58.2	ok	38.7	3.1	38.8	1420	184	ok
243	-38.7	-1420	3.08	-24	-0	53.6	413.5		795.0	safe	296.5	ok	53.6	1420	26.5	ok	24.4	1420	58.2	ok	38.7	3.1	38.8	1420	184	ok
244	-38.7	-1420	3.08	-24	-0	53.6	413.5		795.0	safe	296.5	ok	53.6	1420	26.5	ok	24.4	1420	58.2	ok	38.7	3.1	38.8	1420	184	ok
245	-38.7	-1420	3.1	-24	-0	53.5	413.5		795.0	safe	296.6	ok	53.5	1420	26.5	ok	24.4	1420	58.2	ok	38.7	3.1	38.8	1420	183	ok
246	-38.7	-1420	3.11	-24	-0	53.5	413.5		795.0	safe	296.6	ok	53.5	1420	26.5	ok	24.4	1420	58.3	ok	38.7	3.1	38.8	1420	183	ok

Design pressure for Strength Calculation

Foundation Mkd. F3 STAAD node33	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment			P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
	Fx	Fy	Fz	Mx	My	Mz				
L/C	KN	KN	KN	KN-m	KN-m	KN-m	KN/m^2			
101	-59.1	-918	-35	-31	0	43.7	285.5	220	239.2	173.7
102	-53.1	-1919	12.8	-35	-0	81.6	567.2	445	514.9	392.5
103	-57.1	-1915	12.7	-34	-0	92.1	573.5	435	522.3	384.1
104	-53.1	-1918	12.8	-35	-0	81.7	567.1	445	514.8	392.2
105	-57.1	-1914	12.7	-34	-0	92.2	573.3	435	522.2	383.8
106	-65.6	-921	-37	-37	0	60.9	303.8	212	248.0	156.7
107	-59.6	-1922	10.9	-41	-0	98.9	585.5	437	523.8	375.5
108	-63.6	-1917	10.9	-40	-0	109	591.8	428	531.1	367.1
109	-59.6	-1921	10.9	-41	-0	99	585.4	437	523.7	375.2
110	-63.6	-1917	10.9	-40	-0	109	591.6	427	531.0	366.8
111	-69.6	-780	-29	-24	0	85.9	277.4	149	241.3	112.6
112	-18.7	-779	-28	-19	0	-43	241.5	177	213.0	148.3
113	-44.2	-799	-43	-66	0	20.4	265.0	234	165.4	134.8

114	-46.8	-760	-15	30.1	0	27.3	233.2	192	188.0	147.2
115	-92.3	-1070	-16	-28	0	160	408.5	168	366.5	126.5
116	9.591	-1069	-13	-18	0	-98	354.3	208	327.3	180.5
117	-41.6	-1109	-42	-113	0	29.1	383.8	340	214.6	170.9
118	-46.7	-1031	12.7	80.2	0	42.9	350.2	286	229.8	165.6
119	-88.9	-1090	-23	-52	-0	153	426.4	196	348.6	118.5
120	-2.48	-1070	-7.3	2.7	0	-73	324.4	215	320.4	210.6
121	-65.8	-1109	-35	-92	-0	93.5	416.3	276	278.4	138.1
122	-25.6	-1051	5.03	42.8	0	-13	304.8	285	240.7	220.7
123	-82.8	-579	-23	-22	0	143	268.4	53.6	236.1	21.4
124	19.12	-579	-20	-12	0	-115	239.4	67.4	222.2	50.1
125	-32	-619	-49	-106	0	12.2	243.7	225	84.2	65.8
126	-37.2	-541	5.75	86.7	0	26	219.8	181	89.7	50.7
127	-77.1	-598	-29	-43	0	130	279.8	84.1	215.0	19.3
128	9.333	-578	-14	11.4	0	-96	225.2	81	208.2	63.9
129	-54	-617	-42	-83	0	70.5	269.6	164	144.7	38.9
130	-13.7	-559	-1.3	51.4	0	-36	205.6	151	128.5	74.1
131	-45.7	-787	-28	-21	0	24.4	230.8	194	199.4	162.8
132	-53.1	-1919	12.8	-35	-0	81.6	567.2	445	514.9	392.6
133	-53.1	-1920	12.8	-35	-0	81.6	567.3	445	515.0	392.6
134	-53.1	-1920	12.8	-35	-0	81.7	567.4	445	515.1	392.6
135	-53.1	-1920	12.8	-35	-0	81.7	567.4	445	515.2	392.6
136	-53.1	-1920	12.8	-35	-0	81.6	567.4	445	515.1	392.6
137	-53.1	-1920	12.8	-35	-0	81.6	567.4	445	515.0	392.6
138	-53.1	-1920	12.8	-35	-0	81.6	567.3	445	515.0	392.5
139	-53.1	-1919	12.8	-35	-0	81.7	567.3	445	515.0	392.5
140	-53.1	-1919	12.8	-35	-0	81.7	567.3	445	515.0	392.4
141	-53.1	-1920	12.8	-35	-0	81.7	567.3	445	515.1	392.6
142	-53.1	-1920	12.8	-35	-0	81.7	567.4	445	515.1	392.6

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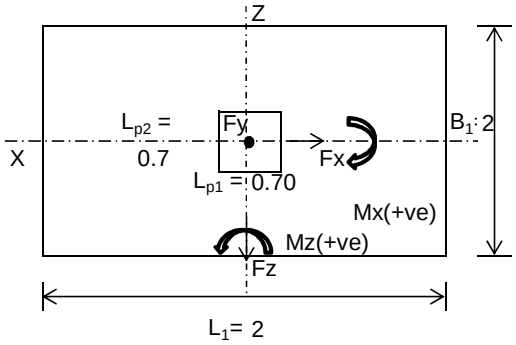


PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	16	219.3	200	16	209.3	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2	m
Footing dimension along Z direction, B_1	=	2	m
Depth, D_1	=	0.45	m
Clear cover	=	75	mm

Column Dimension :

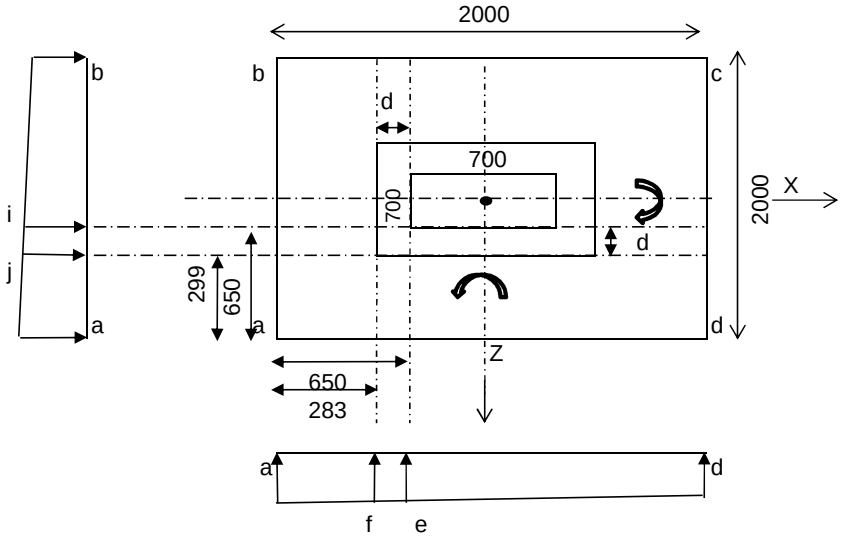
Pedestal dimension along X direction,Lp1	=	0.7	m
Pedestal dimension along Z direction,Lp2	=	0.7	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 591.8 KN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*591.8]*2*0.65*0.65	=	250.0	KN-m
Shear force at Face "f"	= 591.8*0.283*2	=	335.0	KN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*591.8]*2*0.65*0.65	=	250.0	KN-m
Shear force at Face "j"	= 591.8*0.299*2	=	353.9	KN

Design for Bottom Reinforcement :

In X Direction (Along Length 2000 mm)

Cantilever Length		=	650	mm
Dia of Bar		d_b =	16	mm
Effective Depth		d =	367	mm
Bending Moment at "e"		M_u =	250.0	KN-m
C = $0.85f_c'ba$ and T = A_sf_y	Therefore 'a' = $f_y/0.85f_c'b * A_s$	a =	0.0071	$*A_s$
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(367-0.0071As/2)As=250*10^6	A_s =	1834.7	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*2000*450	$A_{s,min}$ =	1620	mm^2
Spacing required	2000*22/28*16^2/1834.7	S_{reqd} =	219.3	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2000*22/28*16^2/200	$A_{s,pro}$ =	2011.5	mm^2
$a = A_sf_y/0.85f'_cb$	2011.5*420/(0.85*35*2000)	a =	14.20	mm
$c = a/\beta_1$	14.2/0.8	c =	17.75	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/17.75*(367-17.75)	ε_t =	0.059	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	335.0	kn
Moment at "f"	= [0.5*591.8]*2*0.283*0.283	M_u =	47.4	kn-m
ρ_w provided	2011.5/2000/367	$\rho_{w,pro}$ =	0.0027	%
Effective Shear Area	=367*2000/10^6	bd =	0.734	m^2

V_c = lesser of

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$ =	783.5	KN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	729.0	KN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	1259.3	KN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	2696.2	KN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 2000 mm)

Cantilever Length		=	650	mm
Dia of Bar		d_b =	16	mm
Effective Depth		d =	351	mm
Bending Moment at "i"		M_u =	250.0	KN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0071	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(351-0.0071As/2)As=250.04*10^6	A_s =	1921.9	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2000*450	$A_{s,min}$ =	1620	mm^2
Spacing required	2000*22/28*16^2/1921.9	S_{reqd} =	209.32	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2000*22/28*16^2/200	$A_{s,pro}$ =	2011.5	mm^2
a = $A_{st} f_y$/0.85f'$_c$b	2011.5*420/(0.85*35*2000)	a =	14.20	mm
c = a/β_1	14.2/0.8	c =	17.75	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/17.75*(351-17.75)	ε_t =	0.056	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	353.9	KN
Moment at "j"	= [0.5*591.8]*2*0.299*0.299	M_u =	52.9	KN-m
ρ_w provided	= 2011.5/2000/351	$\rho_{w,pro}$ =	0.0029	%
Effective Shear Area	=351*2000/10^6	bd =	0.702	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	744.8	KN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	698.7	KN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	1204.4	KN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	2579.8	KN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	591.80	Kn/sqm
Shear force,V_u	= 591.8*[2*2-(0.7+0.367)*(0.7+0.351)]	=	1703.55	KN
Shear peremeter,b_0	= 2*[(0.7+0.367)+(0.7+0.351)]	=	4.24	m
V_c =0.17(1+2/β)$\sqrt{f_c'}b_0d$	= 0.51 $\sqrt{f_c'}b_0d$			
V_c = 0.083(α_sd/b_0 + 2)$\sqrt{f_c'}b_0d$	= 0.38 $\sqrt{f_c'}b_0d$	For Edge column	α_s	30
V_c = 0.33$\sqrt{f_c'}b_0d$	= 0.33 $\sqrt{f_c'}b_0d$			
φV_c	= 0.75*0.33*SQRT(35)*4236*367/1000	=	2276.3	KN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

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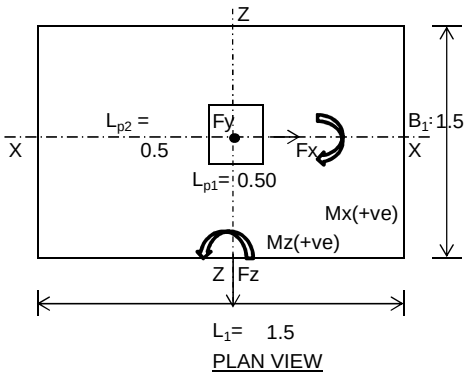
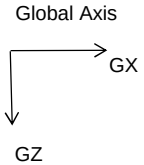
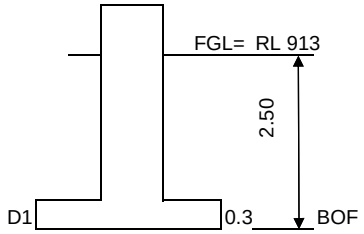
Foundation MKD. **F4**



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

EL of Finish Ground Level ,FGL =	RL	913.0	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		750	kN/m^2

Material Properties :

Unit Weight of Soil =		18.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	1.5	m
Footing dimension along Z direction,B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap =	0.5 x 0.5	=	0.25 m^2
Area of Footing , A =	1.50 x 1.50	=	2.25 m^2
Zz-z =	1.5 x 1.50 x 1.50 / 6	=	0.56 m^3
Zx-x =	1.5 x 1.5 x 1.5 / 6	=	0.56 m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	16.9	kN
Wt of soil above ftg =	79.2	kN
Total weight,W=	96	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.5	m
Pedestal dimension along Z direction,Lp2=	0.5	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 18 = 795 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F4 STAAD node60	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:									Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis										
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zzz+Mx/Zxx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	KN	KN	KN	KN-m	KN-m	KN-m	Kn/m^2	Kn/m^2		Kn/m^2		Kn-m	Kn-m			Kn-m	Kn-m			Kn	Kn	Kn	Kn		
201	2.734	-172	-8.3	-9.5	0	-3.3	99.4	795.0	safe	53.9	ok	3.3	129	38.9	ok	9.5	129	13.6	ok	2.7	8.3	8.7	172	8.34	ok
202	3.78	-155	-17	-21	0	-4.6	114.7	795.0	safe	23.5	ok	4.6	117	25.5	ok	21.1	117	5.53	ok	3.8	17.5	17.9	155	3.56	ok
203	1.575	-156	-17	-21	0	-1.9	109.4	795.0	safe	29.2	ok	1.9	117	60.1	ok	20.6	117	5.68	ok	1.6	17.1	17.1	156		ok
204	3.519	-160	-15	-18	0	-4.3	110.9	795.0	safe	31.1	ok	4.3	120	28.1	ok	18.2	120	6.59	ok	3.5	15.2	15.6	160	4.21	ok
205	1.865	-160	-15	-18	0	-2.3	106.9	795.0	safe	35.4	ok	2.3	120	52.4	ok	17.8	120	6.73	ok	1.9	14.9	15.0	160	4.31	ok
206	-0.77	-172	-11	-13	0	0.86	101.6	795.0	safe	51.6	ok	0.9	129	150	ok	13.2	129	9.79	ok	0.8	11.4	11.4	172	6.04	ok
207	0.274	-155	-21	-25	0	-0.4	113.9	795.0	safe	24.3	ok	0.4	117	304	ok	24.8	117	4.7	ok	0.3	20.6	20.6	155	3.02	ok
208	-1.93	-156	-20	-24	0	2.24	116.5	795.0	safe	22.1	ok	2.2	117	52.2	ok	24.3	117	4.81	ok	1.9	20.2	20.3	156	3.09	ok
209	0.012	-160	-18	-22	0	-0.1	110.1	795.0	safe	31.9	ok	0.1	120	1686	ok	21.9	120	5.47	ok	0.0	18.3	18.3	160	3.49	ok
210	-1.64	-160	-18	-22	0	1.9	112.8	795.0	safe	29.5	ok	1.9	120	63.3	ok	21.5	120	5.57	ok	1.6	18.0	18.1	160	3.56	ok
211	-28.7	-174	-7.6	-8.8	0	34.4	153.9	795.0	safe	0.5	ok	34.4	130	3.79	ok	8.8	130	14.9	ok	28.7	7.6	29.7	174	9.09	ok
212	31.53	-172	-9.1	-10	0	-38	162.5	795.0	safe	-9.4	-	37.9	129	3.41	ok	10.5	129	12.4	ok	31.5	9.1	32.8	172	7.57	ok
213	3.376	-172	-35	-41	0	-4.1	156.4	795.0	safe	-3.3	-	4.1	129	31.5	ok	40.8	129	3.17	ok	3.4	34.7	34.9	172	1.99	ok
214	1.618	-173	18.7	22.4	0	-2	120.3	795.0	safe	33.5	ok	2.0	130	65	ok	22.4	130	5.79	ok	1.6	18.7	18.7	173	3.71	ok
215	-22.4	-172	-20	-23	0	26.7	165.0	795.0	safe	-11.7	-	26.7	129	4.85	ok	23.0	129	5.63	ok	22.4	19.7	29.8	172	3.51	ok
216	27.84	-172	3.13	4.01	0	-33	142.9	795.0	safe	10.2	ok	33.3	129	3.88	ok	4.0	129	32.2	ok	27.8	3.1	28.0	172	22	ok
217	-8.95	-172	-32	-38	0	10.6	162.6	795.0	safe	-9.4	-	10.6	129	12.2	ok	37.7	129	3.43	ok	9.0	32.1	33.3	172	2.15	ok
218	14.42	-172	15.6	18.7	0	-17	140.7	795.0	safe	12.6	ok	17.3	129	7.49	ok	18.7	129	6.9	ok	14.4	15.6	21.2	172	4.43	ok
219	-20.1	-161	-15	-18	0	24	145.5	795.0	safe	-2.8	-	24.0	120	5.02	ok	17.7	120	6.82	ok	20.1	14.7	24.9	161	4.37	ok
220	25.13	-160	-16	-19	0	-30	158.3	795.0	safe	-16.5	-	30.2	120	3.97	ok	18.9	120	6.32	ok	25.1	15.8	29.7	160	4.04	ok
221	4.018	-160	-35	-42	0	-4.9	153.7	795.0	safe	-11.8	-	4.9	120	24.6	ok	41.7	120	2.88	ok	4.0	35.0	35.2	160	1.83	ok
222	2.7	-160	5.02	5.73	0	-3.3	87.2	795.0	safe	55.2	ok	3.3	120	36.6	ok	5.7	120	21	ok	2.7	5.0	5.7	160	12.8	ok
223	-15.3	-160	-24	-28	0	18.2	153.8	795.0	safe	-11.8	-	18.2	120	6.58	ok	28.3	120	4.23	ok	15.3	23.7	28.2	160	2.7	ok
224	22.37	-160	-6.6	-8.1	0	-27	132.9	795.0	safe	9.0	ok	26.8	120	4.47	ok	8.1	120	14.8	ok	22.4	6.6	23.3	160	9.62	ok
225	-5.23	-160	-33	-39	0	6.18	152.0	795.0	safe	-10.1	-	6.2	120	19.4	ok	39.4	120	3.04	ok	5.2	33.1	33.5	160	1.94	ok

226	12.3	-160	2.7	2.96	0	-15	102.5		795.0	safe	39.5	ok	14.7	120	8.13	ok	3.0	120	40.5	ok	12.3	2.7	12.6	160	23.6	ok
227	-27.8	-143	-2.3	-2.6	0	33.3	127.5		795.0	safe	-0.2	-	33.3	107	3.23	ok	2.6	107	41.7	ok	27.8	2.3	27.9	143	24.8	ok
228	32.47	-142	-3.8	-4.3	0	-39	139.9		795.0	safe	-13.8	-	38.9	106	2.74	ok	4.3	106	24.9	ok	32.5	3.8	32.7	142	15.1	ok
229	4.316	-142	-29	-35	0	-5.2	133.8		795.0	safe	-7.6	-	5.2	106	20.6	ok	34.6	106	3.08	ok	4.3	29.4	29.7	142	1.94	ok
230	2.558	-143	24	28.6	0	-3.1	119.7		795.0	safe	7.1	ok	3.1	107	34.9	ok	28.6	107	3.74	ok	2.6	24.0	24.1	143	2.38	ok
231	-21.4	-142	-14	-17	0	25.6	138.6		795.0	safe	-12.3	-	25.6	107	4.17	ok	16.8	107	6.35	ok	21.4	14.3	25.8	142	3.97	ok
232	28.78	-142	8.46	10.2	0	-34	142.3		795.0	safe	-16.3	-	34.4	106	3.1	ok	10.2	106	10.4	ok	28.8	8.5	30.0	142	6.71	ok
233	-8.01	-142	-27	-32	0	9.56	136.2		795.0	safe	-10.0	-	9.6	106	11.2	ok	31.5	106	3.38	ok	8.0	26.8	28.0	142	2.12	ok
234	15.36	-142	20.9	24.9	0	-18	140.1		795.0	safe	-13.9	-	18.3	106	5.81	ok	24.9	106	4.28	ok	15.4	20.9	25.9	142	2.72	ok
235	5.798	-170	-5.8	-6.7	0	-7.3	100.6		795.0	safe	51.0	ok	7.3	128	17.6	ok	6.7	128	19.1	ok	5.8	5.8	8.2	170	11.7	ok
236	3.787	-155	-17	-21	0	-4.6	114.8		795.0	safe	23.4	ok	4.6	117	25.4	ok	21.1	117	5.52	ok	3.8	17.5	17.9	155	3.56	ok
237	3.812	-155	-18	-21	0	-4.6	114.9		795.0	safe	23.3	ok	4.6	116	25.3	ok	21.2	116	5.51	ok	3.8	17.5	17.9	155	3.55	ok
238	3.838	-155	-18	-21	0	-4.6	115.0		795.0	safe	23.1	ok	4.6	116	25.1	ok	21.2	116	5.5	ok	3.8	17.6	18.0	155	3.54	ok
239	3.854	-155	-18	-21	0	-4.7	115.0		795.0	safe	23.1	ok	4.7	116	25	ok	21.2	116	5.49	ok	3.9	17.6	18.0	155	3.54	ok
240	3.841	-155	-18	-21	0	-4.6	114.9		795.0	safe	23.2	ok	4.6	116	25.2	ok	21.2	116	5.51	ok	3.8	17.5	17.9	155	3.55	ok
241	3.829	-155	-17	-21	0	-4.6	114.9		795.0	safe	23.4	ok	4.6	117	25.3	ok	21.1	117	5.52	ok	3.8	17.5	17.9	155	3.56	ok
242	3.815	-155	-17	-21	0	-4.6	114.8		795.0	safe	23.4	ok	4.6	117	25.3	ok	21.1	117	5.52	ok	3.8	17.5	17.9	155	3.56	ok
243	3.8	-155	-17	-21	0	-4.6	114.8		795.0	safe	23.4	ok	4.6	117	25.4	ok	21.1	117	5.52	ok	3.8	17.5	17.9	155	3.56	ok
244	3.792	-155	-17	-21	0	-4.6	114.8		795.0	safe	23.4	ok	4.6	117	25.4	ok	21.1	117	5.52	ok	3.8	17.5	17.9	155	3.56	ok
245	3.832	-155	-18	-21	0	-4.6	114.9		795.0	safe	23.2	ok	4.6	116	25.2	ok	21.2	116	5.5	ok	3.8	17.5	18.0	155	3.55	ok
246	3.838	-155	-18	-21	0	-4.6	115.0		795.0	safe	23.1	ok	4.6	116	25.1	ok	21.2	116	5.5	ok	3.8	17.6	18.0	155	3.54	ok

Tension at top face . Provide top reinforcement for tension.

Design pressure for Strength Calculation

Foundation Mkd. F4 STAAD node60	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment			P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
	Fx	Fy	Fz	Mx	My	Mz				
L/C	KN	KN	KN	KN-m	KN-m	KN-m	KN/m^2			
101	3.827	-107	-12	-13	0	-4.7	79.4	62.9	32.2	15.6
102	5.647	-78	-27	-33	0	-6.8	105.6	81.4	-12.0	-36.2
103	1.897	-79	-27	-32	0	-2.4	96.6	88.2	-18.1	-26.5
104	5.614	-78	-27	-33	0	-6.8	105.5	81.4	-12.0	-36.1
105	1.864	-79	-27	-32	0	-2.3	96.5	88.2	-18.1	-26.4
106	-1.08	-107	-16	-18	0	1.21	82.6	78.3	16.8	12.5
107	0.737	-78	-32	-38	0	-1	104.5	101	-31.7	-35.1
108	-3.01	-79	-31	-37	0	3.5	107.9	95.4	-25.3	-37.8
109	0.704	-78	-32	-38	0	-0.9	104.4	101	-31.7	-34.9
110	-3.05	-79	-31	-37	0	3.54	107.9	95.3	-25.2	-37.8
111	-17.8	-93	-5.4	-6	0	21.4	90.1	14.1	68.7	-7.5
112	30.38	-92	-6.5	-7.4	0	-36	118.7	-10.9	92.5	-37.1

113	7.861	-92	-27	-32	0	-9.4	113.9	80.4	1.4	-32.2
114	6.455	-92	15.7	18.9	0	-7.7	88.5	61.1	21.2	-6.4
115	-42.5	-86	-9.5	-11	0	50.9	148.6	-32.5	108.6	-72.6
116	53.95	-83	-12	-14	0	-65	176.9	-53.2	127.2	-103
117	8.91	-83	-53	-62	0	-11	167.2	129	-55.1	-93
118	6.098	-84	32.6	38.7	0	-7.3	119.3	93.3	-18.3	-44.2
119	-29.7	-84	-28	-34	0	35.4	160.1	34.1	40.3	-85.8
120	42.06	-83	4.11	4.87	0	-50	135.0	-43.8	117.7	-61.2
121	-10.5	-83	-46	-55	0	12.5	156.6	112	-38.1	-82.6
122	22.88	-83	21.9	25.9	0	-27	131.8	34.5	39.6	-57.7
123	-44.8	-71	-3.4	-3.8	0	53.7	133.9	-57.3	120.4	-70.8
124	51.58	-69	-5.7	-6.5	0	-62	152.1	-67.9	129.0	-91
125	6.538	-69	-47	-55	0	-7.8	142.4	115	-53.3	-81.1
126	3.726	-70	38.7	46.1	0	-4.5	121.0	105	-43.0	-58.9
127	-30.4	-69	-21	-24	0	36.3	138.4	9.4	52.2	-76.9
128	41.38	-69	11.9	14.3	0	-49	144.0	-31.9	93.0	-82.9
129	-11.2	-69	-38	-45	0	13.3	134.9	87.5	-26.2	-73.6
130	22.2	-69	29.7	35.4	0	-27	140.7	46.5	15.0	-79.4
131	6.891	-90	-6.7	-7.7	0	-8.6	68.8	38.4	41.6	11.2
132	5.617	-78	-27	-33	0	-6.8	105.6	81.5	-12.0	-36.2
133	5.661	-78	-27	-33	0	-6.8	105.7	81.4	-12.1	-36.4
134	5.705	-78	-27	-33	0	-6.9	105.9	81.4	-12.2	-36.7
135	5.731	-78	-27	-33	0	-6.9	105.9	81.3	-12.2	-36.8
136	5.709	-78	-27	-33	0	-6.9	105.8	81.4	-12.1	-36.5
137	5.689	-78	-27	-33	0	-6.8	105.7	81.4	-11.9	-36.2
138	5.665	-78	-27	-33	0	-6.8	105.7	81.4	-12.0	-36.2
139	5.641	-78	-27	-33	0	-6.8	105.6	81.4	-12.0	-36.2
140	5.626	-78	-27	-33	0	-6.8	105.6	81.4	-12.0	-36.2
141	5.695	-78	-27	-33	0	-6.9	105.8	81.4	-12.2	-36.6
142	5.704	-78	-27	-33	0	-6.9	105.9	81.4	-12.2	-36.7

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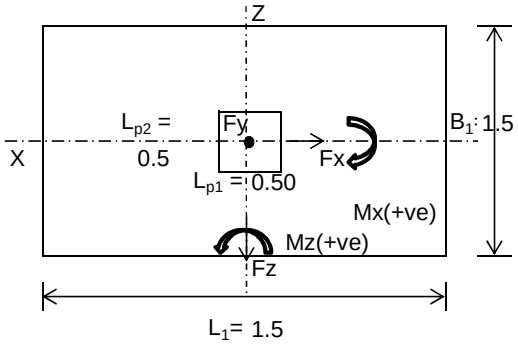


PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	12	209.5	200	12	209.5	200

Footing Dimension :

Footing dimension along X direction, L_1	=	1.5	m
Footing dimension along Z direction, B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

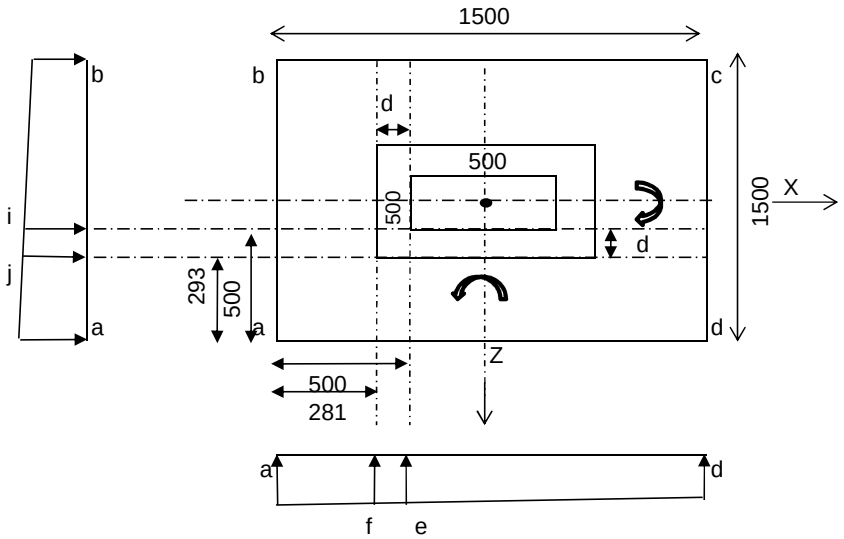
Pedestal dimension along X direction,Lp1	=	0.5	m
Pedestal dimension along Z direction,Lp2	=	0.5	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 176.9 KN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*176.9]*1.5*0.5*0.5	=	33.2	KN-m
Shear force at Face "f"	= 176.9*0.281*1.5	=	74.6	KN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*176.9]*1.5*0.5*0.5	=	33.2	KN-m
Shear force at Face "j"	= 176.9*0.293*1.5	=	77.7	KN

Design for Bottom Reinforcement :

In X Direction (Along Length 1500 mm)

Cantilever Length		=	500	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	219	mm
Bending Moment at "e"		M_u =	33.2	KN-m
C = $0.85f_c'ba$ and T = A_sf_y	Therefore 'a' = $f_y/0.85f_c'b * A_s$	a =	0.0094	$*A_s$
$\varphi M_n \geq M_u = \varphi A_sf_y(d-a/2)$	0.9*420*(219-0.0094As/2)As=33.2*10^6	A_s =	404.6	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*1500*300	$A_{s,min}$ =	810	mm^2
Spacing required	1500*22/28*12^2/810	S_{reqd} =	209.5	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	1500*22/28*12^2/200	$A_{s,pro}$ =	848.6	mm^2
$a = A_sf_y/0.85f'_cb$	848.6*420/(0.85*35*1500)	a =	7.99	mm
$c = a/\beta_1$	7.99/0.8	c =	9.99	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/9.99*(219-9.99)	ε_t =	0.063	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	74.6	kn
Moment at "f"	= [0.5*176.9]*1.5*0.281*0.281	M_u =	10.5	kn-m
ρ_w provided	848.6/1500/219	$\rho_{w,pro}$ =	0.0026	%
Effective Shear Area	=219*1500/10^6	bd =	0.329	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$ =	333.4	KN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	325.4	KN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	563.6	KN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	1206.0	KN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 1500 mm)

Cantilever Length		=	500	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	207	mm
Bending Moment at "i"		M_u =	33.2	KN-m
C = $0.85f_c'ba$ and T = A_sf_y	Therefore 'a' = $f_y/0.85f_c'b * A_s$	a =	0.0094	$*A_s$
$\varphi M_n \geq M_u = \varphi A_sf_y(d-a/2)$	0.9*420*(207-0.0094As/2)As=33.17*10^6	A_s =	428.1	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*1500*300	$A_{s,min}$ =	810	mm^2
Spacing required	1500*22/28*12^2/810	S_{reqd} =	209.52	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	1500*22/28*12^2/200	$A_{s,pro}$ =	848.6	mm^2
$a = A_{st}f_y/0.85f'_cb$	848.6*420/(0.85*35*1500)	a =	7.99	mm
$c = a/\beta_1$	7.99/0.8	c =	9.99	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/9.99*(207-9.99)	ε_t =	0.059	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	77.7	KN
Moment at "j"	= [0.5*176.9]*1.5*0.293*0.293	M_u =	11.4	KN-m
ρ_w provided	= 848.6/1500/207	$\rho_{w,pro}$ =	0.0027	%
Effective Shear Area	=207*1500/10^6	bd =	0.311	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$ =	314.3	KN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	308.3	KN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	532.7	KN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	1140.5	KN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	176.90	Kn/sqm
Shear force,V_u	= 176.9*[1.5*1.5-(0.5+0.219)*(0.5+0.207)]	=	308.11	KN
Shear peremeter,b_0	= 2*[(0.5+0.219)+(0.5+0.207)]	=	2.85	m

$V_c = 0.17(1+2/\beta)\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$

$V_c = 0.083(\alpha_sd/b_0 + 2)\sqrt{f_c'}b_0d$ = 0.36 $\sqrt{f_c'}b_0d$

For Edge column	α_s	30

$V_c = 0.33\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$

φV_c	= 0.75*0.33*SQRT(35)*2852*219/1000	=	914.5	KN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

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ATTACHMENT 1 OTE INPUT (CIVIL WORKS GUIDE DRAWING)

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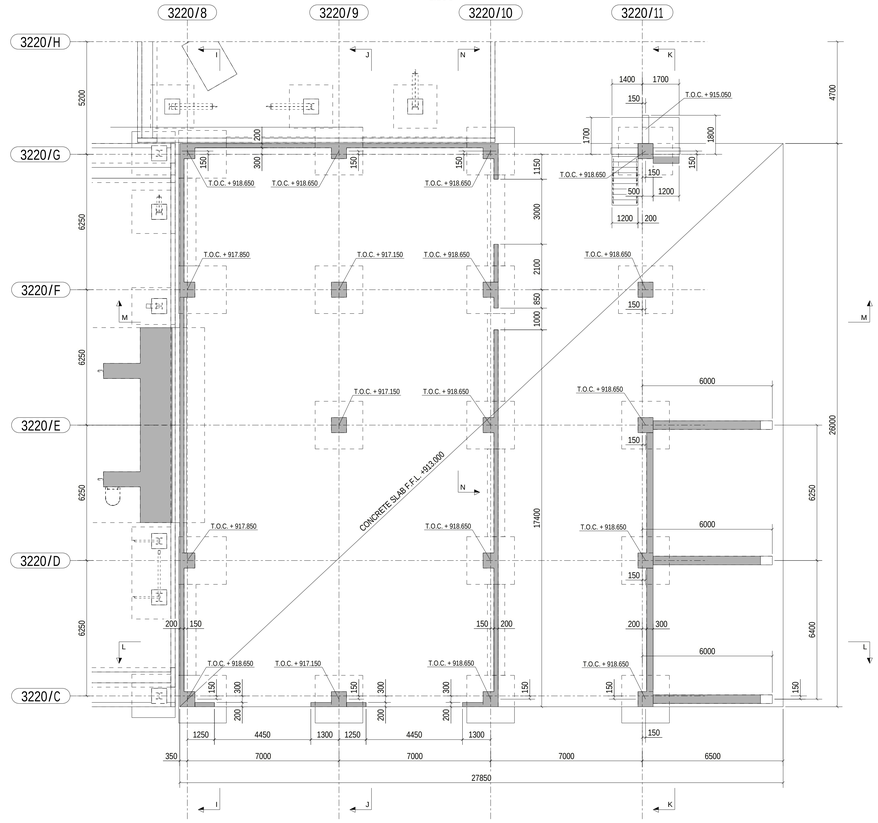
DRAWING NUMBER MD-216-3220-EG-CV-GAD-1003 REV A01

LEVEL +913.000
1:100



LEVEL +919.050
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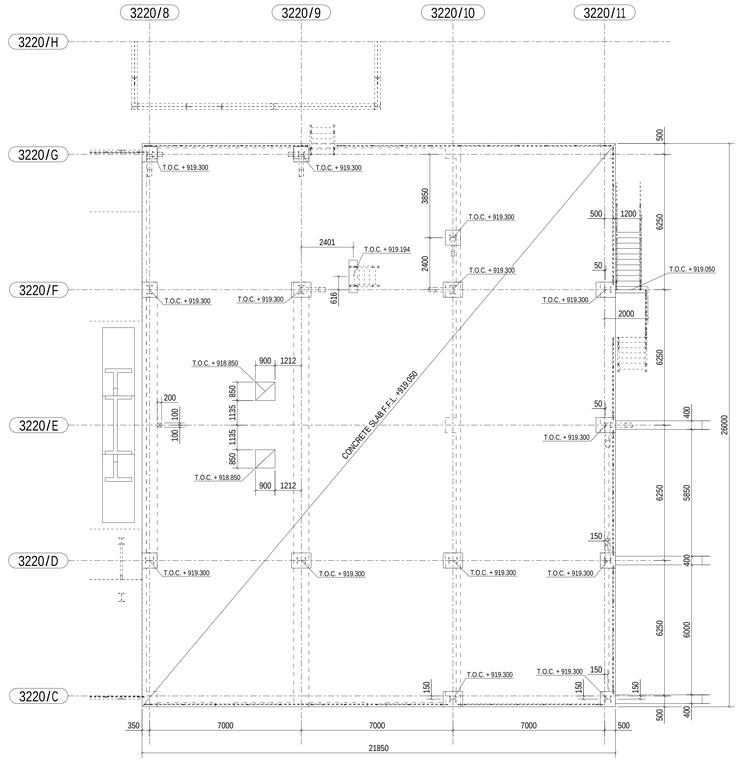

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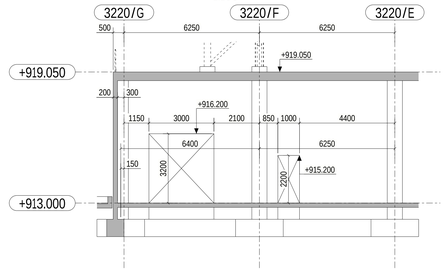

I - I
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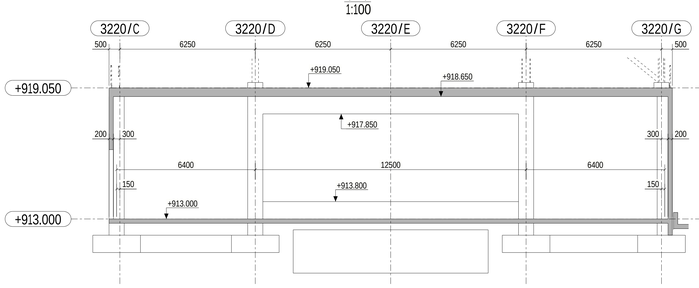

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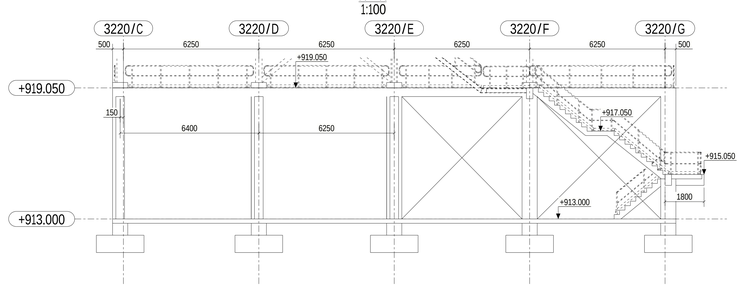

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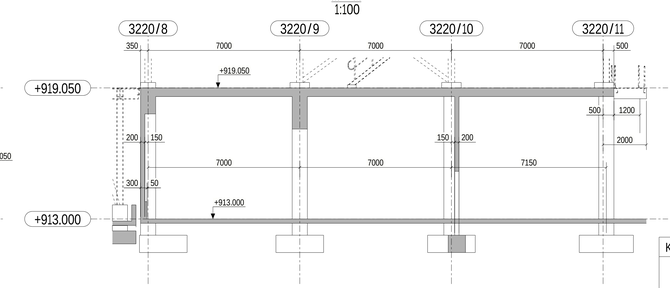

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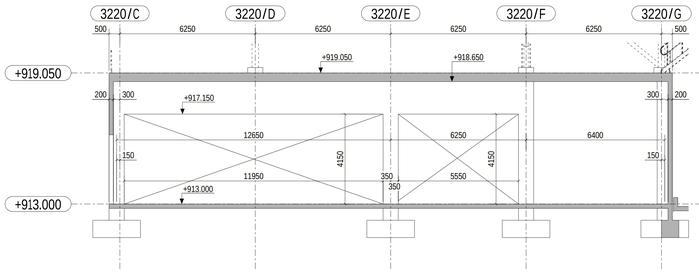
1:100



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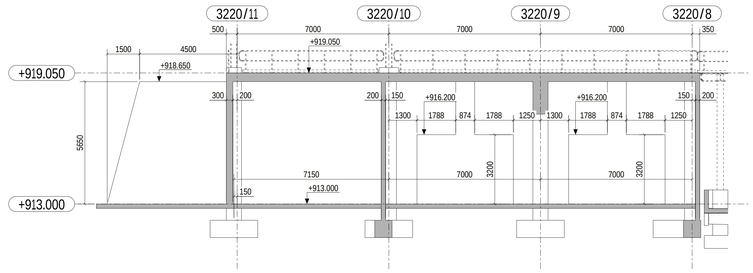


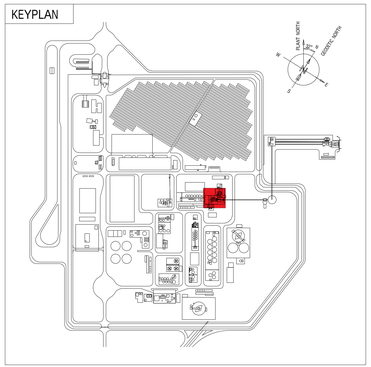
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3220 = RL(+)913.000 m

CHECKERED PLATE
GRATING
F GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3220-EG-CV-GAD-1005	PLAN AT EL. +913.000 - GRID G-J/7-10	MD-216-3220-EG-ST-GAD-1003	PLAN AT EL. +924.550 - GRID D-H/8-11
MD-216-3220-EG-CV-GAD-1002	SECTIONS A-A TO H-H & EQUIP. FOUNDATION	MD-216-3220-EG-CV-GAD-1004	PLAN AT EL. +913.000 - GRID A-C/1-8
MD-216-3220-EG-ME-GAD-1075	SECTION F-F	MD-216-3220-EG-CV-GAD-1001	PLAN AT EL. +913.000 - GRID C-G/1-8
MD-216-3220-EG-ME-GAD-1073	SECTION D-D	MD-216-3220-EG-ME-GAD-1074	SECTION E-E
MD-216-3220-EG-ME-GAD-1071	SECTION B-B	MD-216-3220-EG-ME-GAD-1072	SECTION C-C
MD-216-3220-EG-ME-GAD-1069	PLAN VIEW	MD-216-3220-EG-ME-GAD-1070	SECTION A-A
MD-216-3220-EG-ME-GAD-1067	PLAN AT EL. +922.180	MD-216-3220-EG-ME-GAD-1068	PLAN AT EL. +924.580
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3220-EG-ME-GAD-1066	PLAN AT EL. +913.000

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	11.12.2019	ISSUED FOR APPROVAL	NikJan	MikLai	LuiPiz	SouCha	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER: 09045_450_C_G_24903_001 — REVISION: 0

OUTOTEC DRAWING NUMBER: 900707_GAU01_ZG01_CTA04_32203 — DOCUMENT ID: OU100667742 — REVISION: 0

DRAWN	NikJan	11.12.2019
CHECKED	MikLai	11.12.2019
ENGINEER	LuiPiz	11.12.2019
APPROVED	SouCha	11.12.2019
EMPLOYER		
SCALE	1:100	A0

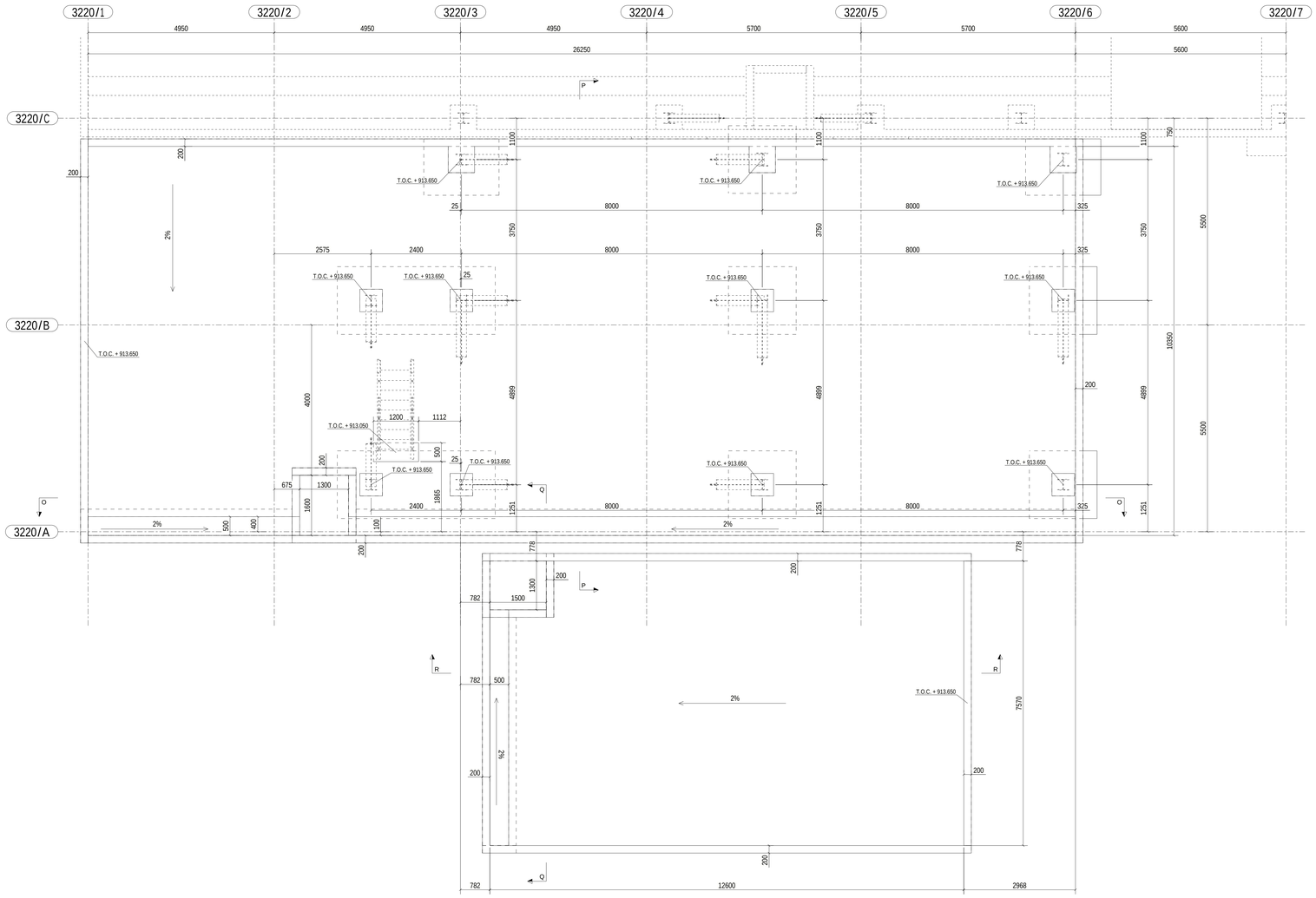
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
GRINDING AND CLASSIFICATION - BALL MILL
PLAN AT EL. +913.000 - GRID C-G/8-11

DRAWING NUMBER: MD-216-3220-EG-CV-GAD-1003 — REV: A01

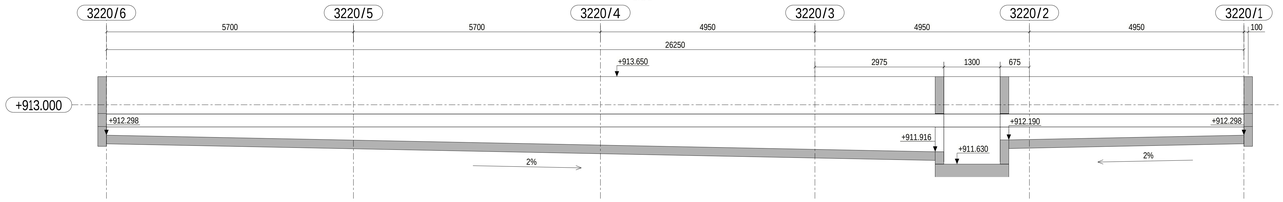
Tekla Structures

DRAWING NUMBER
MD-216-3220-EG-CV-GAD-1004
REV
A01

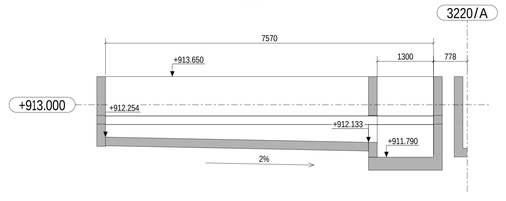
LEVEL +913.000
1:50



0 - 0
1:50



Q - Q
1:50



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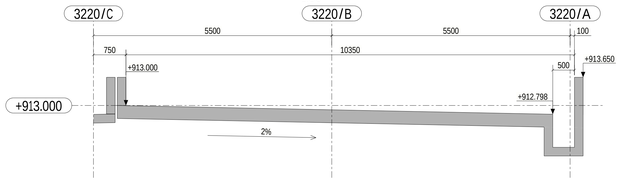
1:50



R - R
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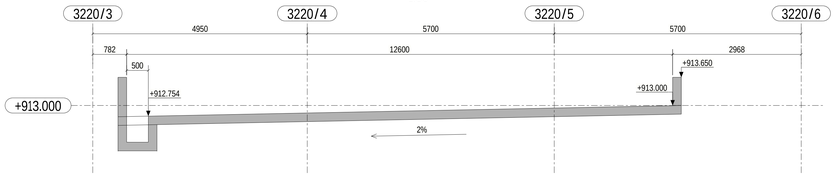

THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3230 & 3620 = RL(+)913.000 m

CHECKERED PLATE
GRATING
GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
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5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
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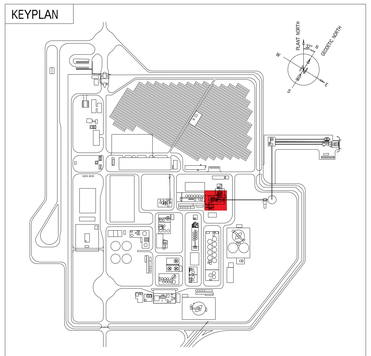
KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3220-EG-CV-GAD-1005	PLAN AT EL. +913.000 - GRID G-J/7-10	MD-216-3220-EG-ST-GAD-1005	PLAN AT EL. +922.150 - GRID A-C/2-6
MD-216-3220-EG-CV-GAD-1002	SECTIONS A-A TO H-H & EQUIP. FOUNDATION	MD-216-3220-EG-CV-GAD-1003	PLAN AT EL. +913.000 - GRID C-G/8-12
MD-216-3220-EG-ME-GAD-1075	SECTION F-F	MD-216-3220-EG-CV-GAD-1001	PLAN AT EL. +913.000 - GRID C-G/1-8
MD-216-3220-EG-ME-GAD-1073	SECTION D-D	MD-216-3220-EG-ME-GAD-1074	SECTION E-E
MD-216-3220-EG-ME-GAD-1071	SECTION B-B	MD-216-3220-EG-ME-GAD-1072	SECTION C-C
MD-216-3220-EG-ME-GAD-1069	PLAN VIEW	MD-216-3220-EG-ME-GAD-1070	SECTION A-A
MD-216-3220-EG-ME-GAD-1067	PLAN AT EL. +922.180	MD-216-3220-EG-ME-GAD-1068	PLAN AT EL. +924.580
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3220-EG-ME-GAD-1066	PLAN AT EL. +913.000

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	11.12.2019	ISSUED FOR APPROVAL	NikJan	MikLai	LuiPiz	SouCha	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER: 09045_450_C_G_24904_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707_GAU01_ZG01_CTA04_32204 — DOCUMENT ID: OU100667743 — REVISION: 0

DRAWN	NikJan	11.12.2019
CHECKED	MikLai	11.12.2019
ENGINEER	LuiPiz	11.12.2019
APPROVED	SouCha	11.12.2019
EMPLOYER		
SCALE	1:50	A0

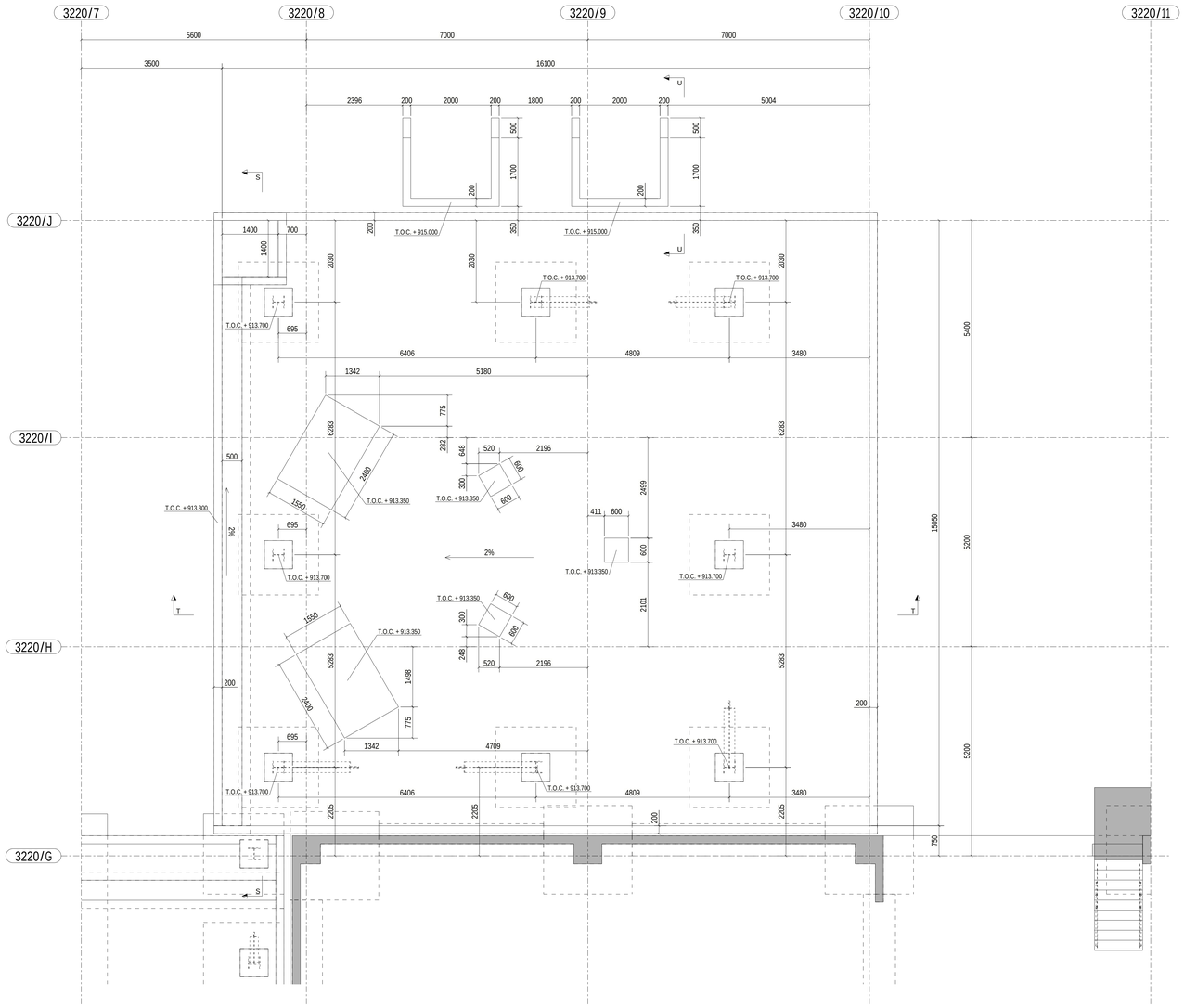
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
GRINDING AND CLASSIFICATION - BALL MILL
PLAN AT EL. +913.000 - GRID A-C/1-6

DRAWING NUMBER: MD-216-3220-EG-CV-GAD-1004 — REV: A01

Tekla Structures

DRAWING NUMBER: MD-216-3220-EG-CV-GAD-1005
REV: A01

LEVEL +913.000
1:50



THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3220 = RL(+)913.000 m

CHECKERED PLATE
GRATING
GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
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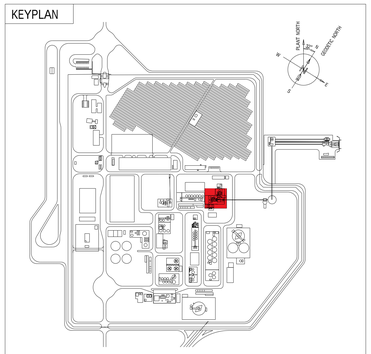
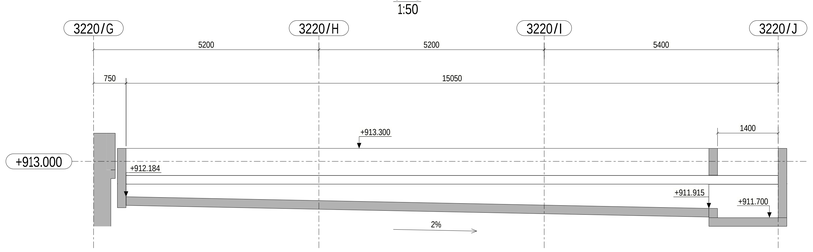
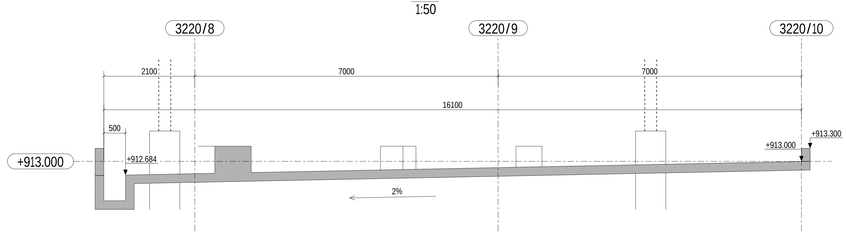
KEYPLAN



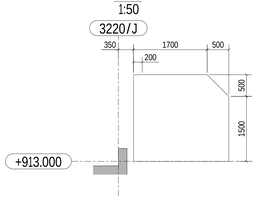
S - S
1:50



T - T
1:50



U - U
1:50



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3220-EG-CV-GAD-1004	PLAN AT EL +913.000 - GRID A-C/1-6	MD-216-3220-EG-ST-GAD-1006	PLAN AT EL +920.690 - GRID G-J/8-10
MD-216-3220-EG-CV-GAD-1002	SECTIONS A-A TO H-H & EQUIP. FOUNDATION	MD-216-3220-EG-CV-GAD-1003	PLAN AT EL +913.000 - GRID C-G/8-11
MD-216-3220-EG-ME-GAD-1075	SECTION F-F	MD-216-3220-EG-CV-GAD-1001	PLAN AT EL +913.000 - GRID C-G/1-8
MD-216-3220-EG-ME-GAD-1073	SECTION D-D	MD-216-3220-EG-ME-GAD-1074	SECTION E-E
MD-216-3220-EG-ME-GAD-1071	SECTION B-B	MD-216-3220-EG-ME-GAD-1072	SECTION C-C
MD-216-3220-EG-ME-GAD-1069	PLAN VIEW	MD-216-3220-EG-ME-GAD-1070	SECTION A-A
MD-216-3220-EG-ME-GAD-1067	PLAN AT EL +922.180	MD-216-3220-EG-ME-GAD-1068	PLAN AT EL +924.580
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESGN BASIS	MD-216-3220-EG-ME-GAD-1066	PLAN AT EL +913.000

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	11.12.2019	ISSUED FOR APPROVAL	NikJan	MikLai	LuiPiz	SouCha	

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LARSEN & TOUBRO DRAWING NUMBER: 09045_450_C_G_24905_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707_GAU01_ZG01_CTA04_32205 — DOCUMENT ID: OU100667744 — REVISION: 0

DRAWN	NikJan	11.12.2019
CHECKED	MikLai	11.12.2019
ENGINEER	LuiPiz	11.12.2019
APPROVED	SouCha	11.12.2019
EMPLOYER		
SCALE	1:50	A0

MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
GRINDING AND CLASSIFICATION - BALL MILL
PLAN AT EL. +913.000 - GRID G-J/7-10

DRAWING NUMBER: MD-216-3220-EG-CV-GAD-1005 — REV: A01

Tekla Structures

ATTACHMENT 2 OTE INPUT (LOAD PLAN)

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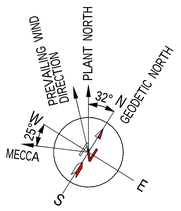
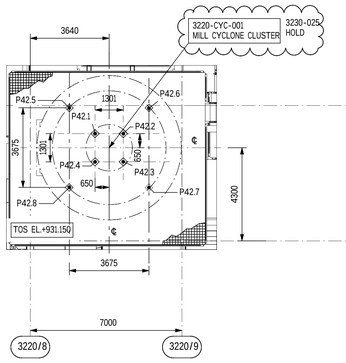
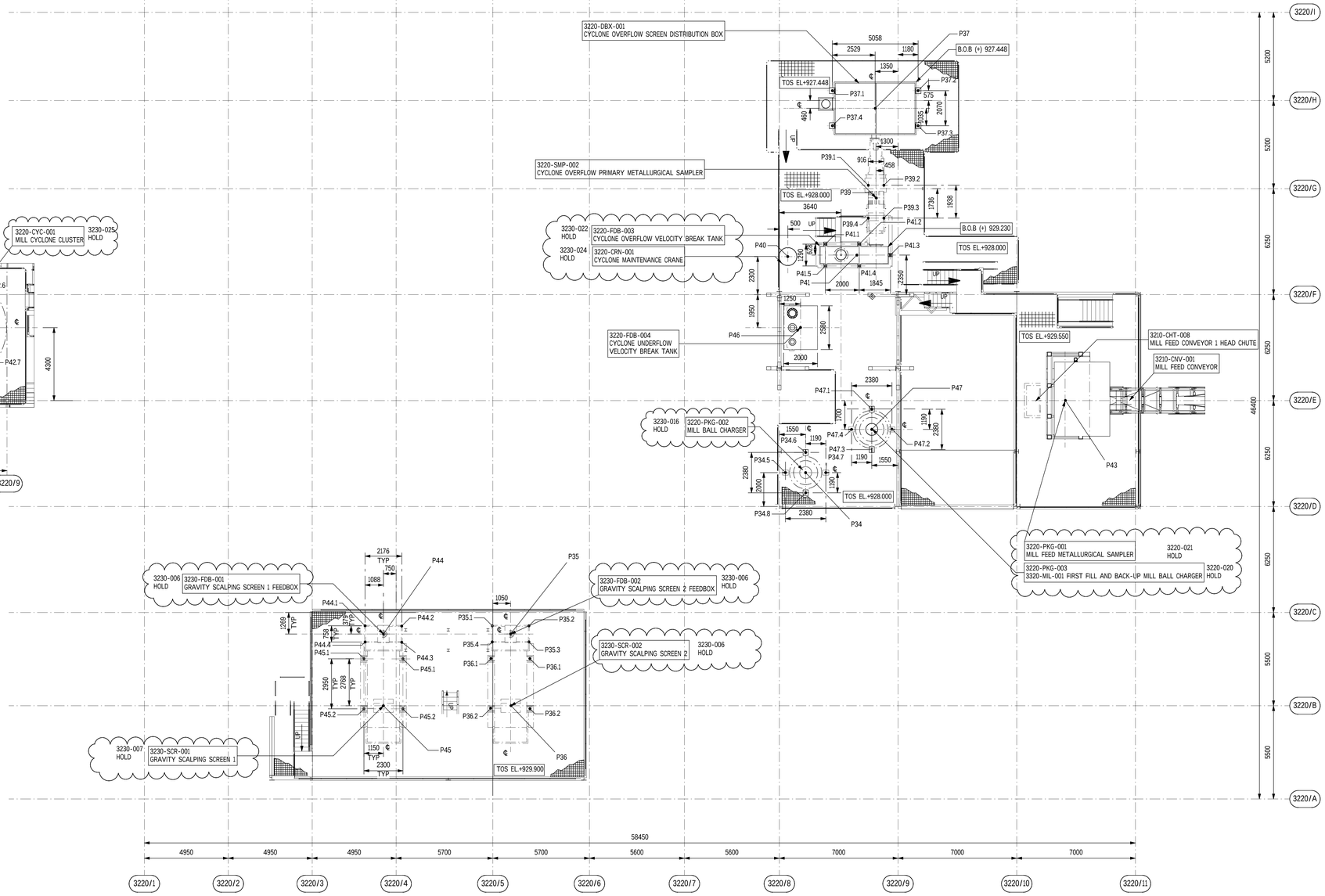
DRAWING NUMBER: MD-216-3220-EG-ME-GAD-1088 | REV: A02





PLAN AT EL.(+)931.150



PLAN AT EL.(+)928.000

NOTE:
FOR LOAD TABLE REFER DRAWING NUMBER MD-216-3220-EG-ME-GAD-1089.

0 1 10m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CHECKERED PLATE
GRATING
GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +Y, +X, +My, +Mx

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m²
GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

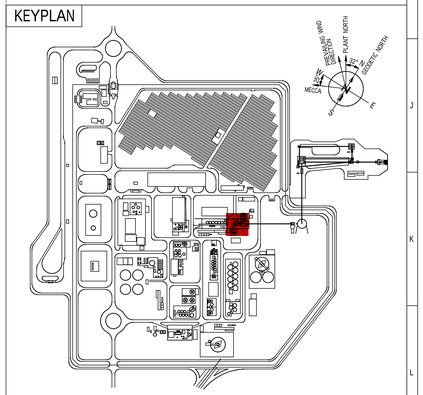
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3220-EG-ME-GAD-1066	GENERAL ARRANGEMENT DRAWING, PLAN AT EL(+) 913.000
MD-216-3220-EG-ME-GAD-1089	LOAD TABLE
MD-216-3220-EG-ME-GAD-1087	PLAN AT EL (+)924.550
MD-216-3220-EG-ME-GAD-1086	PLAN AT EL (+)922.150
MD-216-3220-EG-ME-GAD-1085	PLAN AT EL (+)913.100

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	04-02-2020	ISSUED FOR DESIGN	SF	TV	SC	RL	
A01	03-01-2020	ISSUED FOR APPROVAL	BR	TV	SC	RL	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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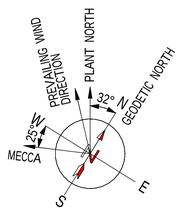
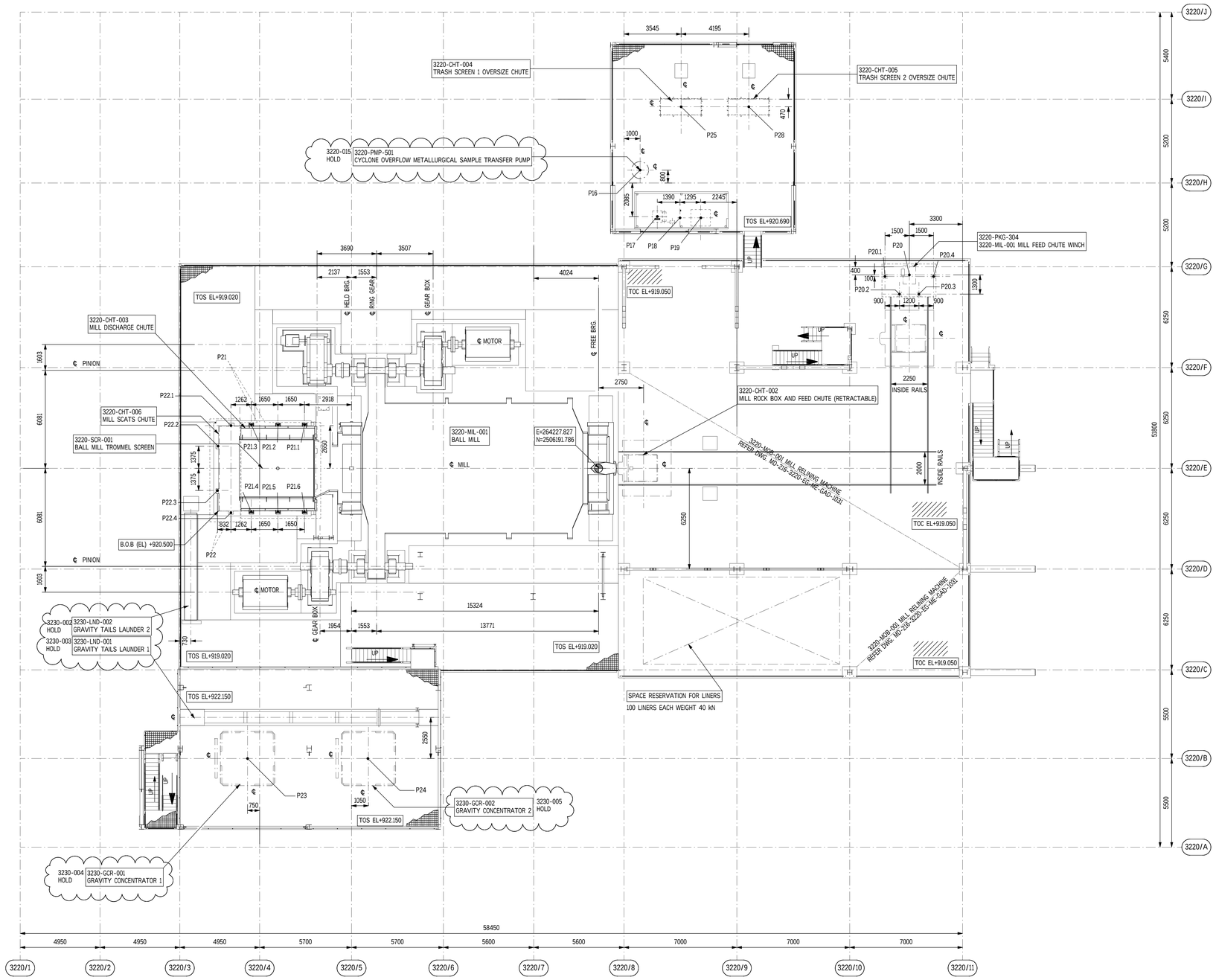
LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_450_M_D_02372_004			1
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707_GAU01_ZG01_MLD14_33204		OU100667808	1
DRAWN	AV	03/01/2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	03/01/2020	LOAD PLAN
ENGINEER	SC	03/01/2020	GRINDING AND CLASSIFICATION - BALL MILL
APPROVED	RL	03/01/2020	PLANT AT EL.(+) 931.150
EMPLOYER			DRAWING NUMBER: MD-216-3220-EG-ME-GAD-1088
SCALE 1:100	A0		REV: A02

DRAWING NUMBER: MD-216-3220-EG-ME-GAD-1086 | REV: A02





PLAN AT EL.(+)922.150

0 1 10m

NOTE:
FOR LOAD TABLE REFER DRAWING NUMBER MD-216-3220-EG-ME-GAD-1089.

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- A — CONCRETE FLOOR WITH ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- F — GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +Y, +X, +My, +Mx

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m² | GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

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FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

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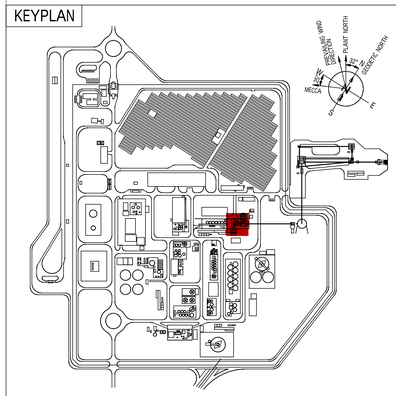
12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3220-EG-ME-GAD-1066	GENERAL ARRANGEMENT DRAWING, PLAN AT EL.(+) 913.000
MD-216-3220-EG-ME-GAD-1089	LOAD TABLE
MD-216-3220-EG-ME-GAD-1088	PLAN AT EL. (+)931.150
MD-216-3220-EG-ME-GAD-1087	PLAN AT EL. (+)924.550
MD-216-3220-EG-ME-GAD-1085	PLAN AT EL. (+)913.000

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	04-02-2020	ISSUED FOR DESIGN	SF	TV	SC	RL	
A01	03-01-2020	ISSUED FOR APPROVAL	BR	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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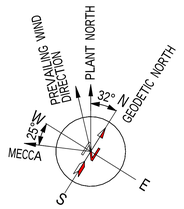
LARSEN & TOUBRO DRAWING NUMBER	O19045_450_M_D_02372_002
REVISION	1
OUTOTEC DRAWING NUMBER	900707 _GAU01 _ZG01 _MLD14 _32202
DOCUMENT ID	OU100667806
REVISION	1

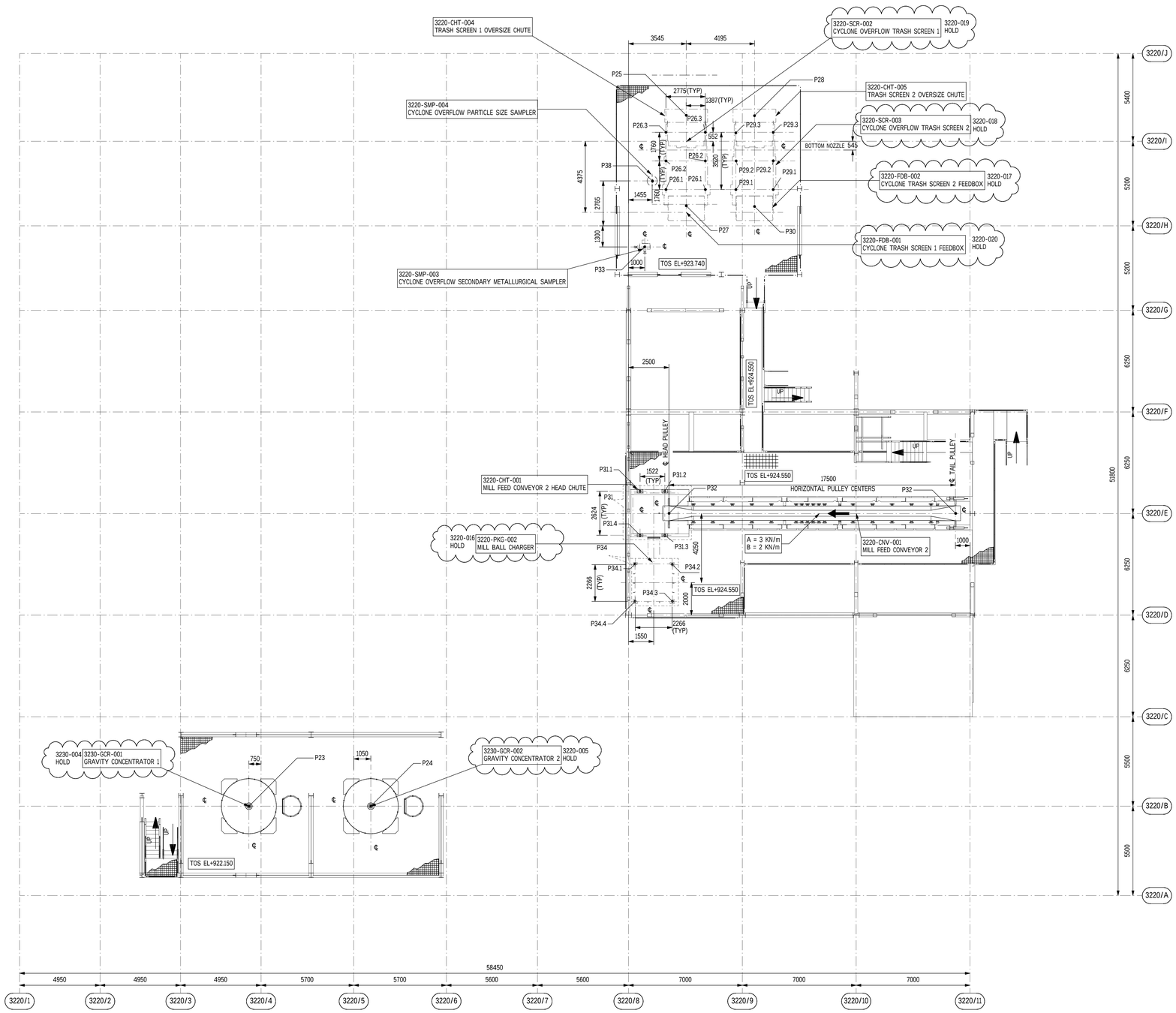
DRAWN	AV	03/01/2020
CHECKED	TV	03/01/2020
ENGINEER	SC	03/01/2020
APPROVED	RL	03/01/2020
EMPLOYER		
SCALE	1:100	A0

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
GRINDING AND CLASSIFICATION - BALL MILL
PLANT AT EL.(+) 922.150

DRAWING NUMBER: MD-216-3220-EG-ME-GAD-1086 | REV: A02

DRAWING NUMBER MD-216-3220-EG-ME-GAD-1087 REV A02





PLAN AT EL.(+)924.550

	CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
	CONCRETE FLOOR
A	CONCRETE FLOOR WITH ACID RESISTANT COATING
	CHECKERED PLATE
	GRATING
F	GRATING - FRP
	ACID RESISTANT BRICKLINING
	BRICK WALL
	CONCRETE WALL

GENERAL NOTES, LEGEND

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03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

DIRECTION OF FORCES AND MOMENTS

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
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HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

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DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

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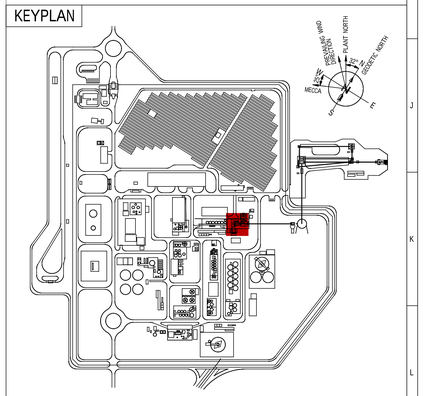
12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - CONVEYOR LINE LOAD A = DEAD WEIGHT, B = MATERIAL WEIGHT.

15 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



NOTE:
FOR LOAD TABLE REFER DRAWING NUMBER MD-216-3220-EG-ME-GAD-1089.

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3220-EG-ME-GAD-1066	GENERAL ARRANGEMENT DRAWING, PLAN AT EL.(+) 913.000
MD-216-3220-EG-ME-GAD-1089	LOAD TABLE
MD-216-3220-EG-ME-GAD-1088	PLAN AT EL. (+)931.150
MD-216-3220-EG-ME-GAD-1086	PLAN AT EL. (+)922.150
MD-216-3220-EG-ME-GAD-1085	PLAN AT EL. (+)913.100

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	04-02-2020	ISSUED FOR DESIGN	SF	TV	SC	RL	
		A01	03-01-2020	ISSUED FOR APPROVAL	BR	TV	SC	RL	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_450_M_D_02372_003		1
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707 _GAU01 _ZG01 _MLD14 _32203	0U100667807	1

DRAWN	AV	03/01/2020
CHECKED	TV	03/01/2020
ENGINEER	SC	03/01/2020
APPROVED	RL	03/01/2020
EMPLOYER		
SCALE	1:100	A0

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
GRINDING AND CLASSIFICATION - BALL MILL
PLANT AT EL.(+) 924.550

DRAWING NUMBER MD-216-3220-EG-ME-GAD-1087 REV A02

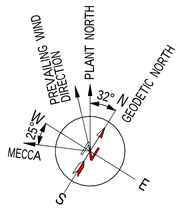
DRAWING NUMBER: MD-216-3220-EG-ME-GAD-1089 | REV: A02

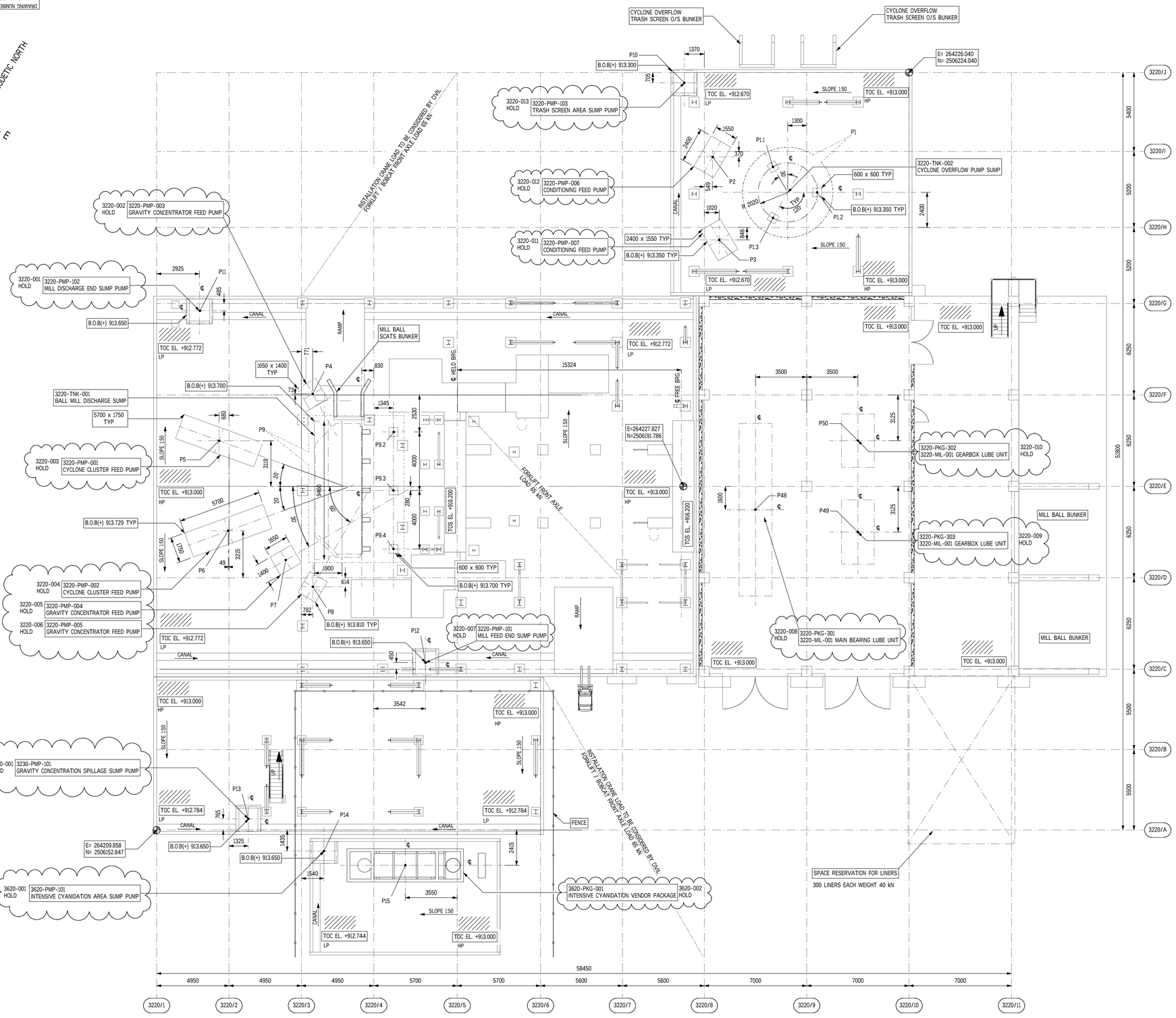
	LOADING POINT	P1 *				P2				P3				P4				P5				P6				P7				P8				P9.1/2/3/4 *				P10				P11				P12				P13				P14				P15			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT					- 49				- 49				- 31				- 279				- 279				- 31				- 31								- 12				- 12				- 13				- 12				- 9				- 95			
2	DEAD LOAD (SECONDARY EQUIPMENT)																																																										
3	OPERATION LOAD 1. FILLING																																																										
4	OPERATION LOAD 2. PRESSURE																																																										
5	OPERATION LOAD 3. TEMPERATURE																																																										
6	OPERATION LOAD 4. DYNAMIC					- 73				- 73				- 47				- 418				- 418				- 47				- 47								- 18				- 17				- 19				- 17				- 13							
7	OPERATION LOAD 5. OTHERS																																																										
8	TEST LOAD FILLING / EMERGENCY																																																							- 400			
9	LIVE LOAD																																																							- 340			
10	WIND ±X - DIRECTION					-	2	-		-	2	-		-	1	-		-	1	-		-	1	-		-	1	-		-	1	-						-	-	-		-	-	-		-	-	-		-	-	-		-	10	-	
11	WIND ±Y - DIRECTION					-	-	1		-	-	1		-	-	1		-	-	6		-	-	6		-	-	1		-	-	1						-	-	-		-	-	-		-	-	-		-	-	-		-	-	32	
12	EARTHQUAKE ±X - DIRECTION					-	3	3		-	3	3		-	2	2		-	14	14		-	14	14		-	2	2		-	2	2						-	1	1		-	1	1		-	1	1		-	1	1		-	20	20	
13	EARTHQUAKE ±Y - DIRECTION					-	3	3		-	3	3		-	2	2		-	14	14		-	14	14		-	2	2		-	2	2						-	1	1		-	1	1		-	1	1		-	1	1		-	20	20	
	EQUIPMENT - NO.	3220-TNK-002				3220-PMP-006				3220-PMP-007				3220-PMP-003				3220-PMP-001				3220-PMP-002				3220-PMP-004				3220-PMP-005				3220-TNK-001				3220-PMP-103				3220-PMP-102				3220-PMP-101				3230-PMP-101				3620-PMP-101				3620-PKG-001			
	DESCRIPTION	CYCLONE OVERFLOW PUMP SUMP				CONDITIONING FEED PUMP				CONDITIONING FEED PUMP				GRAVITY CONCENTRATOR FEED PUMP				CYCLONE CLUSTER FEED PUMP				CYCLONE CLUSTER FEED PUMP				GRAVITY CONCENTRATOR FEED PUMP				GRAVITY CONCENTRATOR FEED PUMP				BALL MILL DISCHARGE SUMP				TRASH SCREEN AREA SUMP PUMP				MILL DISCHARGE END SUMP PUMP				MILL FEED END SUMP PUMP				GRAVITY SEPERATION SPILLAGE SUMP PUMP				INTENSIVE CYANIDATION AREA SUMP PUMP				INTENSIVE CYANIDATION VENDOR PACKAGE			

	LOADING POINT	P16				P17				P18				P19				P20.1/2/3/4 *				P21.1/2/3/4 /5/6				P22.1/2/3/4				P23				P24			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT	-	8			-	3			-	2			-	2							-	30			-	30			-	60			-	60		
2	DEAD LOAD (SECONDARY EQUIPMENT)																																				
3	OPERATION LOAD 1. FILLING																																				
4	OPERATION LOAD 2. PRESSURE																																				
5	OPERATION LOAD 3. TEMPERATURE																																				
6	OPERATION LOAD 4. DYNAMIC	-	11							-	3																										
7	OPERATION LOAD 5. OTHERS																																				
8	TEST LOAD FILLING / EMERGENCY																					-	80			-	80			-	120			-	120		
9	LIVE LOAD					-	6							-	15																						
10	WIND ±X - DIRECTION		"	1	-		"	1	-		"	1	-		"	1	-						-	7.6	-		-	7.6	-		-	12	-		-	12	-
11	WIND ±Y - DIRECTION		"	-	1		-	-	1		-	-	1		-	-	1						-	-	3.1		-	-	3.1		-	-	12		-	-	12
12	EARTHQUAKE ±X - DIRECTION		"	1	1		-	1	1		-	-	-		1	-	-						-	4	4		-	4	4		-	6	6		-	6	6
13	EARTHQUAKE ±Y - DIRECTION		"	1	1		-	1	1		-	-	-		1	-	-						-	4	4		-	4	4		-	6	6		-	6	6
	EQUIPMENT - NO.	3220-PMP-501				3310-ANL-002				3310-ANL-002				3310-ANL-002				3220-PKG-304				3220-CHT-003				3220-CHT-006				3230-GCR-001				3230-GCR-002			
	DESCRIPTION	CYCLONE OVERFLOW METALLURGICAL SAMPLE TRANSFER PUMP				PARTICLE SIZE ANALYZER				WATER PUMP FOR PSI				WATER TANK FOR PSI				3220-ANL-001 MILL FEED CHUTE WINCH				MILL DISCHARGE CHUTE				MILL SCATS CHUTE				GRAVITY CONCENTRATOR 1				GRAVITY CONCENTRATOR 2			

	LOADING POINT	P25				P26.1				P26.2				P26.3				P27				P28				P29.1				P29.2				P29.3				P30				P31.1/2/3/4				P32				P33				P34.1/2/3/4				P35.1/2/3/4				P36.1				P36.2			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT	-	45			-	65	± 7	± 7	-	12	± 2	± 2	-	65	± 7	± 7	-	45			-	25			-	65	± 7	± 7	-	12	± 2	± 2	-	65	± 7	± 7	-	45			-	17			-	45			-	1.3			-	25			-	12			-	15			-	15		
2	DEAD LOAD (SECONDARY EQUIPMENT)																		

DRAWING NUMBER MD-216-3220-EG-ME-GAD-1085 REV A02





PLAN AT EL.(+)913.000

0 1 10m

NOTE:
FOR LOAD TABLE REFER DRAWING NUMBER MD-216-3220-EG-ME-GAD-1089.

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- A — CONCRETE FLOOR WITH ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- F — GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m² GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

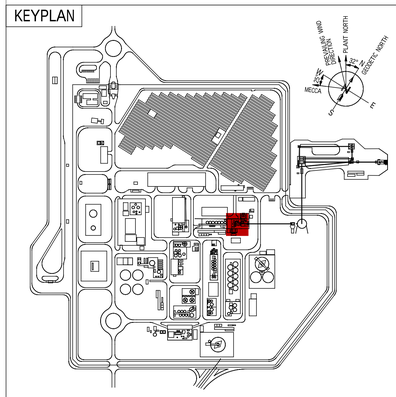
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3220-EG-ME-GAD-1066	GENERAL ARRANGEMENT DRAWING, PLAN AT EL(+) 913.000
MD-216-3220-EG-ME-GAD-1089	LOAD TABLE
MD-216-3220-EG-ME-GAD-1088	PLAN AT EL (+)931.150
MD-216-3220-EG-ME-GAD-1087	PLAN AT EL (+)924.550
MD-216-3220-EG-ME-GAD-1086	PLAN AT EL (+)922.150

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	04-02-2020	ISSUED FOR DESIGN	SF	TV	SC	RL	
A01	03-01-2020	ISSUED FOR APPROVAL	BR	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_450_M_D_02372_001			1
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _GAU01 _ZG01 _ MLD14 _32201		OU100667805	1
DRAWN	BR	03/01/2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	03/01/2020	LOAD PLAN
ENGINEER	SC	03/01/2020	GRINDING AND CLASSIFICATION - BALL MILL
APPROVED	RL	03/01/2020	PLAN AT EL.(+) 913.000
EMPLOYER			
SCALE 1:100	A0	DRAWING NUMBER MD-216-3220-EG-ME-GAD-1085	REV A02

ATTACHMENT 3 STAAD ISOMETRIC VIEW

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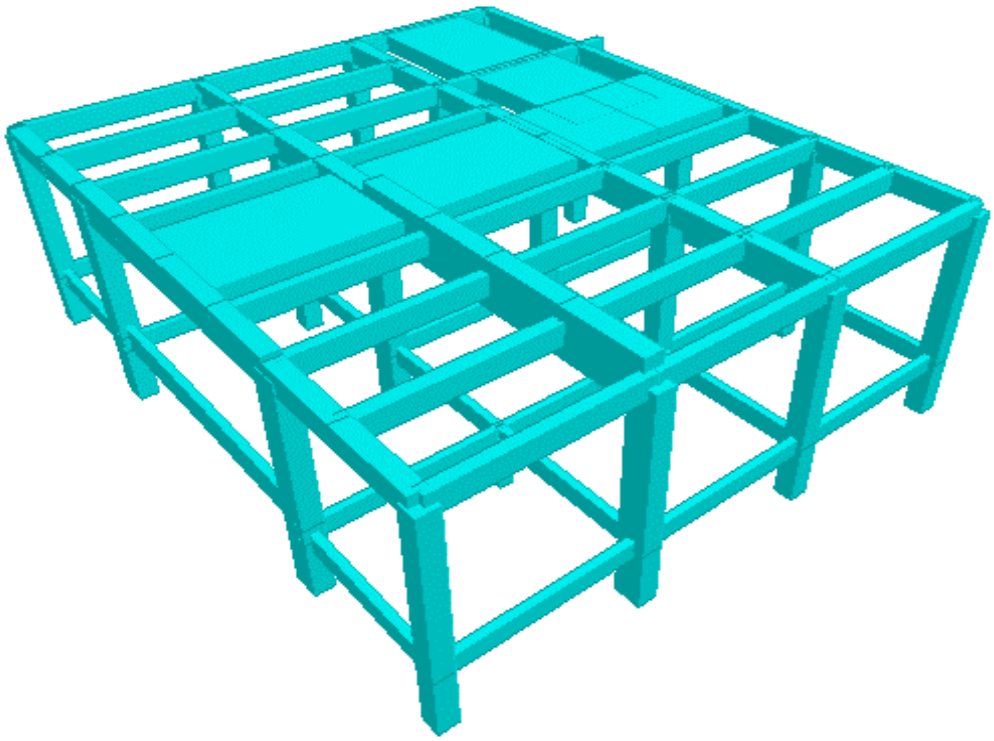

ATTACHMENT 4 STAAD Output

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    FEB 29, 2020                *
*           Time=    15:51: 8                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
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    1. STAAD SPACE
INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\7. 3220 RELINING MACHINE\CIVIL\LATEST\design\STAAD\Relini.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 12-FEB-20
    4. JOB NAME MMGP
    5. JOB CLIENT MAADEN
    6. JOB NO O19045
    7. JOB PART 3220 - RELINING MACHINE
    8. ENGINEER NAME STY
    9. CHECKER NAME BDA
   10. APPROVED NAME AG
   11. END JOB INFORMATION
   12. INPUT WIDTH 79
   13. UNIT METER KN
   14. JOINT COORDINATES
   15. 1 0 912.3 0; 2 7 912.3 0; 3 14 912.3 0; 4 21.15 912.3 0; 5 0 912.3 6.4
   16. 6 7 912.3 6.4; 7 14 912.3 6.4; 8 21.15 912.3 6.4; 9 7 912.3 12.65
   17. 10 14 912.3 12.65; 11 21.15 912.3 12.65; 12 14 912.3 18.9; 13 21.15 912.3 18.9
   18. 14 14 912.3 25.3; 15 21.15 912.3 25.3; 16 7 912.3 25.3; 17 0 912.3 25.3
   19. 18 0 912.3 18.9; 19 7 912.3 18.9; 20 0 910 0; 21 7 910 0; 22 14 910 0
   20. 23 21.15 910 0; 24 0 910 6.4; 25 7 910 6.4; 26 14 910 6.4; 27 21.15 910 6.4
   21. 28 7 910 12.65; 29 14 910 12.65; 30 21.15 910 12.65; 31 14 910 18.9
   22. 32 21.15 910 18.9; 33 14 910 25.3; 34 21.15 910 25.3; 35 7 910 25.3
   23. 36 0 910 25.3; 37 0 910 18.9; 39 0 918.55 0; 40 7 918.55 0; 41 14 918.55 0
   24. 42 21.15 918.55 0; 43 0 918.55 6.4; 44 7 918.55 6.4; 45 14 918.55 6.4
   25. 46 21.15 918.55 6.4; 47 7 918.55 12.65; 48 14 918.55 12.65
   26. 49 21.15 918.55 12.65; 50 14 918.55 18.9; 51 21.15 918.55 18.9
   27. 52 14 918.55 25.3; 53 21.15 918.55 25.3; 54 7 918.55 25.3; 55 0 918.55 25.3
   28. 56 0 918.55 18.9; 57 7 918.55 18.9; 58 0 918.55 12.65; 60 7 910 18.9
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   30. 70 17.575 918.55 12.65; 78 14 918.55 4; 79 21.15 914.85 0; 80 22.4 918.55 6.4
   31. 81 22.4 914.85 0; 82 19.9 914.85 0; 83 0 918.55 3.2; 84 7 918.55 3.2
   32. 85 14 918.55 3.2; 86 0 918.55 22.1; 87 7 918.55 22.1; 88 14 918.55 22.1
   33. 89 0 918.55 9.4; 90 7 918.55 9.4; 91 14 918.55 9.4; 92 7 918.55 15.9
   34. 93 14 918.55 15.9; 94 0 918.55 15.9; 95 21.15 918.55 15.9
   35. 96 21.15 918.55 22.1; 97 0 912.9 18.9; 98 0 912.9 6.4
   36. MEMBER INCIDENCES
   37. 65 20 1; 66 1 39; 67 21 2; 68 2 40; 69 22 3; 70 3 41; 71 23 4; 72 4 79
   38. 73 24 5; 74 5 98; 75 25 6; 76 6 44; 77 26 7; 78 7 45; 79 27 8; 80 8 46
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39. 81 28 9; 82 9 47; 83 29 10; 84 10 48; 85 30 11; 86 11 49; 87 37 18; 88 18 97
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57. 1094 98 43
58. DEFINE PMEMBER
59. 1001 PMEMBER 1
60. 1002 PMEMBER 2
61. 1003 1004 PMEMBER 3
62. 1018 PMEMBER 4
63. 1019 PMEMBER 5
64. 1020 1021 PMEMBER 6
65. 1033 PMEMBER 7
66. 1034 PMEMBER 8
67. 1035 1036 PMEMBER 9
68. 1048 PMEMBER 10
69. 1049 PMEMBER 11
70. 1050 PMEMBER 12
71. 1090 PMEMBER 13
72. 1091 PMEMBER 14
73. 1092 PMEMBER 15
74. 1005 1008 1015 PMEMBER 16
75. 1023 1030 1037 1044 PMEMBER 17
76. 1051 1058 PMEMBER 18
77. 1006 1009 1016 PMEMBER 19
78. 1024 1031 PMEMBER 20
79. 1038 1045 1052 1059 PMEMBER 21
80. 1007 1014 1017 PMEMBER 22
81. 1025 1032 PMEMBER 23
82. 1039 1046 PMEMBER 24
83. 1053 1060 PMEMBER 25
84. 1013 PMEMBER 26
85. 1029 PMEMBER 27
86. 1040 1047 PMEMBER 28
87. 1054 1061 PMEMBER 29
88. 1042 PMEMBER 30
89. 1012 PMEMBER 31
90. START GROUP DEFINITION
91. FLOOR
92. _FLOOR_5KN 1001 TO 1021 1023 TO 1050 1053 1054 1057 1060 1061 1092
93. _FLOOR_50KN 1048 1049 1051 TO 1053 1055 1056 1058 TO 1060 1090 1091
94. MEMBER
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 95. _TB_2 101 106 TO 114 116 TO 118 120 121 123 TO 135
 96. _FB_1 1001 TO 1004 1090 TO 1092
 97. _FB_2 1018 1019 1048 TO 1050
 98. _FB_2A 1020 TO 1022
 99. _FB_3 1033 TO 1036
100. _FB_5 1051 1058
101. _FB_6 1023 1030 1037 1044
102. _FB_7 1006 1009 1016
103. _FB_8 1024 1031
104. _FB_9 1038 1045 1052 1059
105. _FB_10 1007 1014 1017
106. _FB_11 1025 1032 1039 1046
107. _FB_12 1053 1060
108. _FB_13 1013 1029 1040 1047 1054 1061
109. _FB_4 1008 1015
110. _FB_14 1055 TO 1057
111. _FB_15 1010 1011 1026 1027 1041 TO 1043
112. _FB_16 1012 1028
113. _C_1 89
114. _C_2 65 TO 72 75 TO 80 83 TO 86 90 TO 95 102 TO 105 1062
115. _C_3 73 74 81 82 87 88 96 97 1093 1094
116. _TB_1 115
117. JOINT
118. END GROUP DEFINITION
119. DEFINE MATERIAL START
120. ISOTROPIC CONCRETE
121. E 3E+007
122. POISSON 0.17
123. DENSITY 24
124. ALPHA 1E-005
125. DAMP 0.05
126. TYPE CONCRETE
127. STRENGTH FCU 35000
128. END DEFINE MATERIAL
129. MEMBER PROPERTY AMERICAN
130. *COLUMNS
131. 73 74 81 82 87 88 96 97 1093 1094 PRIS YD 0.7 ZD 1
132. 65 TO 72 75 TO 80 83 TO 86 90 TO 95 102 TO 105 1062 PRIS YD 0.7 ZD 0.7
133. 89 PRIS YD 0.5 ZD 0.5
134. *TIE BEAMS
135. 115 PRIS YD 1 ZD 0.3
136. 101 106 TO 114 116 TO 118 120 121 123 TO 135 PRIS YD 0.7 ZD 0.3
137. *FLOOR BEAMS
138. 1038 1045 1052 1059 PRIS YD 1.8 ZD 0.7
139. 1023 1030 1037 1044 PRIS YD 1.2 ZD 0.6
140. 1024 1031 PRIS YD 1 ZD 0.7
141. 1001 TO 1004 1006 1009 TO 1011 1013 1016 1018 TO 1022 1026 1027 1029 1040 -
142. 1041 TO 1043 1047 TO 1050 1054 1061 1090 TO 1092 PRIS YD 0.8 ZD 0.4
143. 1012 1028 PRIS YD 0.7 ZD 4
144. 1033 TO 1036 PRIS YD 0.7 ZD 4.3
145. *SECONDARY BEAMS
146. 1055 TO 1057 PRIS YD 0.8 ZD 0.5
147. *STAIR BEAMS
148. 1064 1065 PRIS YD 0.5 ZD 0.3
149. MEMBER PROPERTY AMERICAN
150. 1005 1007 1008 1014 1015 1017 1025 1032 1039 1046 1051 1053 1058 -
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  151. 1060 PRIS YD 0.8 ZD 0.6
  152. CONSTANTS
  153. MATERIAL CONCRETE ALL
  154. SUPPORTS
  155. 20 TO 37 60 FIXED
  156. MEMBER RELEASE
  157. 65 TO 97 101 TO 118 120 121 123 TO 135 1001 TO 1021 1023 TO 1061 1090 TO 1093 -
  158. 1094 START MPX 0.99
  159. 65 TO 71 73 TO 97 101 TO 118 120 121 123 TO 135 1001 TO 1021 1023 TO 1062 -
  160. 1090 TO 1094 END MPX 0.99
  161. SLAVE ZX MASTER 48 JOINT 39 TO 58 61 62 65 68 70 78
  162. DEFINE IBC 2003
  163. SDS 0.067 SD1 0.013 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 CT 0.044
  164. SELFWEIGHT 1
  165. MEMBER WEIGHT
  166. *WALL LOAD
  167. 112 113 117 118 124 125 129 131 TO 134 UNI 30
  168. *STAIR LOAD
  169. 1022 1064 1065 UNI 21
  170. *RELINING MACHINE
  171. *ROCK BOX
  172. *WINCH
  173. 1012 CON 10 0.65
  174. FLOOR WEIGHT
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

  175. *SELFWT + 50% LIVE LOAD
  176. _FLOOR_5KN FLOAD 10
  177. _FLOOR_50KN FLOAD 32.5
  178. LOAD 11 SEISMIC LOAD SL(+X)
  179. IBC LOAD X 1
  180. JOINT LOAD
  181. *NODE1
  182. 61 FX 20.388 FY 246.113 FZ 194.574 MX -0 MY -0 MZ -0
  183. *NODE2
  184. 62 FX 16.731 FY 60.501 FZ 92.882 MX -0 MY -0 MZ -0
  185. *NODE3
  186. 78 FX 0.773 FY 16.492 FZ 5.554 MX -0 MY -0 MZ -0
  187. *NODE4
  188. 43 FX -0.296 FY -147.054 FZ 5.194 MX -0 MY -0 MZ -0
  189. *NODE5
  190. 44 FX 141.436 FY 77.203 FZ 0.835 MX -0 MY -0 MZ -0
  191. *NODE6
  192. 45 FX 128.523 FY -132.052 FZ -0.153 MX -0 MY -0 MZ -0
  193. *NODE7
  194. 46 80 FX 0.741 FY -84.817 FZ -0.016 MX -0 MY -0 MZ -0
  195. *NODE8
  196. 49 FX 2.366 FY -25.407 FZ -12.449 MX -0 MY -0 MZ -0
  197. *NODE9
```

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198. 56 FX 19.412 FY 12.834 FZ 0.5 MX -0 MY -0 MZ -0
199. *NODE10
200. 57 FX 20.035 FY -18.983 FZ 0.138 MX -0 MY -0 MZ -0
201. *NODE11
202. 50 FX 19.652 FY -20.417 FZ -0.001 MX -0 MY -0 MZ -0
203. *NODE12
204. 51 FX 2.15 FY 15.588 FZ -10.51 MX -0 MY -0 MZ -0
205. *NODE13
206. 52 FX 0.778 FY 1.046 FZ -0 MX -0 MY -0 MZ -0
207. *NODE14
208. 53 FX 0.778 FY -1.046 FZ -0 MX -0 MY -0 MZ -0
209. LOAD 12 SEISMIC LOAD SL(-X)
210. IBC LOAD X -1.
211. JOINT LOAD
212. *NODE1
213. 61 FX -20.388 FY -246.113 FZ -194.574 MX -0 MY -0 MZ -0
214. *NODE2
215. 62 FX -16.731 FY -60.501 FZ -92.882 MX -0 MY -0 MZ -0
216. *NODE3
217. 78 FX -0.773 FY -16.492 FZ -5.554 MX -0 MY -0 MZ -0
218. *NODE4
219. 43 FX 0.296 FY 147.054 FZ -5.194 MX -0 MY -0 MZ -0
220. *NODE5
221. 44 FX -141.436 FY -77.203 FZ -0.835 MX -0 MY -0 MZ -0
222. *NODE6
223. 45 FX -128.523 FY 132.052 FZ 0.153 MX -0 MY -0 MZ -0
224. *NODE7
225. 46 80 FX -0.741 FY 84.817 FZ 0.016 MX -0 MY -0 MZ -0
226. *NODE8
227. 49 FX -2.366 FY 25.407 FZ 12.449 MX -0 MY -0 MZ -0
228. *NODE9
229. 56 FX -19.412 FY -12.834 FZ -0.5 MX -0 MY -0 MZ -0
230. *NODE10
231. 57 FX -20.035 FY 18.983 FZ -0.138 MX -0 MY -0 MZ -0
232. *NODE11
233. 50 FX -19.652 FY 20.417 FZ 0.001 MX -0 MY -0 MZ -0
234. *NODE12
235. 51 FX -2.15 FY -15.588 FZ 10.51 MX -0 MY -0 MZ -0
236. *NODE13
237. 52 FX -0.778 FY -1.046 FZ -0 MX -0 MY -0 MZ -0
238. *NODE14
239. 53 FX -0.778 FY 1.046 FZ -0 MX -0 MY -0 MZ -0
240. LOAD 13 SEISMIC LOAD SL(+Z)
241. IBC LOAD Z 1
242. JOINT LOAD
243. *NODE1
244. 61 FX 16.497 FY 248.154 FZ 206.874 MX -0 MY -0 MZ -0
245. *NODE2
246. 62 FX 9.715 FY 111.176 FZ 126.718 MX -0 MY -0 MZ -0
247. *NODE3
```

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248. 78 FX -0 FY 44.314 FZ 18.277 MX -0 MY -0 MZ -0
249. *NODE4
250. 43 FX -0.352 FY -161.985 FZ 5.691 MX -0 MY -0 MZ -0
251. *NODE5
252. 44 FX 102.916 FY -3.249 FZ 0.554 MX -0 MY -0 MZ -0
253. *NODE6
254. 45 FX 97.824 FY -125.677 FZ 0.248 MX -0 MY -0 MZ -0
255. *NODE7
256. 46 80 FX 0.317 FY -57.677 FZ 0.255 MX -0 MY -0 MZ -0
257. *NODE8
258. 49 FX 0.915 FY 6.728 FZ 6.666 MX -0 MY -0 MZ -0
259. *NODE9
260. 56 FX 15.023 FY -1.517 FZ 0.749 MX -0 MY -0 MZ -0
261. *NODE10
262. 57 FX 15.513 FY -24.539 FZ 0.248 MX -0 MY -0 MZ -0
263. *NODE11
264. 50 FX 15.171 FY -20.012 FZ 0.131 MX -0 MY -0 MZ -0
265. *NODE12
266. 51 FX 1.813 FY -15.832 FZ 6.976 MX -0 MY -0 MZ -0
267. *NODE13
268. 52 FX 0.599 FY 0.905 FZ 0.039 MX -0 MY -0 MZ -0
269. *NODE14
270. 53 FX 0.598 FY -0.788 FZ 0.039 MX -0 MY -0 MZ -0
271. LOAD 14 SEISMIC LOAD SL(-Z)
272. IBC LOAD Z -1.
273. JOINT LOAD
274. *NODE1
275. 61 FX -16.497 FY -248.154 FZ -206.874 MX -0 MY -0 MZ -0
276. *NODE2
277. 62 FX -9.715 FY -111.176 FZ -126.718 MX -0 MY -0 MZ -0
278. *NODE3
279. 78 FX -0 FY -44.314 FZ -18.277 MX -0 MY -0 MZ -0
280. *NODE4
281. 43 FX 0.352 FY 161.985 FZ -5.691 MX -0 MY -0 MZ -0
282. *NODE5
283. 44 FX -102.916 FY 3.249 FZ -0.554 MX -0 MY -0 MZ -0
284. *NODE6
285. 45 FX -97.824 FY 125.677 FZ -0.248 MX -0 MY -0 MZ -0
286. *NODE7
287. 46 80 FX -0.317 FY 57.677 FZ -0.255 MX -0 MY -0 MZ -0
288. *NODE8
289. 49 FX -0.915 FY -6.728 FZ -6.666 MX -0 MY -0 MZ -0
290. *NODE9
291. 56 FX -15.023 FY 1.517 FZ -0.749 MX -0 MY -0 MZ -0
292. *NODE10
293. 57 FX -15.513 FY 24.539 FZ -0.248 MX -0 MY -0 MZ -0
294. *NODE11
295. 50 FX -15.171 FY 20.012 FZ -0.131 MX -0 MY -0 MZ -0
296. *NODE12
297. 51 FX -1.813 FY 15.832 FZ -6.976 MX -0 MY -0 MZ -0
298. *NODE13
299. 52 FX -0.599 FY -0.905 FZ -0.039 MX -0 MY -0 MZ -0
300. *NODE14
```

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53 FX -0.598 FY 0.788 FZ -0.039 MX -0 MY -0 MZ -0
LOAD 1 DEAD LOAD (DL)
SELFWEIGHT Y -1
SELFWEIGHT Y -1 LIST 115
JOINT LOAD
*NODE1
61 FX -5.699 FY -38.217 FZ 4.037 MX -0 MY -0 MZ -0
*NODE2
62 FX 5.826 FY -72.134 FZ -4.122 MX -0 MY -0 MZ -0
*NODE3
78 FX -0 FY -31.201 FZ -3.734 MX -0 MY -0 MZ -0
*NODE4
43 FX 0.001 FY -182.533 FZ 0.213 MX -0 MY -0 MZ -0
*NODE5
44 FX -7.336 FY -186.061 FZ 0.519 MX -0 MY -0 MZ -0
*NODE6
45 FX 14.471 FY -101.542 FZ 0.356 MX -0 MY -0 MZ -0
*NODE7
46 80 FX -2.649 FY -71.957 FZ 0.026 MX -0 MY -0 MZ -0
*NODE8
49 FX -6.358 FY -121.574 FZ -10.78 MX -0 MY -0 MZ -0
*NODE9
56 FX 0.347 FY -83.889 FZ 0.09 MX -0 MY -0 MZ -0
*NODE10
57 FX 0.596 FY -116.982 FZ 0.097 MX -0 MY -0 MZ -0
*NODE11
50 FX 0.587 FY -104.599 FZ 0.032 MX -0 MY -0 MZ -0
*NODE12
51 FX 0.14 FY -75.418 FZ 13.266 MX -0 MY -0 MZ -0
*NODE13
52 FX -0.136 FY -14.803 FZ -0 MX -0 MY -0 MZ -0
*NODE14
53 FX 0.21 FY -14.908 FZ -0 MX -0 MY -0 MZ -0
*SELF WEIGHT OF SLAB
FLOOR LOAD
_FLOOR_5KN FLOAD -7.5 GY
_FLOOR_50KN FLOAD -7.5 GY
*WALL LOAD
MEMBER LOAD
112 113 117 124 129 131 133 134 UNI GY -28.75
115 UNI GY -28.75 6.25 12.5
115 UNI GY -28.75 0 6.25
125 132 UNI GY -43.125
*STAIR LOAD
1022 1064 1065 UNI GY -16
LOAD 2 LIVE LOAD (LL)
JOINT LOAD
*NODE1
61 FX -11.077 FY -47.386 FZ 13.548 MX -0 MY -0 MZ -0
*NODE2
62 FX 10.892 FY -187.481 FZ -13.082 MX -0 MY -0 MZ -0
*NODE3
78 FX -0 FY -74.057 FZ -11.716 MX -0 MY -0 MZ -0
```

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354. *NODE4
355. 43 FX 0.068 FY -339.566 FZ 0.485 MX -0 MY -0 MZ -0
356. *NODE5
357. 44 FX -12.85 FY -450.83 FZ 1.351 MX -0 MY -0 MZ -0
358. *NODE6
359. 45 FX 35.338 FY -249.457 FZ 0.996 MX -0 MY -0 MZ -0
360. *NODE7
361. 46 80 FX -7.676 FY -147.767 FZ 0.076 MX -0 MY -0 MZ -0
362. *NODE8
363. 49 FX -19.029 FY -218.659 FZ -12.62 MX -0 MY -0 MZ -0
364. *NODE9
365. 56 FX 0.842 FY -161.447 FZ 0.193 MX -0 MY -0 MZ -0
366. *NODE10
367. 57 FX 1.625 FY -244.489 FZ 0.255 MX -0 MY -0 MZ -0
368. *NODE11
369. 50 FX 1.371 FY -198.437 FZ 0.068 MX -0 MY -0 MZ -0
370. *NODE12
371. 51 FX 0.3 FY -118.51 FZ 20.445 MX -0 MY -0 MZ -0
372. *NODE13
373. 52 FX 0.098 FY -6.165 FZ -0 MX -0 MY -0 MZ -0
374. *NODE14
375. 53 FX 0.098 FY -6.443 FZ -0 MX -0 MY -0 MZ -0
376. *FLOOR LIVE LOAD
377. FLOOR LOAD
378. _FLOOR_5KN FLOAD -5 GY
379. _FLOOR_50KN FLOAD -50 GY
380. *STAIR LIVE LOAD
381. MEMBER LOAD
382. 1022 1064 1065 UNI GY -10.5
383. LOAD 3 OPERATING FLUID LOAD (FL)
384. JOINT LOAD
385. *NODE1
386. 61 FX -1.651 FY 15.951 FZ 48.758 MX -0 MY -0 MZ -0
387. *NODE2
388. 62 FX 11.74 FY -65.196 FZ 4.045 MX -0 MY -0 MZ -0
389. *NODE3
390. 78 FX -0 FY 6.73 FZ 2.676 MX -0 MY -0 MZ -0
391. *NODE4
392. 43 FX -0.787 FY -416.964 FZ 1.312 MX -0 MY -0 MZ -0
393. *NODE5
394. 44 FX 8.777 FY -279.356 FZ 0.384 MX -0 MY -0 MZ -0
395. *NODE6
396. 45 FX 22.494 FY -23.384 FZ -0.082 MX -0 MY -0 MZ -0
397. *NODE7
398. 46 80 FX 0.028 FY -7.146 FZ -0.001 MX -0 MY -0 MZ -0
399. *NODE8
400. 49 FX 0.153 FY -6.044 FZ -3.103 MX -0 MY -0 MZ -0
401. *NODE9
402. 56 FX 3.209 FY -91.866 FZ 0.265 MX -0 MY -0 MZ -0
403. *NODE10
404. 57 FX 3.223 FY -62.057 FZ 0.026 MX -0 MY -0 MZ -0
405. *NODE11
406. 50 FX 3.225 FY -3.129 FZ 0.016 MX -0 MY -0 MZ -0
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*NODE12
51 FX 0.35 FY 6.714 FZ -3.3 MX -0 MY -0 MZ -0
*NODE13
52 FX 0.117 FY 0.166 FZ -0 MX -0 MY -0 MZ -0
*NODE14
53 FX 0.117 FY -0.166 FZ -0 MX -0 MY -0 MZ -0
LOAD 4 EQP DL (EQDL)
JOINT LOAD
*NODE1
61 FX -7.474 FY -27.078 FZ -0.18 MX -0 MY -0 MZ -0
*NODE2
62 FX 6.798 FY -103.92 FZ -2.356 MX -0 MY -0 MZ -0
*NODE3
78 FX -0 FY -3.22 FZ -0.228 MX -0 MY -0 MZ -0
*NODE4
43 FX 0.016 FY -236.602 FZ -0.122 MX -0 MY -0 MZ -0
*NODE5
44 FX -6.968 FY -196.229 FZ 0.986 MX -0 MY -0 MZ -0
*NODE6
45 FX 6.383 FY -27.734 FZ 0.041 MX -0 MY -0 MZ -0
*NODE7
46 80 FX -0.013 FY -14.751 FZ 0.029 MX -0 MY -0 MZ -0
*NODE8
49 FX -0.022 FY -50.746 FZ -2.93 MX -0 MY -0 MZ -0
*NODE9
56 FX 0.136 FY -183.818 FZ 0.189 MX -0 MY -0 MZ -0
*NODE10
57 FX 0.59 FY -127.785 FZ 0.167 MX -0 MY -0 MZ -0
*NODE11
50 FX 0.465 FY -33.816 FZ 0.055 MX -0 MY -0 MZ -0
*NODE12
51 FX 0.047 FY -16.985 FZ 4.348 MX -0 MY -0 MZ -0
*NODE13
52 FX 0.02 FY 0.029 FZ -0 MX -0 MY -0 MZ -0
*NODE14
53 FX 0.02 FY -0.029 FZ -0 MX -0 MY -0 MZ -0
MEMBER LOAD
*RELINING MACHINE
*ROCK BOX
1033 CON GY -285 1
*WINCH
1012 CON GY -10 0.65
JOINT LOAD
68 FY -168
LOAD 5 EQP LL (EQLL)
JOINT LOAD
*NODE1
61 FX -24.147 FY -90.225 FZ 28.819 MX -0 MY -0 MZ -0
*NODE2
62 FX 26.411 FY -213.691 FZ -4.22 MX -0 MY -0 MZ -0
*NODE3
78 FX -0 FY 6.166 FZ -0.186 MX -0 MY -0 MZ -0
*NODE4
43 FX -0.109 FY -449.503 FZ -0.294 MX -0 MY -0 MZ -0
*NODE5
44 FX 36.367 FY -430.19 FZ 0.981 MX -0 MY -0 MZ -0
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463. *NODE6
464. 45 FX 60.889 FY -113.844 FZ 0.147 MX -0 MY -0 MZ -0
465. *NODE7
466. 46 80 FX 0.073 FY -101.299 FZ 0.067 MX -0 MY -0 MZ -0
467. *NODE8
468. 49 FX -0.982 FY -162.527 FZ -14.258 MX -0 MY -0 MZ -0
469. *NODE9
470. 56 FX 6.538 FY -71.799 FZ 0.072 MX -0 MY -0 MZ -0
471. *NODE10
472. 57 FX 6.946 FY -101.056 FZ 0.125 MX -0 MY -0 MZ -0
473. *NODE11
474. 50 FX 6.67 FY -100.174 FZ -0.039 MX -0 MY -0 MZ -0
475. *NODE12
476. 51 FX 0.589 FY -72.943 FZ 12.125 MX -0 MY -0 MZ -0
477. *NODE13
478. 52 FX 0.297 FY 0.42 FZ -0 MX -0 MY -0 MZ -0
479. *NODE14
480. 53 FX 0.296 FY -0.42 FZ -0 MX -0 MY -0 MZ -0
481. *RELINING MACHINE
482. LOAD 6 STARTING BELT TENSION (SBT)
483. JOINT LOAD
484. *NODE1
485. 61 FX 3.092 FY 36.598 FZ 30.968 MX -0 MY -0 MZ -0
486. *NODE2
487. 62 FX 2.959 FY -5.66 FZ 5.134 MX -0 MY -0 MZ -0
488. *NODE3
489. 78 FX -0 FY 15.133 FZ 0.598 MX -0 MY -0 MZ -0
490. *NODE4
491. 43 FX -0.314 FY -22.644 FZ 0.875 MX -0 MY -0 MZ -0
492. *NODE5
493. 44 FX 69.452 FY 83.466 FZ 0.253 MX -0 MY -0 MZ -0
494. *NODE6
495. 45 FX 56.712 FY -43.565 FZ 0.032 MX -0 MY -0 MZ -0
496. *NODE7
497. 46 80 FX 0.125 FY -80.658 FZ 0.02 MX -0 MY -0 MZ -0
498. *NODE8
499. 49 FX -1.287 FY -72.62 FZ -14.185 MX -0 MY -0 MZ -0
500. *NODE9
501. 56 FX 8.117 FY 23.065 FZ -0.171 MX -0 MY -0 MZ -0
502. *NODE10
503. 57 FX 8.442 FY -10.092 FZ 0.047 MX -0 MY -0 MZ -0
504. *NODE11
505. 50 FX 8.291 FY -7.038 FZ -0.126 MX -0 MY -0 MZ -0
506. *NODE12
507. 51 FX 0.675 FY -21.898 FZ -1.869 MX -0 MY -0 MZ -0
508. *NODE13
509. 52 FX 0.321 FY 0.454 FZ -0 MX -0 MY -0 MZ -0
510. *NODE14
511. 53 FX 0.321 FY -0.454 FZ -0 MX -0 MY -0 MZ -0
512. LOAD 7 WIND LOAD (+X) WL(+X)
513. JOINT LOAD
514. *NODE1
515. 61 FX 25.5 FY 109.961 FZ 62.618 MX -0 MY -0 MZ -0
516. *NODE2
517. 62 FX 24.656 FY -57.144 FZ -3.838 MX -0 MY -0 MZ -0
518. *NODE3
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519. 78 FX 1.347 FY -21.555 FZ -10.682 MX -0 MY -0 MZ -0
520. *NODE4
521. 43 FX 8.16 FY -33.907 FZ 1.732 MX -0 MY -0 MZ -0
522. *NODE5
523. 44 FX 150.424 FY 189.033 FZ 0.02 MX -0 MY -0 MZ -0
524. *NODE6
525. 45 FX 131.324 FY -106.458 FZ 0.36 MX -0 MY -0 MZ -0
526. *NODE7
527. 46 80 FX 2.554 FY -77.129 FZ -0.006 MX -0 MY -0 MZ -0
528. *NODE8
529. 49 FX 5.069 FY -45.883 FZ -25.014 MX -0 MY -0 MZ -0
530. *NODE9
531. 56 FX 22.193 FY 27.995 FZ 0.222 MX -0 MY -0 MZ -0
532. *NODE10
533. 57 FX 22.562 FY -3.795 FZ -0.002 MX -0 MY -0 MZ -0
534. *NODE11
535. 50 FX 22.698 FY -1.466 FZ -0.178 MX -0 MY -0 MZ -0
536. *NODE12
537. 51 FX 6.283 FY 37.936 FZ -22.789 MX -0 MY -0 MZ -0
538. *NODE13
539. 52 FX 2.678 FY 5.999 FZ -0 MX -0 MY -0 MZ -0
540. *NODE14
541. 53 FX 2.913 FY 4.764 FZ -0 MX -0 MY -0 MZ -0
542. MEMBER LOAD
543. 66 95 UNI GX 3.16
544. 74 88 1093 1094 UNI GX 8.41
545. LOAD 8 WIND LOAD (-X) WL(-X)
546. JOINT LOAD
547. *NODE1
548. 61 FX -25.476 FY -109.744 FZ -62.424 MX -0 MY -0 MZ -0
549. *NODE2
550. 62 FX -24.664 FY 58.405 FZ 4.884 MX -0 MY -0 MZ -0
551. *NODE3
552. 78 FX -1.347 FY 23.4 FZ 11.412 MX -0 MY -0 MZ -0
553. *NODE4
554. 43 FX -8.158 FY 33.752 FZ -1.727 MX -0 MY -0 MZ -0
555. *NODE5
556. 44 FX -150.507 FY -189.518 FZ -0.022 MX -0 MY -0 MZ -0
557. *NODE6
558. 45 FX -131.265 FY 105.464 FZ -0.383 MX -0 MY -0 MZ -0
559. *NODE7
560. 46 80 FX -2.556 FY 77.114 FZ 0.006 MX -0 MY -0 MZ -0
561. *NODE8
562. 49 FX -5.069 FY 48.866 FZ 27.408 MX -0 MY -0 MZ -0
563. *NODE9
564. 56 FX -22.194 FY -27.932 FZ -0.221 MX -0 MY -0 MZ -0
565. *NODE10
566. 57 FX -22.56 FY 9.95 FZ -0 MX -0 MY -0 MZ -0
567. *NODE11
568. 50 FX -22.695 FY 17.991 FZ 0.136 MX -0 MY -0 MZ -0
569. *NODE12
570. 51 FX -6.28 FY -30.161 FZ 23.374 MX -0 MY -0 MZ -0
571. *NODE13
572. 52 FX -2.677 FY 4.765 FZ -0 MX -0 MY -0 MZ -0
573. *NODE14
574. 53 FX -2.913 FY 5.999 FZ -0 MX -0 MY -0 MZ -0
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575. MEMBER LOAD
576. 72 105 1062 UNI GX -3.16
577. 80 93 UNI GX -5.63
578. 86 UNI GX -5.56
579. LOAD 9 WIND LOAD (+Z) WL(+Z)
580. JOINT LOAD
581. *NODE1
582. 61 FX 1.331 FY 104.464 FZ 107.933 MX -0 MY -0 MZ -0
583. *NODE2
584. 62 FX -1.892 FY 122.992 FZ 119.96 MX -0 MY -0 MZ -0
585. *NODE3
586. 78 FX -0 FY 92.272 FZ 43.393 MX -0 MY -0 MZ -0
587. *NODE4
588. 43 FX 0.064 FY -88.488 FZ 5.053 MX -0 MY -0 MZ -0
589. *NODE5
590. 44 FX -4.877 FY -104.901 FZ 5.95 MX -0 MY -0 MZ -0
591. *NODE6
592. 45 FX 5.495 FY -76.645 FZ 1.014 MX -0 MY -0 MZ -0
593. *NODE7
594. 46 80 FX -0.15 FY 22.638 FZ 2.098 MX -0 MY -0 MZ -0
595. *NODE8
596. 49 FX 0.071 FY 99.173 FZ 55.088 MX -0 MY -0 MZ -0
597. *NODE9
598. 56 FX -0.101 FY -16.604 FZ 2.348 MX -0 MY -0 MZ -0
599. *NODE10
600. 57 FX -0.025 FY -10.39 FZ 3.473 MX -0 MY -0 MZ -0
601. *NODE11
602. 50 FX -0.22 FY -4.302 FZ 3.358 MX -0 MY -0 MZ -0
603. *NODE12
604. 51 FX 0.199 FY -115.873 FZ 58.746 MX -0 MY -0 MZ -0
605. *NODE13
606. 52 FX 0.053 FY 6.206 FZ 1.242 MX -0 MY -0 MZ -0
607. *NODE14
608. 53 FX 0.052 FY 6.057 FZ 1.242 MX -0 MY -0 MZ -0
609. MEMBER LOAD
610. 66 UNI GZ 3.43
611. 68 UNI GZ 6.23
612. 70 UNI GZ 6.3
613. 72 1062 UNI GZ 0.349
614. LOAD 10 WIND LOAD (-Z) WL(-Z)
615. JOINT LOAD
616. *NODE1
617. 61 FX -1.293 FY -104.145 FZ -107.639 MX -0 MY -0 MZ -0
618. *NODE2
619. 62 FX 1.872 FY -121.047 FZ -118.487 MX -0 MY -0 MZ -0
620. *NODE3
621. 78 FX -0 FY -89.667 FZ -42.482 MX -0 MY -0 MZ -0
622. *NODE4
623. 43 FX -0.062 FY 88.261 FZ -5.046 MX -0 MY -0 MZ -0
624. *NODE5
625. 44 FX 4.777 FY 104.391 FZ -5.963 MX -0 MY -0 MZ -0
626. *NODE6
627. 45 FX -5.432 FY 75.585 FZ -1.041 MX -0 MY -0 MZ -0
628. *NODE7
629. 46 80 FX 0.148 FY -22.61 FZ -2.097 MX -0 MY -0 MZ -0
630. *NODE8
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631. 49 FX -0.072 FY -95.69 FZ -52.331 MX -0 MY -0 MZ -0
632. *NODE9
633. 56 FX 0.109 FY 16.787 FZ -2.345 MX -0 MY -0 MZ -0
634. *NODE10
635. 57 FX 0.029 FY 21.432 FZ -3.478 MX -0 MY -0 MZ -0
636. *NODE11
637. 50 FX 0.224 FY 27.051 FZ -3.41 MX -0 MY -0 MZ -0
638. *NODE12
639. 51 FX -0.196 FY 124.454 FZ -58.014 MX -0 MY -0 MZ -0
640. *NODE13
641. 52 FX -0.052 FY 5.824 FZ -1.242 MX -0 MY -0 MZ -0
642. *NODE14
643. 53 FX -0.052 FY 5.973 FZ -1.242 MX -0 MY -0 MZ -0
644. MEMBER LOAD
645. 95 UNI GZ -0.343
646. 97 UNI GZ -6.23
647. 103 UNI GZ -6.3
648. 105 UNI GZ -3.49
649. LOAD 15 TEMPERATURE LOAD (TL)
650. JOINT LOAD
651. *NODE1
652. 61 FX -21.247 FY -5.278 FZ 6.949 MX -0 MY -0 MZ -0
653. *NODE2
654. 62 FX 12.231 FY -7.433 FZ -24.035 MX -0 MY -0 MZ -0
655. *NODE3
656. 78 FX -0 FY -30.396 FZ -18.405 MX -0 MY -0 MZ -0
657. *NODE4
658. 43 FX -1.006 FY -3.96 FZ 4.297 MX -0 MY -0 MZ -0
659. *NODE5
660. 44 FX -49.677 FY 18.988 FZ 2.222 MX -0 MY -0 MZ -0
661. *NODE6
662. 45 FX 51.32 FY 29.749 FZ 2.073 MX -0 MY -0 MZ -0
663. *NODE7
664. 46 80 FX 0.769 FY 2.077 FZ -0.32 MX -0 MY -0 MZ -0
665. *NODE8
666. 49 FX 0.766 FY 16.891 FZ -39.777 MX -0 MY -0 MZ -0
667. *NODE9
668. 56 FX -2.931 FY -0.541 FZ 0.897 MX -0 MY -0 MZ -0
669. *NODE10
670. 57 FX 1.703 FY 3.571 FZ 0.854 MX -0 MY -0 MZ -0
671. *NODE11
672. 50 FX 5.815 FY -0.977 FZ 0.595 MX -0 MY -0 MZ -0
673. *NODE12
674. 51 FX 1.386 FY -22.69 FZ 64.651 MX -0 MY -0 MZ -0
675. *NODE13
676. 52 FX 0.4 FY 0.616 FZ -0 MX -0 MY -0 MZ -0
677. *NODE14
678. 53 FX 0.471 FY -0.616 FZ -0 MX -0 MY -0 MZ -0
679. LOAD 16 PIPE LATERAL LOAD (PL)
680. JOINT LOAD
681. *NODE1
682. 61 FX 6.264 FY 66.701 FZ 48.85 MX -0 MY -0 MZ -0
683. *NODE2
684. 62 FX 5.645 FY -7.08 FZ 8.122 MX -0 MY -0 MZ -0
685. *NODE3
686. 78 FX -0 FY -0.813 FZ -0.726 MX -0 MY -0 MZ -0

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687. *NODE4
688. 43 FX -0.176 FY -41.819 FZ 1.117 MX -0 MY -0 MZ -0
689. *NODE5
690. 44 FX 17.078 FY 14.146 FZ 0.171 MX -0 MY -0 MZ -0
691. *NODE6
692. 45 FX 12.85 FY -9.932 FZ 0.027 MX -0 MY -0 MZ -0
693. *NODE7
694. 46 80 FX 0.045 FY -12.078 FZ -0.003 MX -0 MY -0 MZ -0
695. *NODE8
696. 49 FX 0.152 FY -6.575 FZ -3.457 MX -0 MY -0 MZ -0
697. *NODE9
698. 56 FX 2.808 FY -1.396 FZ 0.107 MX -0 MY -0 MZ -0
699. *NODE10
700. 57 FX 2.932 FY -4.887 FZ 0.021 MX -0 MY -0 MZ -0
701. *NODE11
702. 50 FX 2.913 FY -1.132 FZ -0.007 MX -0 MY -0 MZ -0
703. *NODE12
704. 51 FX 0.282 FY 4.865 FZ -3.227 MX -0 MY -0 MZ -0
705. *NODE13
706. 52 FX 0.101 FY 0.143 FZ -0 MX -0 MY -0 MZ -0
707. *NODE14
708. 53 FX 0.101 FY -0.143 FZ -0 MX -0 MY -0 MZ -0
709. LOAD 17 EQP TEST LOAD (EQTL)
710. JOINT LOAD
711. *NODE1
712. 61 FX -19.154 FY -68.647 FZ -14.123 MX -0 MY -0 MZ -0
713. *NODE2
714. 62 FX 16.312 FY -261.784 FZ 4.384 MX -0 MY -0 MZ -0
715. *NODE3
716. 78 FX -0 FY 19.362 FZ 9.941 MX -0 MY -0 MZ -0
717. *NODE4
718. 43 FX -0.054 FY -1246.3 FZ -1.726 MX -0 MY -0 MZ -0
719. *NODE5
720. 44 FX -22.255 FY -687.581 FZ 3.325 MX -0 MY -0 MZ -0
721. *NODE6
722. 45 FX 17.173 FY -27.748 FZ -0.329 MX -0 MY -0 MZ -0
723. *NODE7
724. 46 80 FX -0.084 FY 6.935 FZ 0.019 MX -0 MY -0 MZ -0
725. *NODE8
726. 49 FX -0.106 FY -2.94 FZ -1.202 MX -0 MY -0 MZ -0
727. *NODE9
728. 56 FX 1.699 FY -932.705 FZ 1.306 MX -0 MY -0 MZ -0
729. *NODE10
730. 57 FX 3.27 FY -472.672 FZ 0.625 MX -0 MY -0 MZ -0
731. *NODE11
732. 50 FX 2.803 FY -5.159 FZ 0.117 MX -0 MY -0 MZ -0
733. *NODE12
734. 51 FX 0.257 FY 11.553 FZ -2.337 MX -0 MY -0 MZ -0
735. *NODE13
736. 52 FX 0.07 FY 0.099 FZ -0 MX -0 MY -0 MZ -0
737. *NODE14
738. 53 FX 0.07 FY -0.099 FZ -0 MX -0 MY -0 MZ -0
739. LOAD 18 MONORAIL LOAD (ML1)
740. JOINT LOAD
741. *NODE1
742. 61 FX 0.039 FY 1.375 FZ -0.379 MX -0 MY -0 MZ -0
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743. *NODE2
744. 62 FX -0.116 FY 0.542 FZ 0.291 MX -0 MY -0 MZ -0
745. *NODE3
746. 78 FX -0 FY 0.101 FZ 0.039 MX -0 MY -0 MZ -0
747. *NODE4
748. 43 FX 0.002 FY -7.359 FZ -0.06 MX -0 MY -0 MZ -0
749. *NODE5
750. 44 FX -0.116 FY -0.404 FZ -0 MX -0 MY -0 MZ -0
751. *NODE6
752. 45 FX -0.023 FY -0.17 FZ -0.001 MX -0 MY -0 MZ -0
753. *NODE7
754. 46 80 FX -0 FY 0.207 FZ -0 MX -0 MY -0 MZ -0
755. *NODE8
756. 49 FX 0.004 FY 0.044 FZ 0.034 MX -0 MY -0 MZ -0
757. *NODE9
758. 56 FX 0.077 FY -22.698 FZ 0.083 MX -0 MY -0 MZ -0
759. *NODE10
760. 57 FX 0.07 FY -1.171 FZ -0.001 MX -0 MY -0 MZ -0
761. *NODE11
762. 50 FX 0.053 FY -0.073 FZ -0 MX -0 MY -0 MZ -0
763. *NODE12
764. 51 FX 0.009 FY 0.185 FZ -0.007 MX -0 MY -0 MZ -0
765. *NODE13
766. 52 FX 0.001 FY 0.001 FZ -0 MX -0 MY -0 MZ -0
767. *NODE14
768. 53 FX 0.001 FY -0.001 FZ -0 MX -0 MY -0 MZ -0
769. LOAD 19 MONORAIL LOAD (ML2)
770. JOINT LOAD
771. *NODE1
772. 61 FX 0.015 FY 0.905 FZ -0.173 MX -0 MY -0 MZ -0
773. *NODE2
774. 62 FX -0.037 FY -0.283 FZ 0.156 MX -0 MY -0 MZ -0
775. *NODE3
776. 78 FX -0 FY 0.06 FZ 0.05 MX -0 MY -0 MZ -0
777. *NODE4
778. 43 FX 0.001 FY -4.506 FZ -0.034 MX -0 MY -0 MZ -0
779. *NODE5
780. 44 FX -0.068 FY -1.066 FZ 0.035 MX -0 MY -0 MZ -0
781. *NODE6
782. 45 FX 0.014 FY -0.126 FZ -0.002 MX -0 MY -0 MZ -0
783. *NODE7
784. 46 80 FX -0 FY 0.009 FZ -0 MX -0 MY -0 MZ -0
785. *NODE8
786. 49 FX -0.002 FY -0.048 FZ -0.016 MX -0 MY -0 MZ -0
787. *NODE9
788. 56 FX 0.012 FY -14.288 FZ 0.005 MX -0 MY -0 MZ -0
789. *NODE10
790. 57 FX 0.031 FY -10.136 FZ 0.007 MX -0 MY -0 MZ -0
791. *NODE11
792. 50 FX 0.032 FY -0.104 FZ 0.001 MX -0 MY -0 MZ -0
793. *NODE12
794. 51 FX 0.002 FY 0.162 FZ -0.031 MX -0 MY -0 MZ -0
795. *NODE13
796. 52 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
797. *NODE14
798. 53 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
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799. LOAD 20 MONORAIL LOAD (ML3)
800. JOINT LOAD
801. *NODE1
802. 61 FX -0.007 FY 0.418 FZ -0.047 MX -0 MY -0 MZ -0
803. *NODE2
804. 62 FX 0.038 FY -1.091 FZ 0.029 MX -0 MY -0 MZ -0
805. *NODE3
806. 78 FX -0 FY 0.029 FZ 0.062 MX -0 MY -0 MZ -0
807. *NODE4
808. 43 FX -0 FY -1.674 FZ -0.012 MX -0 MY -0 MZ -0
809. *NODE5
810. 44 FX -0.027 FY -1.737 FZ 0.07 MX -0 MY -0 MZ -0
811. *NODE6
812. 45 FX 0.053 FY -0.087 FZ -0.003 MX -0 MY -0 MZ -0
813. *NODE7
814. 46 80 FX -0 FY -0.164 FZ 0.001 MX -0 MY -0 MZ -0
815. *NODE8
816. 49 FX -0.007 FY -0.139 FZ -0.066 MX -0 MY -0 MZ -0
817. *NODE9
818. 56 FX -0.051 FY -5.832 FZ 0.003 MX -0 MY -0 MZ -0
819. *NODE10
820. 57 FX -0.004 FY -19.12 FZ 0.014 MX -0 MY -0 MZ -0
821. *NODE11
822. 50 FX 0.013 FY -0.141 FZ 0.003 MX -0 MY -0 MZ -0
823. *NODE12
824. 51 FX -0.006 FY 0.118 FZ -0.053 MX -0 MY -0 MZ -0
825. *NODE13
826. 52 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
827. *NODE14
828. 53 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
829. LOAD 21 MONORAIL LOAD (ML4)
830. JOINT LOAD
831. *NODE1
832. 61 FX -0.042 FY 0.233 FZ 0.151 MX -0 MY -0 MZ -0
833. *NODE2
834. 62 FX 0.075 FY -2.009 FZ -0.342 MX -0 MY -0 MZ -0
835. *NODE3
836. 78 FX -0 FY 0.184 FZ 0.073 MX -0 MY -0 MZ -0
837. *NODE4
838. 43 FX -0 FY -0.04 FZ 0.003 MX -0 MY -0 MZ -0
839. *NODE5
840. 44 FX -0.02 FY -2.078 FZ 0.096 MX -0 MY -0 MZ -0
841. *NODE6
842. 45 FX 0.044 FY -0.062 FZ -0.002 MX -0 MY -0 MZ -0
843. *NODE7
844. 46 80 FX -0 FY -0.069 FZ -0 MX -0 MY -0 MZ -0
845. *NODE8
846. 49 FX -0.003 FY -0.028 FZ -0.03 MX -0 MY -0 MZ -0
847. *NODE9
848. 56 FX -0.065 FY -1.073 FZ 0.022 MX -0 MY -0 MZ -0
849. *NODE10
850. 57 FX -0.004 FY -23.909 FZ 0.01 MX -0 MY -0 MZ -0
851. *NODE11
852. 50 FX 0.023 FY -0.337 FZ -0 MX -0 MY -0 MZ -0
853. *NODE12
854. 51 FX -0.008 FY -0.231 FZ 0.019 MX -0 MY -0 MZ -0
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855. *NODE13
856. 52 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
857. *NODE14
858. 53 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
859. LOAD 22 MONORAIL LOAD (ML5)
860. JOINT LOAD
861. *NODE1
862. 61 FX -0.018 FY 0.052 FZ 0.021 MX -0 MY -0 MZ -0
863. *NODE2
864. 62 FX 0.04 FY -0.87 FZ -0.044 MX -0 MY -0 MZ -0
865. *NODE3
866. 78 FX -0 FY -0.955 FZ 0.06 MX -0 MY -0 MZ -0
867. *NODE4
868. 43 FX 0.001 FY -0.062 FZ 0.001 MX -0 MY -0 MZ -0
869. *NODE5
870. 44 FX -0.046 FY -1.035 FZ 0.049 MX -0 MY -0 MZ -0
871. *NODE6
872. 45 FX 0.1 FY -1.027 FZ -0.006 MX -0 MY -0 MZ -0
873. *NODE7
874. 46 80 FX 0.001 FY -0.145 FZ 0.001 MX -0 MY -0 MZ -0
875. *NODE8
876. 49 FX -0 FY -0.148 FZ -0.059 MX -0 MY -0 MZ -0
877. *NODE9
878. 56 FX -0.041 FY -0.436 FZ -0.002 MX -0 MY -0 MZ -0
879. *NODE10
880. 57 FX -0.017 FY -12.217 FZ 0.011 MX -0 MY -0 MZ -0
881. *NODE11
882. 50 FX -0.023 FY -12.575 FZ 0.015 MX -0 MY -0 MZ -0
883. *NODE12
884. 51 FX -0.001 FY -0.003 FZ -0.045 MX -0 MY -0 MZ -0
885. *NODE13
886. 52 FX 0.001 FY 0.002 FZ -0 MX -0 MY -0 MZ -0
887. *NODE14
888. 53 FX 0.001 FY -0.002 FZ -0 MX -0 MY -0 MZ -0
889. LOAD 23 MONORAIL LOAD (ML6)
890. JOINT LOAD
891. *NODE1
892. 61 FX -0.011 FY 0.003 FZ 0.057 MX -0 MY -0 MZ -0
893. *NODE2
894. 62 FX -0.004 FY -0.074 FZ -0.034 MX -0 MY -0 MZ -0
895. *NODE3
896. 78 FX -0 FY -2.398 FZ -0.061 MX -0 MY -0 MZ -0
897. *NODE4
898. 43 FX 0.003 FY -0.067 FZ 0.002 MX -0 MY -0 MZ -0
899. *NODE5
900. 44 FX -0.071 FY 0.099 FZ 0.002 MX -0 MY -0 MZ -0
901. *NODE6
902. 45 FX 0.12 FY -1.884 FZ -0.006 MX -0 MY -0 MZ -0
903. *NODE7
904. 46 80 FX 0.003 FY 0.065 FZ -0 MX -0 MY -0 MZ -0
905. *NODE8
906. 49 FX 0.011 FY 0.09 FZ 0.005 MX -0 MY -0 MZ -0
907. *NODE9
908. 56 FX 0.002 FY 0.189 FZ -0.002 MX -0 MY -0 MZ -0
909. *NODE10
910. 57 FX -0.018 FY -0.423 FZ 0.001 MX -0 MY -0 MZ -0
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911. *NODE11
912. 50 FX -0.048 FY -24.725 FZ 0.021 MX -0 MY -0 MZ -0
913. *NODE12
914. 51 FX 0.007 FY -0.294 FZ 0.014 MX -0 MY -0 MZ -0
915. *NODE13
916. 52 FX 0.003 FY 0.004 FZ -0 MX -0 MY -0 MZ -0
917. *NODE14
918. 53 FX 0.003 FY -0.004 FZ -0 MX -0 MY -0 MZ -0
919. LOAD 24 MONORAIL LOAD (ML7)
920. JOINT LOAD
921. *NODE1
922. 61 FX -0.017 FY 0.016 FZ 0.067 MX -0 MY -0 MZ -0
923. *NODE2
924. 62 FX -0.007 FY -0.218 FZ -0.148 MX -0 MY -0 MZ -0
925. *NODE3
926. 78 FX -0 FY -1.134 FZ -0.017 MX -0 MY -0 MZ -0
927. *NODE4
928. 43 FX 0.002 FY -0.08 FZ 0.003 MX -0 MY -0 MZ -0
929. *NODE5
930. 44 FX -0.058 FY 0.091 FZ 0.003 MX -0 MY -0 MZ -0
931. *NODE6
932. 45 FX 0.038 FY -0.934 FZ -0.003 MX -0 MY -0 MZ -0
933. *NODE7
934. 46 80 FX 0.001 FY -0.514 FZ 0.015 MX -0 MY -0 MZ -0
935. *NODE8
936. 49 FX 0.006 FY -3.226 FZ -0.784 MX -0 MY -0 MZ -0
937. *NODE9
938. 56 FX 0.02 FY 0.242 FZ -0.004 MX -0 MY -0 MZ -0
939. *NODE10
940. 57 FX 0.009 FY -0.201 FZ 0.002 MX -0 MY -0 MZ -0
941. *NODE11
942. 50 FX -0.001 FY -12.519 FZ 0.015 MX -0 MY -0 MZ -0
943. *NODE12
944. 51 FX 0.006 FY -10.941 FZ 0.852 MX -0 MY -0 MZ -0
945. *NODE13
946. 52 FX 0.001 FY 0.002 FZ -0 MX -0 MY -0 MZ -0
947. *NODE14
948. 53 FX 0.001 FY -0.002 FZ -0 MX -0 MY -0 MZ -0
949. LOAD 25 MONORAIL LOAD (ML8)
950. JOINT LOAD
951. *NODE1
952. 61 FX -0.003 FY 0.038 FZ 0.05 MX -0 MY -0 MZ -0
953. *NODE2
954. 62 FX -0.005 FY -0.048 FZ -0.1 MX -0 MY -0 MZ -0
955. *NODE3
956. 78 FX -0 FY -0.183 FZ -0.024 MX -0 MY -0 MZ -0
957. *NODE4
958. 43 FX -0.001 FY -0.04 FZ 0.001 MX -0 MY -0 MZ -0
959. *NODE5
960. 44 FX -0.011 FY 0.076 FZ -0.003 MX -0 MY -0 MZ -0
961. *NODE6
962. 45 FX -0.019 FY 0.04 FZ -0 MX -0 MY -0 MZ -0
963. *NODE7
964. 46 80 FX -0.002 FY -1.232 FZ 0.03 MX -0 MY -0 MZ -0
965. *NODE8
966. 49 FX -0.006 FY -6.637 FZ -1.596 MX -0 MY -0 MZ -0
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 967. *NODE9
 968. 56 FX 0.013 FY 0.154 FZ -0.001 MX -0 MY -0 MZ -0
 969. *NODE10
 970. 57 FX 0.01 FY -0.173 FZ -0.001 MX -0 MY -0 MZ -0
 971. *NODE11
 972. 50 FX 0.021 FY -0.021 FZ 0.002 MX -0 MY -0 MZ -0
 973. *NODE12
 974. 51 FX 0.004 FY -21.394 FZ 1.643 MX -0 MY -0 MZ -0
 975. *NODE13
 976. 52 FX -0.001 FY -0.001 FZ -0 MX -0 MY -0 MZ -0
 977. *NODE14
 978. 53 FX -0.001 FY 0.001 FZ -0 MX -0 MY -0 MZ -0
 979. LOAD 26 MONORAIL LOAD (ML9)
 980. JOINT LOAD
 981. *NODE1
 982. 61 FX 0.006 FY 0.051 FZ 0.04 MX -0 MY -0 MZ -0
 983. *NODE2
 984. 62 FX -0.004 FY 0.049 FZ -0.073 MX -0 MY -0 MZ -0
 985. *NODE3
 986. 78 FX -0 FY 0.359 FZ -0.028 MX -0 MY -0 MZ -0
 987. *NODE4
 988. 43 FX -0.002 FY -0.016 FZ 0.001 MX -0 MY -0 MZ -0
 989. *NODE5
 990. 44 FX 0.016 FY 0.068 FZ -0.007 MX -0 MY -0 MZ -0
 991. *NODE6
 992. 45 FX -0.052 FY 0.595 FZ 0.002 MX -0 MY -0 MZ -0
 993. *NODE7
 994. 46 80 FX -0.003 FY -1.642 FZ 0.039 MX -0 MY -0 MZ -0
 995. *NODE8
 996. 49 FX -0.013 FY -8.582 FZ -2.06 MX -0 MY -0 MZ -0
 997. *NODE9
 998. 56 FX 0.009 FY 0.104 FZ -0 MX -0 MY -0 MZ -0
 999. *NODE10
1000. 57 FX 0.011 FY -0.157 FZ -0.002 MX -0 MY -0 MZ -0
1001. *NODE11
1002. 50 FX 0.034 FY 7.103 FZ -0.006 MX -0 MY -0 MZ -0
1003. *NODE12
1004. 51 FX 0.002 FY -27.352 FZ 2.094 MX -0 MY -0 MZ -0
1005. *NODE13
1006. 52 FX -0.002 FY -0.003 FZ -0 MX -0 MY -0 MZ -0
1007. *NODE14
1008. 53 FX -0.002 FY 0.003 FZ -0 MX -0 MY -0 MZ -0
1009. LOAD 27 MONORAIL LOAD (ML10)
1010. JOINT LOAD
1011. *NODE1
1012. 61 FX -0.047 FY 0.385 FZ -0.218 MX -0 MY -0 MZ -0
1013. *NODE2
1014. 62 FX 0.127 FY -2.85 FZ 0.199 MX -0 MY -0 MZ -0
1015. *NODE3
1016. 78 FX -0 FY -0.014 FZ 0.127 MX -0 MY -0 MZ -0
1017. *NODE4
1018. 43 FX 0.001 FY -3.094 FZ -0.017 MX -0 MY -0 MZ -0
1019. *NODE5
1020. 44 FX -0.06 FY -3.174 FZ 0.176 MX -0 MY -0 MZ -0
1021. *NODE6
1022. 45 FX 0.1 FY -0.203 FZ -0.006 MX -0 MY -0 MZ -0
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*NODE7
46 80 FX -0.001 FY -0.349 FZ 0.002 MX -0 MY -0 MZ -0
*NODE8
49 FX -0.015 FY -0.346 FZ -0.149 MX -0 MY -0 MZ -0
*NODE9
56 FX -0.064 FY -4.267 FZ 0.007 MX -0 MY -0 MZ -0
*NODE10
57 FX -0.021 FY -16.081 FZ 0.035 MX -0 MY -0 MZ -0
*NODE11
50 FX -0.014 FY -0.004 FZ 0.007 MX -0 MY -0 MZ -0
*NODE12
51 FX -0.005 FY 0.576 FZ -0.163 MX -0 MY -0 MZ -0
*NODE13
52 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
*NODE14
53 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
LOAD 28 MONORAIL LOAD (ML11)
JOINT LOAD
*NODE1
61 FX -0.01 FY 0.415 FZ -0.06 MX -0 MY -0 MZ -0
*NODE2
62 FX 0.045 FY -1.229 FZ 0.042 MX -0 MY -0 MZ -0
*NODE3
78 FX -0 FY 0.026 FZ 0.067 MX -0 MY -0 MZ -0
*NODE4
43 FX -0 FY -1.786 FZ -0.013 MX -0 MY -0 MZ -0
*NODE5
44 FX -0.03 FY -1.85 FZ 0.078 MX -0 MY -0 MZ -0
*NODE6
45 FX 0.056 FY -0.096 FZ -0.003 MX -0 MY -0 MZ -0
*NODE7
46 80 FX -0 FY -0.178 FZ 0.001 MX -0 MY -0 MZ -0
*NODE8
49 FX -0.008 FY -0.155 FZ -0.072 MX -0 MY -0 MZ -0
*NODE9
56 FX -0.052 FY -5.709 FZ 0.003 MX -0 MY -0 MZ -0
*NODE10
57 FX -0.006 FY -18.881 FZ 0.016 MX -0 MY -0 MZ -0
*NODE11
50 FX 0.011 FY -0.13 FZ 0.003 MX -0 MY -0 MZ -0
*NODE12
51 FX -0.006 FY 0.154 FZ -0.062 MX -0 MY -0 MZ -0
*NODE13
52 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
*NODE14
53 FX -0 FY -0 FZ -0 MX -0 MY -0 MZ -0
********************************************************************************
* FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4(DL+EQDL+FL)
1 1.4 4 1.4 3 1.4
LOAD COMB 102 1.4(DL+EQDL+FL)+1.7(LL+EQLL)
1 1.4 4 1.4 3 1.4 2 1.7 5 1.7
LOAD COMB 103 1.4(DL+EQDL+FL)+1.7(LL+SBT)
1 1.4 4 1.4 3 1.4 2 1.7 6 1.7
LOAD COMB 104 1.4(DL+EQDL+FL+TL)+1.7(LL+EQLL)
1 1.4 4 1.4 3 1.4 15 1.4 2 1.7 5 1.7
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1079. LOAD COMB 105 1.4(DL+EQDL+FL+TL)+1.7(LL+SBT)
1080. 1 1.4 4 1.4 3 1.4 15 1.4 2 1.7 6 1.7
1081. LOAD COMB 106 1.4(DL+EQDL+FL+PL)
1082. 1 1.4 4 1.4 3 1.4 16 1.4
1083. LOAD COMB 107 1.4(DL+EQDL+FL+PL)+1.7(LL+EQLL)
1084. 1 1.4 4 1.4 3 1.4 16 1.4 2 1.7 5 1.7
1085. LOAD COMB 108 1.4(DL+EQDL+FL+PL)+1.7(LL+SBT)
1086. 1 1.4 4 1.4 3 1.4 16 1.4 2 1.7 6 1.7
1087. LOAD COMB 109 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+EQLL)
1088. 1 1.4 4 1.4 3 1.4 16 1.4 15 1.4 2 1.7 5 1.7
1089. LOAD COMB 110 1.4(DL+EQDL+FL+PL+TL)+1.7(LL+SBT)
1090. 1 1.4 4 1.4 3 1.4 16 1.4 15 1.4 2 1.7 6 1.7
1091. LOAD COMB 111 1.2(DL+EQDL)+0.8WL(+X)
1092. 1 1.2 4 1.2 7 0.8
1093. LOAD COMB 112 1.2(DL+EQDL)+0.8WL(-X)
1094. 1 1.2 4 1.2 8 0.8
1095. LOAD COMB 113 1.2(DL+EQDL)+0.8WL(+Z)
1096. 1 1.2 4 1.2 9 0.8
1097. LOAD COMB 114 1.2(DL+EQDL)+0.8WL(-Z)
1098. 1 1.2 4 1.2 10 0.8
1099. LOAD COMB 115 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
1100. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
1101. LOAD COMB 116 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
1102. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
1103. LOAD COMB 117 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
1104. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
1105. LOAD COMB 118 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
1106. 1 1.2 4 1.2 10 1.6 2 0.5 5 0.5
1107. LOAD COMB 119 1.2(DL+EQDL)+1.0SL(+X)+0.5(LL+EQLL)+0.5FL
1108. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
1109. LOAD COMB 120 1.2(DL+EQDL)+1.0SL(-X)+0.5(LL+EQLL)+0.5FL
1110. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
1111. LOAD COMB 121 1.2(DL+EQDL)+1.0SL(+Z)+0.5(LL+EQLL)+0.5FL
1112. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
1113. LOAD COMB 122 1.2(DL+EQDL)+1.0SL(-Z)+0.5(LL+EQLL)+0.5FL
1114. 1 1.2 4 1.2 14 1.0 2 0.5 5 0.5 3 0.5
1115. LOAD COMB 123 0.9(DL+EQDL)+1.6WL(+X)
1116. 1 0.9 4 0.9 7 1.6
1117. LOAD COMB 124 0.9(DL+EQDL)+1.6WL(-X)
1118. 1 0.9 4 0.9 8 1.6
1119. LOAD COMB 125 0.9(DL+EQDL)+1.6WL(+Z)
1120. 1 0.9 4 0.9 9 1.6
1121. LOAD COMB 126 0.9(DL+EQDL)+1.6WL(-Z)
1122. 1 0.9 4 0.9 10 1.6
1123. LOAD COMB 127 0.9(DL+EQDL)+1.0SL(+X)
1124. 1 0.9 4 0.9 11 1.0
1125. LOAD COMB 128 0.9(DL+EQDL)+1.0SL(-X)
1126. 1 0.9 4 0.9 12 1.0
1127. LOAD COMB 129 0.9(DL+EQDL)+1.0SL(+Z)
1128. 1 0.9 4 0.9 13 1.0
1129. LOAD COMB 130 0.9(DL+EQDL)+1.0SL(-Z)
1130. 1 0.9 4 0.9 14 1.0
1131. LOAD COMB 131 1.2DL+EQTL
1132. 1 1.2 17 1.0
1133. LOAD COMB 132 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML1
1134. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 18 1.7
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1135. LOAD COMB 133 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML2
1136. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 19 1.7
1137. LOAD COMB 134 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML3
1138. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 20 1.7
1139. LOAD COMB 135 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML4
1140. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 21 1.7
1141. LOAD COMB 136 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML5
1142. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 22 1.7
1143. LOAD COMB 137 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML6
1144. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 23 1.7
1145. LOAD COMB 138 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML7
1146. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 24 1.7
1147. LOAD COMB 139 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML8
1148. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 25 1.7
1149. LOAD COMB 140 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML9
1150. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 26 1.7
1151. LOAD COMB 141 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML10
1152. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 27 1.7
1153. LOAD COMB 142 1.4(DL+EQDL+FL)+1.7(LL+EQLL)+1.7ML11
1154. 1 1.4 4 1.4 3 1.4 2 1.7 5 1.7 28 1.7
1155. * UNFACTORED LOAD COMBINATIONS FOR DEFLECTION CHECK
1156. LOAD COMB 201 DL+EQDL+FL
1157. 1 1.0 4 1.0 3 1.0
1158. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
1159. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 15 1.0
1160. LOAD COMB 203 DL+EQDL+FL+LL+SBT+TL
1161. 1 1.0 4 1.0 3 1.0 2 1.0 6 1.0 15 1.0
1162. LOAD COMB 204 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
1163. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 15 0.75
1164. LOAD COMB 205 DL+EQDL+FL+0.75(LL+SBT)+0.75TL
1165. 1 1.0 4 1.0 3 1.0 2 0.75 6 0.75 15 0.75
1166. LOAD COMB 206 DL+EQDL+FL+PL
1167. 1 1.0 4 1.0 3 1.0 16 1.0
1168. LOAD COMB 207 DL+EQDL+FL+PL+LL+EQLL+TL
1169. 1 1.0 4 1.0 3 1.0 16 1.0 2 1.0 5 1.0 15 1.0
1170. LOAD COMB 208 DL+EQDL+FL+PL+LL+SBT+TL
1171. 1 1.0 4 1.0 3 1.0 16 1.0 2 1.0 6 1.0 15 1.0
1172. LOAD COMB 209 DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL
1173. 1 1.0 4 1.0 3 1.0 16 1.0 2 0.75 5 0.75 15 0.75
1174. LOAD COMB 210 DL+EQDL+FL+PL+0.75(LL+SBT)+0.75TL
1175. 1 1.0 4 1.0 3 1.0 16 1.0 2 0.75 6 0.75 15 0.75
1176. LOAD COMB 211 DL+EQDL+FL+WL(+X)
1177. 1 1.0 4 1.0 3 1.0 7 1.0
1178. LOAD COMB 212 DL+EQDL+FL+WL(-X)
1179. 1 1.0 4 1.0 3 1.0 8 1.0
1180. LOAD COMB 213 DL+EQDL+FL+WL(+Z)
1181. 1 1.0 4 1.0 3 1.0 9 1.0
1182. LOAD COMB 214 DL+EQDL+FL+WL(-Z)
1183. 1 1.0 4 1.0 3 1.0 10 1.0
1184. LOAD COMB 215 DL+EQDL+FL+0.7SL(+X)
1185. 1 1.0 4 1.0 3 1.0 11 0.7
1186. LOAD COMB 216 DL+EQDL+FL+0.7SL(-X)
1187. 1 1.0 4 1.0 3 1.0 12 0.7
1188. LOAD COMB 217 DL+EQDL+FL+0.7SL(+Z)
1189. 1 1.0 4 1.0 3 1.0 13 0.7
1190. LOAD COMB 218 DL+EQDL+FL+0.7SL(-Z)
```

1191. 1 1.0 4 1.0 3 1.0 14 0.7
1192. LOAD COMB 219 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
1193. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
1194. LOAD COMB 220 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
1195. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
1196. LOAD COMB 221 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
1197. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
1198. LOAD COMB 222 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
1199. 1 1.0 4 1.0 3 1.0 10 0.75 2 0.75 5 0.75
1200. LOAD COMB 223 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
1201. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
1202. LOAD COMB 224 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
1203. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
1204. LOAD COMB 225 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
1205. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
1206. LOAD COMB 226 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
1207. 1 1.0 4 1.0 3 1.0 14 0.525 2 0.75 5 0.75
1208. LOAD COMB 227 0.6(DL+EQDL)+WL(+X)
1209. 1 0.6 4 0.6 7 1.0
1210. LOAD COMB 228 0.6(DL+EQDL)+WL(-X)
1211. 1 0.6 4 0.6 8 1.0
1212. LOAD COMB 229 0.6(DL+EQDL)+WL(+Z)
1213. 1 0.6 4 0.6 9 1.0
1214. LOAD COMB 230 0.6(DL+EQDL)+WL(-Z)
1215. 1 0.6 4 0.6 10 1.0
1216. LOAD COMB 231 0.6(DL+EQDL)+0.7SL(+X)
1217. 1 0.6 4 0.6 11 0.7
1218. LOAD COMB 232 0.6(DL+EQDL)+0.7SL(-X)
1219. 1 0.6 4 0.6 12 0.7
1220. LOAD COMB 233 0.6(DL+EQDL)+0.7SL(+Z)
1221. 1 0.6 4 0.6 13 0.7
1222. LOAD COMB 234 0.6(DL+EQDL)+0.7SL(-Z)
1223. 1 0.6 4 0.6 14 0.7
1224. LOAD COMB 235 DL+EQTL
1225. 1 1.0 17 1.0
1226. LOAD COMB 236 DL+EQDL+FL+LL+EQLL+ML1
1227. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 18 1.0
1228. LOAD COMB 237 DL+EQDL+FL+LL+EQLL+ML2
1229. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 19 1.0
1230. LOAD COMB 238 DL+EQDL+FL+LL+EQLL+ML3
1231. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 20 1.0
1232. LOAD COMB 239 DL+EQDL+FL+LL+EQLL+ML4
1233. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 21 1.0
1234. LOAD COMB 240 DL+EQDL+FL+LL+EQLL+ML5
1235. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 22 1.0
1236. LOAD COMB 241 DL+EQDL+FL+LL+EQLL+ML6
1237. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 23 1.0
1238. LOAD COMB 242 DL+EQDL+FL+LL+EQLL+ML7
1239. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 24 1.0
1240. LOAD COMB 243 DL+EQDL+FL+LL+EQLL+ML8
1241. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 25 1.0
1242. LOAD COMB 244 DL+EQDL+FL+LL+EQLL+ML9
1243. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 26 1.0
1244. LOAD COMB 245 DL+EQDL+FL+LL+EQLL+ML10
1245. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 27 1.0
1246. LOAD COMB 246 DL+EQDL+FL+LL+EQLL+ML11

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1247. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 28 1.0
1248. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**84**	**NUMBER OF MEMBERS**	**135**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**19**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 79/ 16/ 178 DOF
TOTAL PRIMARY LOAD CASES = 28, TOTAL DEGREES OF FREEDOM = 315
TOTAL LOAD COMBINATION CASES = 88 SO FAR.
SIZE OF STIFFNESS MATRIX = 57 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.1/ 197058.8 MB

*WARNING- APPLIED SELFWEIGHT IS MORE THAN TOTAL WEIGHT OF ALL
STRUCTURAL ELEMENTS IN LOAD CASE 1 ALONG Y.
THIS COULD BE DUE TO SELFWEIGHT APPLIED TO SPECIFIC
LIST OF MEMBERS/PLATES/SOLIDS/SURFACES.
TOTAL UNFACTORED WEIGHT OF THE STRUCTURE = 8043.475 KN
TOTAL UNFACTORED WEIGHT OF THE STRUCTURE APPLIED = 8133.475 KN

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS _CHANGE.

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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.018 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.220 T =  0.490 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.374                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.014 X     17665.18   *
*                       =       245.61  KN                 *
************************************************************
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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.018 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.220 T =  0.490 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.374                           *
*  LOAD FACTOR          = -1.000                           *
*  DESIGN BASE SHEAR    = -1.000 X  0.014 X     17665.18   *
*                       =      -245.61  KN                 *
************************************************************
```

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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.018 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.220 T =  0.414 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.374                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.014 X     17665.18   *
*                       =       245.61  KN                 *
************************************************************
```

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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.018 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.220 T =  0.414 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.374                           *
*  LOAD FACTOR          = -1.000                           *
*  DESIGN BASE SHEAR    = -1.000 X  0.014 X     17665.18   *
*                       =      -245.61  KN                 *
************************************************************
```

*** LOADING 11 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 245.61 FORCE-Y = 0.00 FORCE-Z = 0.00
MOM-X = 0.00 MOM-Y = 3411.86 MOM-Z = -225479.86

*** LOADING 12 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = -245.61 FORCE-Y = 0.00 FORCE-Z = 0.00
MOM-X = 0.00 MOM-Y = -3411.86 MOM-Z = 225479.86

*** LOADING 13 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 0.00 FORCE-Y = 0.00 FORCE-Z = 245.61
MOM-X = 225479.86 MOM-Y = -2543.45 MOM-Z = 0.00

*** LOADING 14 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 0.00 FORCE-Y = 0.00 FORCE-Z = -245.61
MOM-X = -225479.86 MOM-Y = 2543.45 MOM-Z = 0.00

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 11
SEISMIC LOAD SL(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.132495372E+02
Y = 0.918347983E+03
Z = 0.161462912E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.713175074E+01
Y = 0.918550012E+03
Z = 0.126503614E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = 619.82
SUMMATION FORCE-Y = -84.82
SUMMATION FORCE-Z = 276.53

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 256803.27 MY= 6136.45 MZ= -574190.21

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = -619.82
SUMMATION FORCE-Y = 84.82
SUMMATION FORCE-Z = -276.53

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -256803.22 MY= -6136.45 MZ= 574190.10

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 11)

	MAXIMUMS	AT NODE
X =	3.02657E-01	55
Y =	-7.46253E-02	80
Z =	1.11183E-01	83
RX=	1.24738E-04	98
RY=	2.48014E-05	91
RZ=	-5.21736E-04	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 12
SEISMIC LOAD SL(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.132495372E+02
Y = 0.918347983E+03
Z = 0.161462912E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.713175074E+01
Y = 0.918550012E+03
Z = 0.126503614E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = -619.82
SUMMATION FORCE-Y = 84.82
SUMMATION FORCE-Z = -276.53

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -256803.27 MY= -6136.45 MZ= 574190.21

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = 619.82
SUMMATION FORCE-Y = -84.82
SUMMATION FORCE-Z = 276.53

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 256803.22 MY= 6136.45 MZ= -574190.10

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 12)

	MAXIMUMS	AT NODE
X =	-3.02657E-01	55
Y =	7.46253E-02	80
Z =	-1.11183E-01	83
RX=	-1.24738E-04	98
RY=	-2.48014E-05	91
RZ=	5.21736E-04	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 13
SEISMIC LOAD SL(+Z)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.140096172E+02
Y = 0.918550012E+03
Z = 0.232371124E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.850627977E+01
Y = 0.918347824E+03
Z = 0.130874815E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = 276.87
SUMMATION FORCE-Y = -57.68
SUMMATION FORCE-Z = 619.33

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 572211.17 MY= -1767.82 MZ= -257859.21

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = -276.87
SUMMATION FORCE-Y = 57.68
SUMMATION FORCE-Z = -619.33

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -572211.07 MY= 1767.82 MZ= 257859.17

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 13)

	MAXIMUMS	AT NODE
X =	1.44500E-01	55
Y =	-4.45717E-02	80
Z =	2.05831E-01	86
RX=	2.84189E-04	79
RY=	1.46420E-05	96
RZ=	-3.07025E-04	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 14
SEISMIC LOAD SL(-Z)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.140096172E+02
Y = 0.918550012E+03
Z = 0.232371124E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.850627977E+01
Y = 0.918347824E+03
Z = 0.130874815E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = -276.87
SUMMATION FORCE-Y = 57.68
SUMMATION FORCE-Z = -619.33

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -572211.17 MY= 1767.82 MZ= 257859.21

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = 276.87
SUMMATION FORCE-Y = -57.68
SUMMATION FORCE-Z = 619.33

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 572211.07 MY= -1767.82 MZ= -257859.17

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 14)

	MAXIMUMS	AT NODE
X =	-1.44500E-01	55
Y =	4.45717E-02	80
Z =	-2.05831E-01	86
RX=	-2.84189E-04	79
RY=	-1.46420E-05	96
RZ=	3.07025E-04	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
DEAD LOAD (DL)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.224000033E+02
Y = 0.918550064E+03
Z = 0.640017228E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106725060E+02
Y = 0.916623264E+03
Z = 0.128573079E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.224177506E+02
Y = 0.966170709E+03
Z = 0.640004478E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -2.65
SUMMATION FORCE-Y = -15931.74
SUMMATION FORCE-Z = 0.03

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 204863.21 MY= -17.54 MZ= -167598.38

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 2.65
SUMMATION FORCE-Y = 15931.74
SUMMATION FORCE-Z = -0.03

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -204863.21 MY= 17.54 MZ= 167598.39

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	4.05834E-02	79
Y =	-4.26684E-01	68
Z =	-1.38027E-02	81
RX=	-7.61011E-04	70
RY=	1.63180E-05	18
RZ=	8.78039E-04	46

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LIVE LOAD (LL)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.224000017E+02
Y = 0.918550033E+03
Z = 0.640074952E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.905630673E+01
Y = 0.918539619E+03
Z = 0.159075924E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.225886242E+02
Y = 0.929554664E+03
Z = 0.631683237E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -7.68
SUMMATION FORCE-Y = -9345.31
SUMMATION FORCE-Z = 0.08

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 148730.30 MY= -50.82 MZ= -77583.21

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 7.68
SUMMATION FORCE-Y = 9345.31
SUMMATION FORCE-Z = -0.08

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -148730.30 MY= 50.82 MZ= 77583.22

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	-2.85160E-02	97
Y =	-5.56494E-01	57
Z =	2.78624E-02	16
RX=	-1.17843E-03	54
RY=	-2.14355E-05	98
RZ=	-2.35546E-03	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
OPERATING FLUID LOAD (FL)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.140046092E+02
Y = 0.918550012E+03
Z = 0.230409982E+02

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.366756913E+01
Y = 0.918550012E+03
Z = 0.814229058E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.359969893E+01
Y = 0.918550012E+03
Z = 0.126501224E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 51.02
SUMMATION FORCE-Y = -932.89
SUMMATION FORCE-Z = 50.99

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 54437.35 MY= 461.70 MZ= -50288.63

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -51.02
SUMMATION FORCE-Y = 932.89
SUMMATION FORCE-Z = -50.99

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -54437.34 MY= -461.70 MZ= 50288.62

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	3.28767E-02	55
Y =	-1.68145E-02	57
Z =	2.36036E-02	39
RX=	4.67839E-05	62
RY=	8.16508E-06	80
RZ=	-4.45522E-05	56

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
EQP DL (EQDL)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.212799597E+02
Y = -0.107140366E+05
Z = 0.843670322E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.625871305E+01
Y = 0.918550012E+03
Z = 0.102285542E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.233445743E+02
Y = 0.108474958E+04
Z = 0.617683814E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -0.02
SUMMATION FORCE-Y = -1500.44
SUMMATION FORCE-Z = 0.03

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15373.00 MY= -0.76 MZ= -9377.01

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.02
SUMMATION FORCE-Y = 1500.44
SUMMATION FORCE-Z = -0.03

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15373.00 MY= 0.76 MZ= 9377.01

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-6.65737E-03	39
Y =	-1.58527E-01	58
Z =	3.26836E-03	97
RX=	-2.88710E-04	94
RY=	2.30835E-06	18
RZ=	1.76672E-04	46

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
EQP LL (EQLL)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.140051133E+02
Y = 0.918550012E+03
Z = 0.117668106E+02

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.874919693E+01
Y = 0.918550012E+03
Z = 0.812582634E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.185292079E+02
Y = 0.918550012E+03
Z = 0.126321091E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 119.91
SUMMATION FORCE-Y = -2002.38
SUMMATION FORCE-Z = 23.41

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 37770.61 MY= 1082.72 MZ= -127663.51

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -119.91
SUMMATION FORCE-Y = 2002.38
SUMMATION FORCE-Z = -23.41

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -37770.61 MY= -1082.72 MZ= 127663.49

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	6.65258E-02	39
Y =	-6.23018E-02	80
Z =	1.60511E-02	80
RX=	7.40100E-05	62
RY=	-7.10013E-06	80
RZ=	-4.18889E-04	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
STARTING BELT TENSION (SBT)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.140066861E+02
Y = 0.918550012E+03
Z = 0.120130499E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.186030571E+02
Y = 0.918550012E+03
Z = 0.126442117E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 157.03
SUMMATION FORCE-Y = -186.57
SUMMATION FORCE-Z = 21.60

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 21914.16 MY= 1583.91 MZ= -149778.39

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -157.03
SUMMATION FORCE-Y = 186.57
SUMMATION FORCE-Z = -21.60

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -21914.15 MY= -1583.91 MZ= 149778.36

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	8.14059E-02	39
Y =	-5.27240E-02	80
Z =	1.48110E-02	80
RX=	2.22449E-05	70
RY=	-6.10371E-06	80
RZ=	-3.57425E-04	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
WIND LOAD (+X) WL(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.121555386E+02
Y = 0.917764746E+03
Z = 0.113785444E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.307705478E+03
Y = 0.918550012E+03
Z = 0.126653876E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 575.54
SUMMATION FORCE-Y = -48.78
SUMMATION FORCE-Z = 2.44

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2154.03 MY= 6514.74 MZ= -532468.48

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = -575.54
SUMMATION FORCE-Y = 48.78
SUMMATION FORCE-Z = -2.44

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2154.03 MY= -6514.74 MZ= 532468.39

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	2.53021E-01	39
Y =	-6.84412E-02	80
Z =	2.54850E-02	80
RX=	3.54734E-05	41
RY=	-1.85643E-05	39
RZ=	-4.78873E-04	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
WIND LOAD (-X) WL(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.149856994E+02
Y = 0.917764749E+03
Z = 0.113979864E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.252881701E+03
Y = 0.918550012E+03
Z = 0.126346874E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = -575.54
SUMMATION FORCE-Y = 105.46
SUMMATION FORCE-Z = 2.45

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1177.89 MY= -6594.35 MZ= 533367.28

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 575.54
SUMMATION FORCE-Y = -105.46
SUMMATION FORCE-Z = -2.45

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1177.89 MY= 6594.35 MZ= -533367.19

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X =	-2.53265E-01	39
Y =	6.35432E-02	80
Z =	-2.25321E-02	80
RX=	-3.58775E-05	41
RY=	1.75010E-05	39
RZ=	4.40723E-04	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
WIND LOAD (+Z) WL(+Z)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.223999919E+02
Y = 0.918550012E+03
Z = -0.131161638E+05

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.967681059E+01
Y = 0.917931411E+03
Z = 0.113615544E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = -0.15
SUMMATION FORCE-Y = 59.24
SUMMATION FORCE-Z = 514.93

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 474919.70 MY= -4983.81 MZ= 1197.91

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 0.15
SUMMATION FORCE-Y = -59.24
SUMMATION FORCE-Z = -514.93

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -474919.62 MY= 4983.81 MZ= -1197.91

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 9)

	MAXIMUMS	AT NODE
X =	-8.32585E-03	55
Y =	3.06168E-02	78
Z =	1.58543E-01	80
RX=	2.57629E-04	79
RY=	-1.83064E-05	80
RZ=	7.88504E-05	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 10
WIND LOAD (-Z) WL(-Z)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.223999865E+02
Y = 0.918550012E+03
Z = -0.119553518E+05

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.104137905E+02
Y = 0.917922361E+03
Z = 0.138946268E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 0.15
SUMMATION FORCE-Y = 13.99
SUMMATION FORCE-Z = -509.18

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -471098.72 MY= 5303.47 MZ= -89.39

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = -0.15
SUMMATION FORCE-Y = -13.99
SUMMATION FORCE-Z = 509.18

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 471098.65 MY= -5303.47 MZ= 89.39

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 10)

	MAXIMUMS	AT NODE
X =	1.49546E-02	55
Y =	-2.70971E-02	78
Z =	-1.63908E-01	80
RX=	-2.64174E-04	79
RY=	2.35005E-05	80
RZ=	-7.60619E-05	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 15
TEMPERATURE LOAD (TL)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.223999460E+02
Y = 0.918550012E+03
Z = 0.638668173E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.223940926E+02
Y = 0.918550012E+03
Z = 0.638043508E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = 0.77
SUMMATION FORCE-Y = 2.08
SUMMATION FORCE-Z = -0.32

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -306.32 MY= 12.07 MZ= -659.82

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = -0.77
SUMMATION FORCE-Y = -2.08
SUMMATION FORCE-Z = 0.32

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 306.32 MY= -12.07 MZ= 659.82

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 15)

	MAXIMUMS	AT NODE
X =	2.65047E-04	52
Y =	-9.80937E-03	85
Z =	-2.86179E-04	3
RX=	-2.82350E-05	45
RY=	2.00882E-06	80
RZ=	-1.78057E-05	85

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 16
PIPE LATERAL LOAD (PL)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.140074055E+02
Y = 0.918550012E+03
Z = 0.230353846E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.359882829E+01
Y = 0.918550012E+03
Z = 0.126503675E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 16)
SUMMATION FORCE-X = 51.04
SUMMATION FORCE-Y = -12.08
SUMMATION FORCE-Z = 50.99

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47359.65 MY= 461.86 MZ= -47597.01

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 16)
SUMMATION FORCE-X = -51.04
SUMMATION FORCE-Y = 12.08
SUMMATION FORCE-Z = -50.99

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47359.64 MY= -461.86 MZ= 47597.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 16)

	MAXIMUMS	AT NODE
X =	3.33712E-02	55
Y =	-8.65487E-03	80
Z =	2.36194E-02	39
RX=	2.65528E-05	97
RY=	7.84554E-06	80
RZ=	-5.92755E-05	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 17
EQP TEST LOAD (EQTL)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.225015057E+02
Y = 0.918550012E+03
Z = 0.624952530E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.263891530E+01
Y = 0.918550012E+03
Z = 0.106305355E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.220725150E+02
Y = 0.918550012E+03
Z = 0.640006606E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 17)
SUMMATION FORCE-X = -0.08
SUMMATION FORCE-Y = -3660.75
SUMMATION FORCE-Z = 0.02

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 38933.20 MY= -0.94 MZ= -9584.17

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 17)
SUMMATION FORCE-X = 0.08
SUMMATION FORCE-Y = 3660.75
SUMMATION FORCE-Z = -0.02

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -38933.20 MY= 0.94 MZ= 9584.17

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 17)

	MAXIMUMS	AT NODE
X =	-3.65763E-03	97
Y =	-1.34031E-01	57
Z =	5.76461E-03	16
RX=	-2.44186E-04	54
RY=	4.02155E-06	14
RZ=	-1.60097E-04	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 18
MONORAIL LOAD (ML1)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.139999996E+02
Y = 0.918550012E+03
Z = 0.158999069E+02

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.158555825E+00
Y = 0.918550012E+03
Z = 0.169740788E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.172499882E+02
Y = 0.918550012E+03
Z = 0.126499998E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 18)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -29.21
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 494.96 MY= 0.03 MZ= 3.71

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 18)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 29.21
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -494.96 MY= -0.03 MZ= -3.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 18)

	MAXIMUMS	AT NODE
X =	-5.16926E-05	52
Y =	-9.10717E-04	56
Z =	1.60718E-05	16
RX=	-1.75179E-06	86
RY=	1.14342E-08	4
RZ=	1.04313E-06	58

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 19
MONORAIL LOAD (ML2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.271902438E+01
Y = 0.918550012E+03
Z = 0.169027366E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.477506074E+01
Y = 0.918550012E+03
Z = 0.126499998E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 19)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -29.41
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 495.31 MY= 0.01 MZ= -79.97

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 19)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 29.41
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -495.31 MY= -0.01 MZ= 79.97

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 19)

	MAXIMUMS	AT NODE
X =	-7.58341E-05	97
Y =	-2.91079E-03	57
Z =	1.23669E-04	16
RX=	-5.29732E-06	54
RY=	8.59785E-08	14
RZ=	-3.68320E-06	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 20
MONORAIL LOAD (ML3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.139999996E+02
Y = 0.918550012E+03
Z = 0.130000686E+02

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.554381929E+01
Y = 0.918550012E+03
Z = 0.168421172E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.178499404E+02
Y = 0.918550012E+03
Z = 0.952499220E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 20)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -29.58
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 500.09 MY= -0.01 MZ= -165.85

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 20)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 29.58
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -500.09 MY= 0.01 MZ= 165.85

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 20)

	MAXIMUMS	AT NODE
X =	-1.48480E-04	97
Y =	-5.44465E-03	57
Z =	2.31747E-04	16
RX=	-9.90628E-06	54
RY=	1.63174E-07	14
RZ=	-7.79153E-06	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 21
MONORAIL LOAD (ML4)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.703556682E+01
Y = 0.918550012E+03
Z = 0.168751623E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 21)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -29.49
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 497.61 MY= -0.01 MZ= -207.46

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 21)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 29.49
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -497.61 MY= 0.01 MZ= 207.46

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 21)

	MAXIMUMS	AT NODE
X =	-1.88572E-04	97
Y =	-6.79552E-03	57
Z =	2.89251E-04	16
RX=	-1.23634E-05	54
RY=	2.04744E-07	14
RZ=	-9.99010E-06	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 22
MONORAIL LOAD (ML5)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.105585703E+02
Y = 0.918550012E+03
Z = 0.168480820E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.113833364E+02
Y = 0.918550012E+03
Z = 0.105666631E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 22)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -29.57
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 500.89 MY= -0.05 MZ= -311.26

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 22)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 29.57
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -500.89 MY= 0.05 MZ= 311.26

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 22)

	MAXIMUMS	AT NODE
X =	-9.18028E-05	97
Y =	-3.52462E-03	57
Z =	1.49699E-04	16
RX=	-6.41999E-06	54
RY=	-1.10526E-07	17
RZ=	-4.98389E-06	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 23
MONORAIL LOAD (ML6)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.223999721E+02
Y = 0.918548188E+03
Z = 0.924998706E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.139809374E+02
Y = 0.918550012E+03
Z = 0.169233032E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.225500225E+02
Y = 0.918550012E+03
Z = 0.126499998E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 23)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -29.35
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 495.85 MY= 0.04 MZ= -413.15

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 23)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 29.35
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -495.85 MY= -0.04 MZ= 413.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 23)

	MAXIMUMS	AT NODE
X =	-6.56076E-05	52
Y =	-1.37949E-03	50
Z =	-2.66593E-05	3
RX=	2.79066E-06	93
RY=	-2.70529E-08	16
RZ=	1.68581E-06	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 24
MONORAIL LOAD (ML7)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.181999978E+02
Y = 0.918550012E+03
Z = -0.367510259E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.176581935E+02
Y = 0.918550012E+03
Z = 0.167205898E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.235250094E+02
Y = 0.918550012E+03
Z = 0.679062759E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 24)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -29.93
SUMMATION FORCE-Z = 0.02

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 515.18 MY= -0.36 MZ= -530.38

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 24)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 29.93
SUMMATION FORCE-Z = -0.02

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -515.18 MY= 0.36 MZ= 530.38

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 24)

	MAXIMUMS	AT NODE
X =	5.28450E-05	39
Y =	-7.03355E-04	50
Z =	1.91211E-05	80
RX=	1.41043E-06	93
RY=	-1.00753E-07	80
RZ=	-1.82331E-06	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 25
MONORAIL LOAD (ML8)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.195999993E+02
Y = 0.918550012E+03
Z = 0.768341445E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.211996968E+02
Y = 0.918550012E+03
Z = 0.164777938E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.220516150E+02
Y = 0.918550012E+03
Z = 0.660161707E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 25)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -30.65
SUMMATION FORCE-Z = 0.03

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 533.55 MY= -0.71 MZ= -647.06

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 25)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 30.65
SUMMATION FORCE-Z = -0.03

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -533.55 MY= 0.71 MZ= 647.06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 25)

	MAXIMUMS	AT NODE
X =	1.45265E-04	65
Y =	-1.21782E-03	51
Z =	2.80596E-05	80
RX=	-2.39955E-06	96
RY=	-1.89271E-07	80
RZ=	-4.52850E-06	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 26
MONORAIL LOAD (ML9)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.223999939E+02
Y = 0.918549615E+03
Z = 0.350018626E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.221769218E+02
Y = 0.918550012E+03
Z = 0.640001211E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 26)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -31.06
SUMMATION FORCE-Z = 0.04

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 543.52 MY= -0.88 MZ= -716.22

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 26)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 31.06
SUMMATION FORCE-Z = -0.04

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -543.52 MY= 0.88 MZ= 716.22

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 26)

	MAXIMUMS	AT NODE
X =	1.99773E-04	65
Y =	-1.55385E-03	51
Z =	3.37781E-05	80
RX=	-3.06420E-06	96
RY=	-2.42895E-07	80
RZ=	-6.07446E-06	80

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 27
MONORAIL LOAD (ML10)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.564875523E+01
Y = 0.918550012E+03
Z = 0.142576635E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.251499294E+02
Y = 0.918552747E+03
Z = 0.640003764E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 27)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -29.77
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 426.29 MY= -0.04 MZ= -168.16

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 27)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 29.77
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -426.29 MY= 0.04 MZ= 168.16

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 27)

	MAXIMUMS	AT NODE
X =	-1.23951E-04	97
Y =	-4.57220E-03	57
Z =	1.94803E-04	16
RX=	-8.31889E-06	54
RY=	1.36510E-07	14
RZ=	-6.57622E-06	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 28
MONORAIL LOAD (ML11)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.555100335E+01
Y = 0.918550012E+03
Z = 0.166381540E+02

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.307001033E+02
Y = 0.918559882E+03
Z = 0.640013590E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 28)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -29.60
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 493.36 MY= -0.03 MZ= -164.29

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 28)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 29.60
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -493.36 MY= 0.03 MZ= 164.29

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 28)

	MAXIMUMS	AT NODE
X =	-1.46701E-04	97
Y =	-5.37603E-03	57
Z =	2.28790E-04	16
RX=	-9.78154E-06	54
RY=	1.61053E-07	14
RZ=	-7.69596E-06	86

********** END OF DATA FROM INTERNAL STORAGE **********

1249. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= FEB 29,2020 TIME= 15:51:11 ****

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* Details about additional assistance from *
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